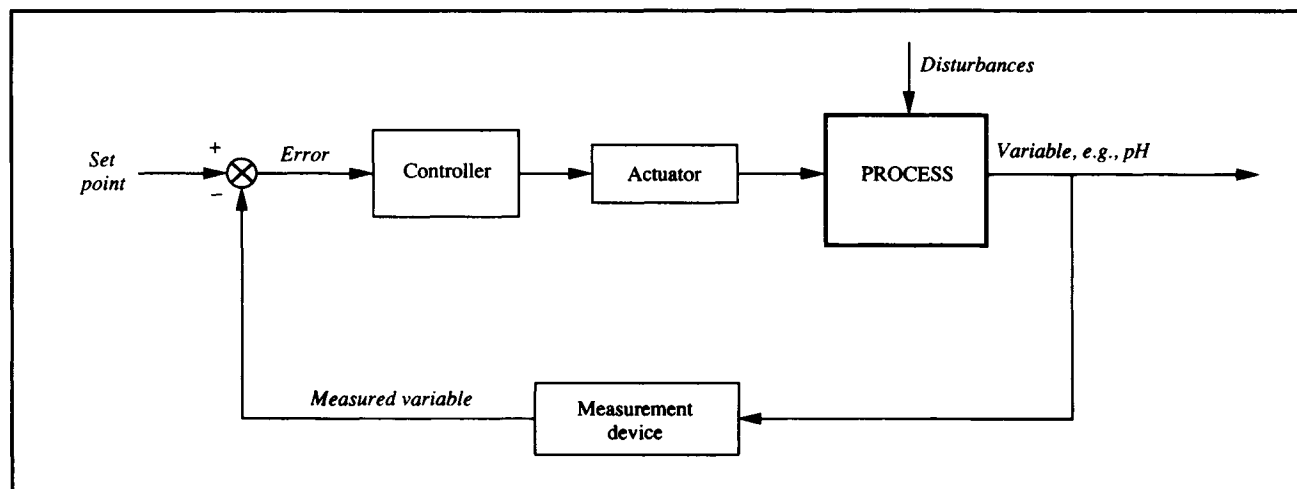


Figure 13.18 Components of a feedback control loop.



major fluctuations occur in operating conditions or if cell properties and model parameters change with time, the model may become inadequate and estimation accuracy will decline. *Adaptive estimators* are used to adjust faulty model parameters as the fermentation proceeds to alleviate problems caused by error in the equations. Off-line measurements can also be incorporated into the estimation procedure as they become available to improve the accuracy and reliability of prediction. Another technique involves 'generic' software sensors which use generally-structured models rather than model equations specific to the process of interest. Process characteristics are then 'learnt' or incorporated into the model structure as information becomes available from on- and off-line measurements. The primary advantage of generic sensors is that the software-development time is significantly reduced [28].

Mass-balancing techniques are another approach to state estimation. As described in Chapter 4, biomass can be estimated from stoichiometry and other process measurements using elemental balances. This method is suitable for fermentation of defined media, but is difficult to apply with complex medium ingredients such as molasses, casein hydrolysate, soybean meal and corn steep liquor which have undefined elemental composition. Another disadvantage is that the accuracy of biomass estimation often depends on measurement of substrate or product concentration which must be done off-line.

13.4.6 Feedback Control

Let us assume we wish to maintain the pH in a bioreactor at a constant value against a variety of disturbances, for example,

from metabolic activity. One of the simplest control schemes is a conventional *feedback control loop*, the basic elements of which are shown in Figure 13.18. A measurement device senses the value of the pH and sends the signal to a controller. At the controller, the measured value is compared with the desired value known as the *set point*. The deviation between measured and desired values is the *error*, which is used by the controller to determine what action must be taken to correct the process. The controller may be a person who monitors the process measurements and decides what to do; more often the controller is an automatic electronic, pneumatic or computer device. The controller produces a signal which is transmitted to the actuator, which executes the control action. In a typical system for pH control, an electrode would serve as the measurement device and a pump connected to a reservoir of acid or alkali as the actuator. Simple *on-off control* is generally sufficient for pH; if the measured pH falls below the set point, the controller switches on the pump which adds alkali to the fermenter. When enough alkali is added and the pH returned to the set value, the pump is switched off. Small deviations from the set point are usually tolerated in on-off control to avoid rapid switching and problems due to measurement delay.

On-off control is used when the actuator is an on-off device, such as a single-speed pump. If function of the actuator covers a continuous range, such as a variable-speed pump for supply of cooling water or a valve determining the rate of air flow, it is common to use *proportional-integral-derivative* or *PID control*. With PID control, the control action is determined in proportion to the error, the integral of the error and the derivative of the error with respect to time. The relative weightings given to these functions determine the response of

the controller and the overall 'strength' of the control action. PID control is used to determine, for example, whether the pump speed for cooling water in the coils of a fermenter should be increased by a small or large amount when a certain increase in reactor temperature is detected above the set point. Proper adjustment of PID controllers usually provides excellent regulation of the measured variable. However, poorly tuned controllers can destabilise a process and cause continuous or accentuated fluctuations in culture conditions. Considerations for adjusting PID controllers are covered in texts on process control, for example [29].

Close fermentation control requires simultaneous monitoring and adjustment of many parameters. Instead of individual controllers for each function, it is becoming commonplace to use a single computer or microprocessor for several feedback control loops. The computer logs measurements from a range of sensors in a time sequence and generates electronic signals which may be used directly or indirectly to drive various actuators. Application of computers requires digitisation of signals from the sensor; after digital-to-analogue conversion of the output, the computer or microprocessor can provide the same PID functions as a conventional analogue controller. If computers are used to drive the actuator devices, the system is said to be under *direct digital control* (DDC).

13.4.7 Indirect Metabolic Control

Maintaining particular values of temperature or pH is a rather indirect approach to bioreactor control; the wider objective is to optimise performance of the catalyst and maximise production of the desired product. Certain derived variables such as the oxygen-uptake rate and respiratory quotient can be calculated from on-line fermentation measurements; these variables reflect to some extent the biological state of the culture. It can be advantageous to base control actions on deviation of these metabolic variables from desired values rather than on environmental conditions.

Indirect metabolic control is often used in fed-batch culture of bakers' yeast. Due to the Crabtree effect, yeast metabolism can switch from respiratory to fermentative mode depending on prevailing glucose and dissolved-oxygen concentrations. Maximum biomass yield from substrate is achieved at relatively low glucose concentrations in the presence of adequate oxygen; fermentative metabolism occurs if the glucose concentration rises above a certain level even though oxygen may be present. Fermentative metabolism should be avoided for biomass production because the yield of cells is reduced as ethanol and carbon dioxide are formed as end-products. The respiratory quotient RQ is a convenient

indicator of metabolic state in this process; RQ values above about 1 indicate ethanol formation. In industrial bakers' yeast production, the feed rate of glucose to the culture is controlled to maintain RQ values within the desired range.

13.4.8 Programmed Control

Because of the inherent time-varying character of batch and fed-batch fermentations, maintaining a constant environment or constant values of metabolic variables is not always the optimal control strategy. Depending on the process, changes in variables such as pH and temperature at critical times can improve production rate and yield. Varying the rate of the feed is important in fed-batch bakers' yeast fermentations to minimise the Crabtree effect and maximise biomass production. Feed rate is also manipulated in *E. coli* fermentations to reduce by-product synthesis. In secondary-metabolite fermentations, specific growth rate should be high at the start of the culture but, at high cell densities, different conditions are required to slow growth and stimulate product formation. Similar strategies are needed to optimise protein synthesis from recombinant organisms. Expression of recombinant product is usually avoided at the start of the culture because cell growth is adversely affected; however, later in the batch an inducer is added to switch on protein synthesis.

For many bioprocesses, a particular time sequence of pH, temperature, dissolved-oxygen tension, feed rate and other variables is required to develop the culture in such a way that productivity is maximised. A control strategy that can accommodate wide-ranging changes in fermentation variables is *programmed control*, also known as *batch fermenter scheduling*. In programmed control, the control policy consists of a schedule of control functions to be implemented at various times during the process. This type of control requires a detailed understanding of the requirements of the process at various stages and a reasonably complete and accurate mathematical model of the system.

13.4.9 Application of Artificial Intelligence in Bioprocess Control

Several different approaches to bioprocess control have been described in the preceding sections. However, fermentation systems are by nature multivariable with non-linear and time-varying properties, and conventional control strategies may not be totally satisfactory. In the industrial environment, additional problems are caused by unexpected disturbances which affect process operation. Most bioprocesses cannot be described exactly by a mathematical model; it is also difficult

to identify beforehand the optimum values of metabolic or environmental parameters and controller response functions. The flexibility of computer software is a significant advantage in complex situations; researchers continue to make progress in developing optimal, adaptive and self-tuning algorithms for bioprocess control. However, although these techniques offer much for improving process performance, the usual mathematical approach may not be the best way to solve bioprocess control problems.

A relatively recent development in control engineering is application of artificial intelligence techniques, especially *knowledge-based expert systems*. The term 'expert system' usually refers to computer software that processes linguistically-formulated 'knowledge' about a particular subject; this knowledge is represented by simple rules. The most important step in building a useful expert system is extraction of heuristic or subjective rules of thumb from experimental data and human experts. Use can be made of the ever-increasing quantity of measured fermentation data from industrial processes to synthesise the wide range of rules required for expert systems. The knowledge available is encoded using a computer as IF/THEN type rules, e.g. IF the cell density is high, THEN dilution with sterile water is recommended, or IF the dissolved-oxygen concentration increases quickly AND the rate of carbon dioxide evolution decreases quickly AND the sugar concentration drops AND the pH increases, THEN broth harvest is advised. Information about the progress of highly productive fermentations is also stored in the knowledge base. As measured data become available from fermentations in progress, pattern recognition techniques are applied to assess the results and, in conjunction with the knowledge base and its rules, handle a variety of operating problems or disturbances. The expert system can also 'learn' new information about process behaviour by upgrading its knowledge base. To maximise the potential of expert systems for intelligent supervisory control of fermentation processes, large and representative data bases of microbiological information and engineering knowledge must be established for use in rule formulation and interpretation of process phenomena. Expert systems can also be used for fault diagnosis, estimating unmeasurable fermentation properties, reconciling contradictory data and computer-aided modelling of metabolism [30–33].

Another area of artificial intelligence with applications in fermentation control is the theory of *neural networks*. Neural networks are particularly suited for extracting useful information from complex and uncertain data such as fermentation measurements, and for formulating generalisations from previous experience. They offer the ability to learn complex, non-linear relationships between variables and may therefore be useful in development of process models and for estimating

unknown fermentation parameters. Neural network technology is based on an analogy with the brain in that information is stored in the form of connected computational elements or weights (synapses) between artificial neurons. The most commonly-used neural structure is the feed-forward network in which neurons are arranged in layers; incoming signals at the input layer are fed forward through the network connections to the output layer. The topology of the network provides it with powerful data processing capabilities. To solve a problem, a network structure is chosen and examples of the knowledge to be acquired are shown to the network which adjusts the synaptic strength of its neural connections so that, in effect, the knowledge is integrated within the structure. When the real data set is presented to the system, the network is able to predict outcomes based on the learning set. For example, information about feed flow rate and substrate and biomass concentrations as a function of time can be used to develop a neural network for analysis of transient behaviour in continuous fermentation which is able to predict future changes in substrate and cell concentrations [34–36]. Although a considerable amount of research remains to be done in this area, neural networks are a promising tool for modelling, estimating and predicting fermentation characteristics.

13.5 Ideal Reactor Operation

So far in this chapter we have considered the configuration of bioreactors and aspects of their construction and control. Another important factor affecting reactor performance is mode of operation. There are three principal modes of bioreactor operation: batch, fed-batch and continuous. Choice of operating strategy has a significant effect on substrate conversion, product concentration, susceptibility to contamination and process reliability.

Characteristics such as final substrate, product and biomass concentrations and the time required for conversion can be determined for different reactor operating schemes using mass balances. For a general reaction system, we can relate rates of change of component masses in the system to the rate of reaction using Eq. (6.5):

$$\frac{dM}{dt} = \hat{M}_i - \hat{M}_o + R_G - R_C \quad (6.5)$$

where M is mass of component A in the vessel, t is time, $\hat{M}_i$ is the mass flow rate of A entering the reactor, $\hat{M}_o$ is the mass flow rate of A leaving, R_G is the mass rate of generation of A by reaction, and R_C is the mass rate of consumption of A by reaction. In this section we consider application of Eq. (6.5) to batch,

fed-batch and continuous reactors for enzyme conversion and fermentation.

13.5.1 Batch Operation of a Mixed Reactor

Batch processes operate in closed systems; substrate is added at the beginning of the process and products removed only at the end. Aerobic reactions are not batch operations in the strictest sense; the low solubility of oxygen in aqueous media means that it must be supplied continuously while carbon dioxide and other off-gases are removed. However, reactors with neither input nor output of liquid are classified as batch. If there are no leaks or evaporation from the vessel, the liquid volume in batch reactors can be considered constant.

Most commercial bioreactors are mixed vessels operated in batch. The classic mixed reactor is the stirred tank; however mixed reactors can also be of bubble column, airlift or other configuration as long as concentrations of substrate, product and catalyst inside the vessel are uniform. The cost of running a batch reactor depends on the time taken to achieve the desired product concentration or level of substrate conversion; operating costs are reduced if the reaction is completed quickly. It is therefore useful to be able to predict the time required for batch reactions.

13.5.1.1 Enzyme reaction

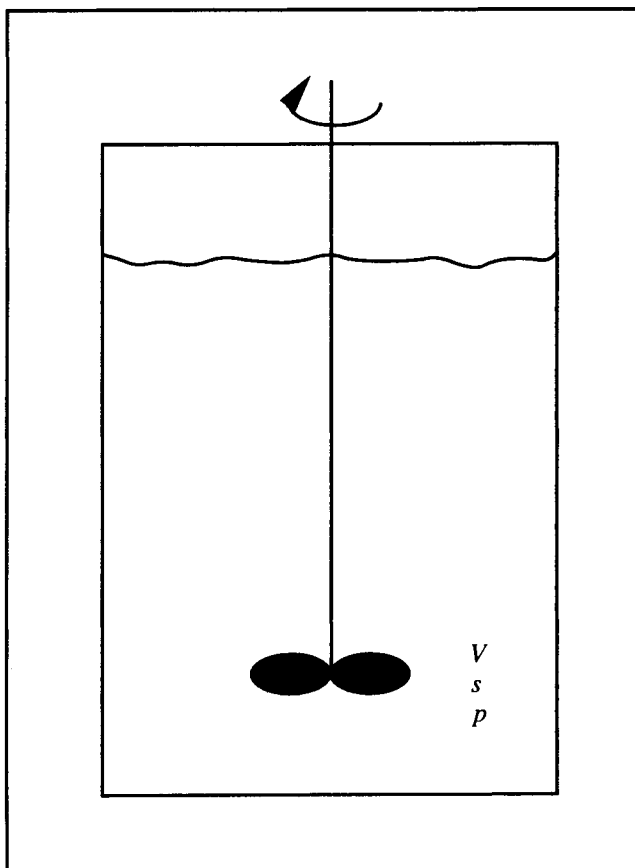
Let us apply Eq. (6.5) to the limiting substrate in a batch enzyme reactor such as that shown in Figure 13.19. $\hat{M}_i = \hat{M}_o = 0$ because there is no substrate flow into or out of the vessel; mass of substrate in the reactor, M , is equal to the substrate concentration s multiplied by the liquid volume V . As substrate is not generated in the reaction, $R_G = 0$. Rate of substrate consumption R_C is equal to the volumetric rate of reaction v multiplied by V ; v is given by Eq. (11.31). Therefore, the mass balance from Eq. (6.5) is:

$$\frac{d(sV)}{dt} = \frac{-v_{\max} s}{K_m + s} V \quad (13.7)$$

where $v_{\max}$ is the maximum rate of enzyme reaction and K_m is the Michaelis constant. Because V is constant in batch reactors, we can take it outside of the differential and cancel through the equation to give:

$$\frac{ds}{dt} = \frac{-v_{\max} s}{K_m + s} \quad (13.8)$$

Figure 13.19 Flowsheet for a stirred batch enzyme-reactor.



Integration of this differential equation provides an expression for batch reaction time. Assuming $v_{\max}$ and K_m are constant during the reaction, separating variables:

$$- \int dt = \int \frac{K_m + s}{v_{\max} s} ds \quad (13.9)$$

and integrating with initial condition $s = s_0$ at $t = 0$ gives:

$$t_b = \frac{K_m}{v_{\max}} \ln \frac{s_0}{s_f} + \frac{s_0 - s_f}{v_{\max}} \quad (13.10)$$

where t_b is the batch reaction time required to reduce the substrate concentration from s_0 to s_f . The batch time required to produce a certain concentration of product can be determined from Eq. (13.10) and stoichiometric relationships.

Example 13.1 Time course for batch enzyme conversion

An enzyme is used to produce a compound used in manufacture of sunscreen lotion. $v_{\max}$ for the enzyme is $2.5 \text{ mmol m}^{-3} \text{ s}^{-1}$; K_m is 8.9 mM . The initial concentration of substrate is 12 mM . Plot the time required for batch reaction as a function of substrate conversion.

Solution:

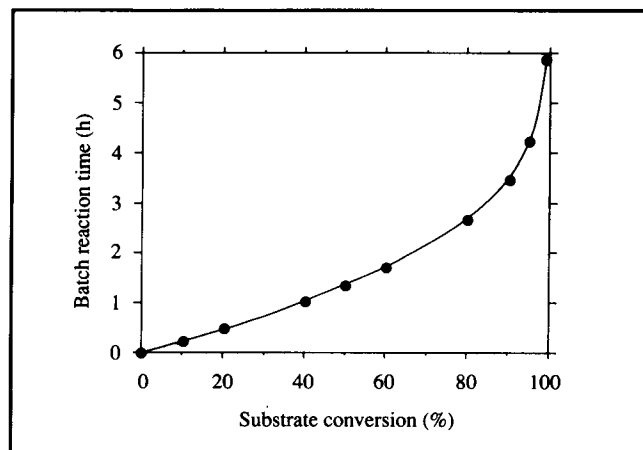
$s_0 = 12 \text{ mM}$. Converting units of $v_{\max}$ to mM h^{-1} :

$$\begin{aligned} v_{\max} &= 2.5 \text{ mmol m}^{-3} \text{ s}^{-1} \cdot \left| \frac{3600 \text{ s}}{1 \text{ h}} \right| \cdot \left| \frac{1 \text{ m}^3}{1000 \text{ l}} \right| \\ &= 9 \text{ mmol l}^{-1} \text{ h}^{-1} \\ &= 9 \text{ mM h}^{-1}. \end{aligned}$$

Results from application of Eq. (13.10) are tabulated below and plotted in Figure 13E1.1.

Substrate conversion (%)	s_f (mM)	t_b (h)
0	12.0	0.00
10	10.8	0.24
20	9.6	0.49
40	7.2	1.04
50	6.0	1.35
60	4.8	1.71
80	2.4	2.66
90	1.2	3.48
95	0.60	4.23
99	0.12	5.87

Figure 13E1.1 Batch reaction time as a function of substrate conversion for a mixed enzyme reactor.



At high substrate conversions, the time required to achieve an incremental increase in conversion is greater than at low conversions. Accordingly, the benefit gained from conversions greater than 80–90% must be weighed against the significantly greater reaction time and reactor operating costs involved.

As discussed in Section 11.5, enzymes are subject to deactivation. Thus, the concentration of active enzyme in the reactor, and therefore the value of $v_{\max}$, may change during reaction. When deactivation is significant, variation of $v_{\max}$ with time can be expressed using Eq. (11.44) so that Eq. (13.8) becomes:

$$\frac{ds}{dt} = \frac{-v_{\max 0} e^{-k_d t} s}{K_m + s} \quad (13.11)$$

where $v_{\max 0}$ is the value of $v_{\max}$ before deactivation occurs and k_d is the first-order deactivation rate constant. Separating variables gives:

$$-\int e^{-k_d t} dt = \int \frac{K_m + s}{v_{\max 0} s} ds. \quad (13.12)$$

Integrating Eq. (13.12) with initial condition $s = s_0$ at $t = 0$ gives:

$$t_b = \frac{-1}{k_d} \ln \left[1 - k_d \left(\frac{K_m}{v_{\max 0}} \ln \frac{s_0}{s_f} + \frac{s_0 - s_f}{v_{\max 0}} \right) \right] \quad (13.13)$$

where t_b is the batch reaction time and s_f is the final substrate concentration.

Example 13.2 Batch reaction time with enzyme deactivation

The enzyme of Example 13.1 deactivates with half-life 4.4 h. Compare with Figure 13E1.1 the batch reaction time required to achieve 90% substrate conversion.

Solution:

$s_0 = 12$ mM; $v_{\max 0} = 9$ mM h⁻¹; $K_m = 8.9$ mM. $s_f = (0.1 s_0) = 1.2$ mM. The deactivation rate constant is calculated from the half-life t_h using Eq. (11.45):

$$k_d = \frac{\ln 2}{t_h} = \frac{\ln 2}{4.4 \text{ h}} = 0.158 \text{ h}^{-1}.$$

Substituting values into Eq. (13.13) gives:

$$t_b = \frac{-1}{0.158 \text{ h}^{-1}} \ln \left[1 - 0.158 \text{ h}^{-1} \left(\frac{8.9 \text{ mM}}{9 \text{ mM h}^{-1}} \ln \frac{12 \text{ mM}}{1.2 \text{ mM}} + \frac{(12 - 1.2) \text{ mM}}{9 \text{ mM h}^{-1}} \right) \right] \\ = 5.0 \text{ h}.$$

With enzyme deactivation, the time required for 90% conversion increases from 3.5 h to 5.0 h.

For reactions with immobilised enzymes, Eq. (13.8) must be modified to account for mass-transfer effects:

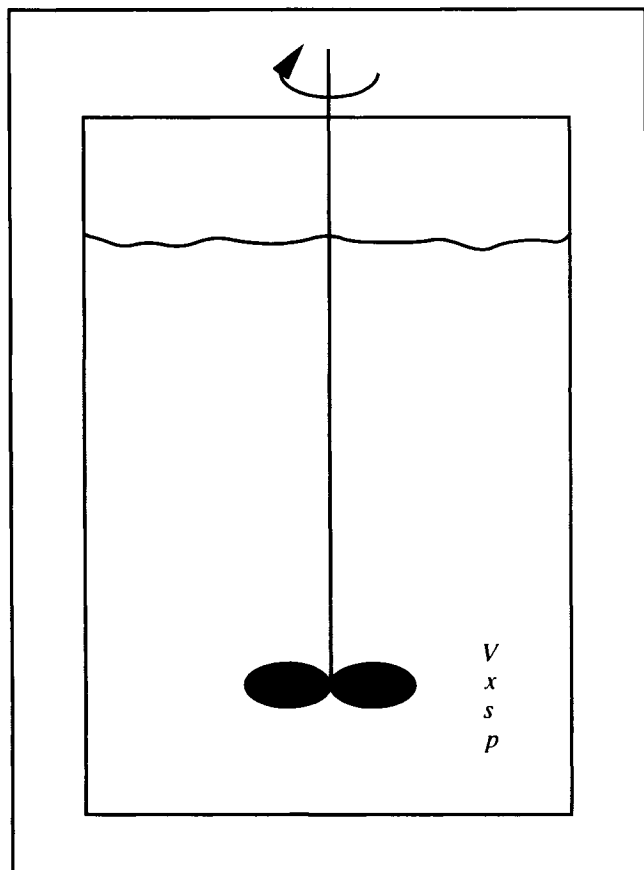
$$\frac{ds}{dt} = -\eta_T \frac{v_{\max} s}{K_m + s} \quad (13.14)$$

where η_T is the total effectiveness factor incorporating internal and external mass-transfer limitations (see Section 12.5), s is the bulk substrate concentration, and $v_{\max}$ and K_m are the intrinsic kinetic parameters. In principal, integration of

Eq. (13.14) allows evaluation of t_b ; however, because η_T is a function of s , integration is not straightforward.

13.5.1.2 Cell culture

Similar analysis can be applied to fermentation processes to evaluate reaction times. Let us perform a mass balance on cells in a well-mixed batch fermenter using Eq. (6.5); a typical flow-sheet for this system is shown in Figure 13.20. $\dot{M}_i = \dot{M}_o = 0$ because cells do not flow into or out of the vessel. Mass of cells in the reactor, M , is equal to the cell concentration x multi-

Figure 13.20 Flowsheet for a stirred batch fermenter.

plied by the liquid volume V . The mass rate of cell growth R_G is equal to $r_X V$ where r_X is the volumetric rate of growth. From Eq. (11.52), $r_X = \mu x$ where μ is the specific growth rate. If cell death takes place in the reactor alongside growth, $R_C = r_d V$ where r_d is the volumetric rate of cell death. From Eq. (11.85), r_d can be expressed using the first-order equation: $r_d = k_d x$ where k_d is the specific death constant. Therefore, Eq. (6.5) for cells in a batch reactor is:

$$\frac{d(xV)}{dt} = \mu x V - k_d x V. \quad (13.15)$$

For V constant, Eq. (13.15) becomes:

$$\frac{dx}{dt} = (\mu - k_d) x. \quad (13.16)$$

Because μ in batch culture remains approximately constant and equal to $\mu_{\max}$ for most of the growth period (see Section 11.7.3), if k_d likewise remains constant, we can integrate Eq. (13.16) directly to find the relationship between batch time and cell concentration. Using the initial condition that $x = x_0$ at $t = 0$:

$$x = x_0 e^{(\mu_{\max} - k_d)t}. \quad (13.17)$$

If x_f is the final biomass concentration after batch culture time t_b , rearrangement of Eq. (13.17) gives:

$$t_b = \frac{1}{\mu_{\max} - k_d} \ln \frac{x_f}{x_0}. \quad (13.18)$$

If the rate of cell death is negligible compared with growth, $k_d \ll \mu_{\max}$ and Eqs (13.17) and (13.18) reduce to:

$$x = x_0 e^{\mu_{\max} t} \quad (13.19)$$

and:

$$t_b = \frac{1}{\mu_{\max}} \ln \frac{x_f}{x_0}. \quad (13.20)$$

Therefore, we can calculate using Eq. (13.18) or Eq. (13.20) the time required to achieve cell density x_f starting from cell density x_0 . Batch culture time can also be related to substrate conversion and product concentration using expressions for rates of substrate uptake and product formation derived in Chapter 11.

Let us apply Eq. (6.5) to the growth-limiting substrate in a batch fermentation. $\hat{M}_i = \hat{M}_o = 0$ because substrate does not flow into or out of the reactor; the mass of substrate in the reactor, M , is equal to sV where s is substrate concentration and V is liquid volume. Substrate is not generated; therefore $R_G = 0$. R_C is equal to $r_s V$ where r_s is the volumetric rate of substrate uptake. As discussed in Section 11.10, the expression for r_s depends on whether extracellular product is formed by the culture and the relationship between product synthesis and energy generation in the cell. If product is formed but not directly coupled with energy metabolism, r_s is given by Eq. (11.76) and:

$$R_C = r_s V = \left(\frac{\mu}{Y_{XS}} + \frac{q_P}{Y_{PS}} + m_S \right) x V \quad (13.21)$$

where μ is the specific growth rate, Y_{XS} is the true biomass yield from substrate, q_P is the specific rate of product formation, Y_{PS} is the true product yield from substrate and m_S is the maintenance coefficient. Therefore, from Eq. (6.5) the mass-balance equation is:

$$\frac{d(sV)}{dt} = -\left(\frac{\mu}{Y_{XS}} + \frac{q_P}{Y_{PS}} + m_S\right)xV. \quad (13.22)$$

For μ equal to $\mu_{\max}$ and V constant, we can write Eq. (13.22) as:

$$\frac{ds}{dt} = -\left(\frac{\mu_{\max}}{Y_{XS}} + \frac{q_P}{Y_{PS}} + m_S\right)x. \quad (13.23)$$

Because x is a function of time, we must substitute an expression for x into Eq. (13.23) before it can be integrated. Assuming cell death is negligible, x is given by Eq. (13.19). Therefore:

$$\frac{ds}{dt} = -\left(\frac{\mu_{\max}}{Y_{XS}} + \frac{q_P}{Y_{PS}} + m_S\right)x_0 e^{\mu_{\max} t}. \quad (13.24)$$

If all the bracketed terms are constant during culture, Eq. (13.24) can be integrated directly with initial condition $s = s_0$ at $t = 0$ to obtain the following equation:

$$t_b = \frac{1}{\mu_{\max}} \ln \left[1 + \frac{s_0 - s_f}{\left(\frac{1}{Y_{XS}} + \frac{q_P}{\mu_{\max} Y_{PS}} + \frac{m_S}{\mu_{\max}} \right) x_0} \right] \quad (13.25)$$

where t_b is the batch culture time and s_f is the final substrate concentration.

For products indirectly coupled or not related at all to energy metabolism, evaluating q_P requires further analysis (see Sections 11.9.2 and 11.9.3). However, Eq. (13.25) can be simplified if no product is formed or if production is directly linked with energy metabolism; in these cases the expression for rate of substrate consumption does not contain a separate term for product synthesis (see Sections 11.10.1 and 11.10.2) and Eq. (13.25) reduces to:

$$t_b = \frac{1}{\mu_{\max}} \ln \left[1 + \frac{s_0 - s_f}{\left(\frac{1}{Y_{XS}} + \frac{m_S}{\mu_{\max}} \right) x_0} \right]. \quad (13.26)$$

If, in addition, maintenance requirements can be neglected:

$$t_b = \frac{1}{\mu_{\max}} \ln \left[1 + \frac{Y_{XS}}{x_0} (s_0 - s_f) \right]. \quad (13.27)$$

To obtain an expression for batch culture time as a function of product concentration, we must apply Eq. (6.5) to the product. Again, $\hat{M}_i = \hat{M}_o = 0$; mass of product in the reactor, M , is equal to pV where p is product concentration and V is liquid volume. Assuming product is not consumed, $R_C = 0$. R_G is equal to $r_P V$ where r_P is the volumetric rate of product formation. According to Eq. (11.67), for all types of product $r_P = q_P x$ where q_P is the specific rate of product formation. Therefore:

$$R_G = r_P V = q_P x V \quad (13.28)$$

and, from Eq. (6.5), the mass-balance equation is:

$$\frac{d(pV)}{dt} = q_P x V. \quad (13.29)$$

If cell death is negligible x is given by Eq. (13.19). Therefore, for V constant, we can write Eq. (13.29) as:

$$\frac{dp}{dt} = q_P x_0 e^{\mu_{\max} t}. \quad (13.30)$$

If q_P is constant, Eq. (13.30) can be integrated directly with initial condition $p = p_0$ at $t = 0$ to obtain the following equation for batch culture time as a function of the final product concentration p_f :

$$t_b = \frac{1}{\mu_{\max}} \ln \left[1 + \frac{\mu_{\max}}{x_0 q_P} (p_f - p_0) \right]. \quad (13.31)$$

The batch culture time needed to achieve a certain biomass

density can be evaluated using Eq. (13.18) or (13.20). If cell death is negligible, the time required for a particular level of substrate conversion can be calculated using Eq. (13.25), (13.26) or (13.27) depending on the type of product and the importance

of maintenance metabolism. For negligible cell death and constant q_p , the batch time required to achieve a particular product concentration can be found from Eq. (13.31). Application of these equations is illustrated in Example 13.3.

Example 13.3 Batch culture time

Zymomonas mobilis is used to convert glucose to ethanol in a batch fermenter under anaerobic conditions. The yield of biomass from substrate is 0.06 g g^{-1} ; Y_{PX} is 7.7 g g^{-1} . The maintenance coefficient is $2.2 \text{ g g}^{-1} \text{ h}^{-1}$; the specific rate of product formation due to maintenance is 1.1 h^{-1} . The maximum specific growth rate of *Z. mobilis* is approximately 0.3 h^{-1} . 5 g bacteria are inoculated into 50 litres of medium containing 12 g l^{-1} glucose. Determine batch culture times required to:

- produce 10 g biomass;
- achieve 90% substrate conversion; and
- produce 100 g ethanol.

Solution:

$Y_{XS} = 0.06 \text{ g g}^{-1}$; $Y_{PX} = 7.7 \text{ g g}^{-1}$; $\mu_{\max} = 0.3 \text{ h}^{-1}$; $m_S = 2.2 \text{ g g}^{-1} \text{ h}^{-1}$; $m_P = 1.1 \text{ h}^{-1}$; $x_0 = 5 \text{ g}/50 \text{ l} = 0.1 \text{ g l}^{-1}$; $s_0 = 12 \text{ g l}^{-1}$.

(a) If 10 g biomass are produced by reaction, the final amount of biomass present is $(10 + 5) \text{ g} = 15 \text{ g}$. Therefore $x_f = 15 \text{ g}/50 \text{ l} = 0.3 \text{ g l}^{-1}$. From Eq. (13.20):

$$t_b = \frac{1}{0.3 \text{ h}^{-1}} \ln \frac{0.3 \text{ g l}^{-1}}{0.1 \text{ g l}^{-1}} = 3.7 \text{ h}.$$

(b) If 90% of the substrate is converted, $s_f = (0.1 s_0) = 1.2 \text{ g l}^{-1}$. Ethanol synthesis is directly coupled to energy metabolism in the cell; therefore, from Eq. (13.26):

$$t_b = \frac{1}{0.3 \text{ h}^{-1}} \ln \left[1 + \frac{(12 - 1.2) \text{ g l}^{-1}}{\left(\frac{1}{0.06 \text{ g g}^{-1}} + \frac{2.2 \text{ g g}^{-1} \text{ h}^{-1}}{0.3 \text{ h}^{-1}} \right) 0.1 \text{ g l}^{-1}} \right] = 5.7 \text{ h}.$$

(c) q_p is calculated using Eq. (11.70). In batch culture with $\mu = \mu_{\max}$:

$$q_p = 7.7 \text{ g g}^{-1} (0.3 \text{ h}^{-1}) + 1.1 \text{ h}^{-1} = 3.4 \text{ h}^{-1}.$$

As no product is present initially, $p_0 = 0$. Production of 100 g ethanol corresponds to $100 \text{ g}/50 \text{ l} = 2 \text{ g l}^{-1} = p_f$. From Eq. (13.31):

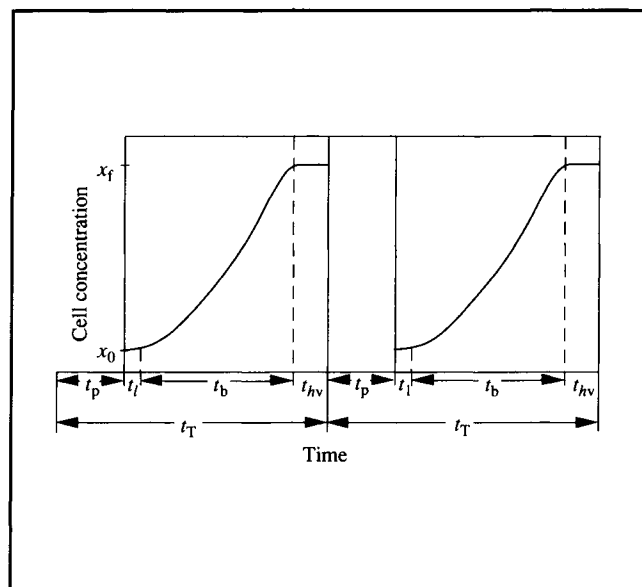
$$t_b = \frac{1}{0.3 \text{ h}^{-1}} \ln \left[1 + \frac{0.3 \text{ h}^{-1}}{(0.1 \text{ g l}^{-1}) (3.4 \text{ h}^{-1})} (2 \text{ g l}^{-1}) \right] = 3.4 \text{ h}.$$

13.5.2 Total Time For Batch Reaction Cycle

In the above analysis, t_b represents the time required for batch cell or enzyme conversion. In practice, batch operations involve lengthy unproductive periods in addition to t_b .

Following the fermentation or enzyme reaction, time t_{hv} is taken to harvest the contents of the reactor and time t_p is needed to clean, sterilise and otherwise prepare the reactor for the next batch. For cell culture, a lag time of duration t_l occurs after inoculation during which no growth or product forma-

Figure 13.21 Preparation, lag, reaction and harvest times in operation of a batch fermenter.



tion occurs. These time periods are illustrated for fermentation processes in Figure 13.21. Therefore, the total *downtime* t_{dn} associated with batch reactor operation is:

$$t_{dn} = t_{hv} + t_p + t_l \quad (13.32)$$

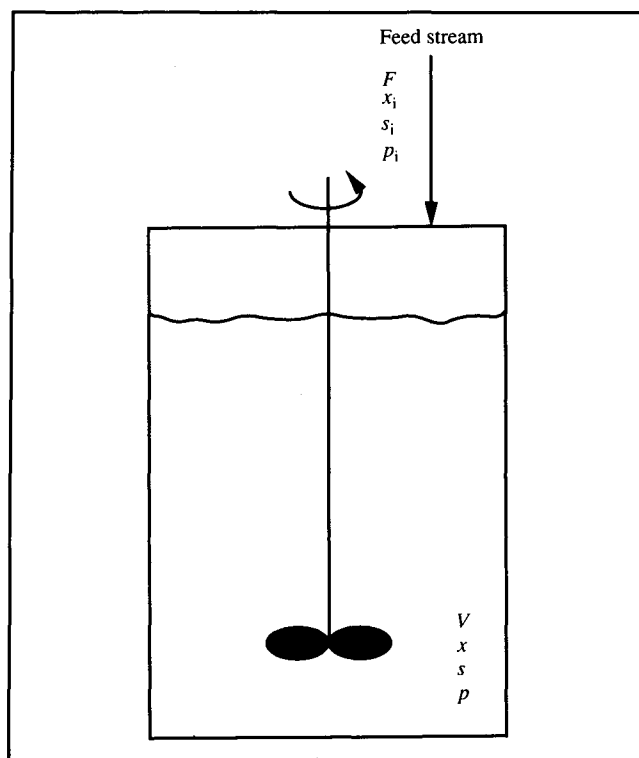
and the total batch reaction time t_T is:

$$t_T = t_b + t_{dn} \quad (13.33)$$

13.5.3 Fed-Batch Operation of a Mixed Reactor

In fed-batch operation, intermittent or continuous feeding of nutrients is used to supplement the reactor contents and provide control over the substrate concentration. By starting with a relatively dilute solution of substrate and adding more nutrients as the conversion proceeds, high growth rates are avoided. This is important, for example, in cultures where the oxygen demand during fast growth is too high for the mass-transfer capabilities of the reactor. Alternatively, high substrate concentrations may be inhibitory or switch on undesirable metabolic pathways. Fed-batch culture is used extensively in production of bakers' yeast to overcome catabolite repression and control oxygen demand; it is also used routinely for peni-

Figure 13.22 Flowsheet for a stirred fed-batch fermenter.



cillin production. Space must be allowed in fed-batch reactors for addition of fresh medium; in some cases a portion of the broth is removed before injection of additional material. The flow rate and timing of the feed are often determined by monitoring parameters such as dissolved-oxygen level or exhaust gas composition. As enzyme reactions are rarely carried out as fed-batch operations, we will consider fed-batch reactors for fermentation only.

The flowsheet for a well-mixed fed-batch fermenter is shown in Figure 13.22. The volumetric flow rate of entering feed is F ; the concentrations of biomass, growth-limiting substrate and product in this stream are x_i , s_i and p_i , respectively. We will assume F is constant. Owing to input of the feed, the liquid volume V is not constant. Equations for fed-batch culture are derived by carrying out unsteady-state mass balances.

The unsteady-state mass-balance equation for total mass in a flow reactor was derived in Chapter 6:

$$\frac{d(\rho V)}{dt} = F_i \rho_i - F_o \rho_o \quad (6.6)$$

where ρ is density of the reactor contents, V is liquid volume in

the reactor, F_i and F_o are input and output mass flow rates, and ρ_i and ρ_o are densities of input and output streams, respectively. For the fed-batch reactor of Figure 13.22, $F_o = 0$ and $F_i = F$. With dilute solutions such as those often used in bioprocessing, we can assume ρ is constant and that $\rho_i = \rho$; density can then be taken outside of the differential and cancelled through the equation. Therefore, Eq. (6.6) for fed-batch fermentation is:

$$\frac{dV}{dt} = F. \quad (13.34)$$

A similar mass balance based on Eq. (6.5) can be performed for cells. In fed-batch operation $\hat{M}_o = 0$; $\hat{M}_i$ is equal to the feed flow rate F multiplied by the cell concentration x_i in the feed. As in Section 13.5.1.2, mass of cells in the reactor, M , is equal to xV where x is cell concentration and V is liquid volume, rate of biomass generation R_G is equal to μxV where μ is the specific growth rate, and rate of cell death R_C is equal to $k_d xV$ where k_d is the specific death constant. Applying these terms in Eq. (6.5) gives:

$$\frac{d(xV)}{dt} = Fx_i + \mu xV - k_d xV. \quad (13.35)$$

Because V in fed-batch culture is a function of time, it cannot be cancelled through Eq. (13.35). Instead, we must expand the differential using the product rule of Eq. (D.22) in Appendix D. After grouping terms this gives:

$$x \frac{dV}{dt} + V \frac{dx}{dt} = Fx_i + (\mu - k_d) xV. \quad (13.36)$$

Applying Eq. (13.34) to Eq. (13.36):

$$xF + V \frac{dx}{dt} = Fx_i + (\mu - k_d) xV. \quad (13.37)$$

Dividing through by V and rearranging gives:

$$\frac{dx}{dt} = \frac{F}{V} x_i + x \left(\mu - k_d - \frac{F}{V} \right).$$

$$(13.38)$$

Let us define the *dilution rate* D with dimensions T^{-1} :

$$D = \frac{F}{V}. \quad (13.39)$$

In fed-batch systems V increases with time; therefore, if F is constant, D decreases as the reaction proceeds. Applying Eq. (13.39) to Eq. (13.38):

$$\frac{dx}{dt} = Dx_i + x(\mu - k_d - D). \quad (13.40)$$

Eq. (13.40) can be simplified for most applications. Usually the feed material is sterile so that $x_i = 0$. If, in addition, the rate of cell death is negligible compared with growth so that $k_d \ll \mu$, Eq. (13.40) becomes:

$$\frac{dx}{dt} = x(\mu - D). \quad (13.41)$$

Let us now apply Eq. (6.5) to the limiting substrate in our fed-batch reactor. In this case, $\hat{M}_o$ and R_G are zero; the mass flow rate of substrate entering the reactor $\hat{M}_i$ is equal to Fs_i . Mass of substrate in the reactor, M , is equal to concentration s multiplied by volume V . For fermentations producing product not directly coupled with energy metabolism, R_C is given by Eq. (13.21). Substituting these terms into Eq. (6.5) gives:

$$\frac{d(sV)}{dt} = Fs_i - \left(\frac{\mu}{Y_{XS}} + \frac{q_P}{Y_{PS}} + m_S \right) xV \quad (13.42)$$

where μ is the specific growth rate, Y_{XS} is the true biomass yield from substrate, q_P is the specific rate of product formation, Y_{PS} is the true product yield from substrate and m_S is the maintenance coefficient. Expanding the differential and applying Eqs (13.34) and (13.39) gives:

$$\frac{ds}{dt} = D(s_i - s) - \left(\frac{\mu}{Y_{XS}} + \frac{q_P}{Y_{PS}} + m_S \right) x. \quad (13.43)$$

Eqs (13.41) and (13.43) are differential equations for rates of change of cell and substrate concentrations in fed-batch reac-

tors. Because D is a function of time, integration of these equations is more complicated than for batch reactors. However, we can derive analytical expressions for fed-batch culture if we simplify Eqs (13.41) and (13.43). Here, we will examine the situation where the reactor is operated first in batch until a high cell density is achieved and the substrate virtually exhausted. When this condition is reached, fed-batch operation is started with medium flow rate F . As a result, cell concentration x is maintained high and approximately constant so that $\frac{dx}{dt} \approx 0$. From Eq. (13.41), if $\frac{dx}{dt} \approx 0$, $\mu \approx D$. Therefore, substituting $\mu \approx D$ into the Monod expression of Eq. (11.60):

$$D \approx \frac{\mu_{\max} s}{K_s + s} \quad (13.44)$$

Rearrangement of Eq. (13.44) gives an expression for substrate concentration as a function of dilution rate:

$$s \approx \frac{DK_s}{\mu_{\max} - D} \quad (13.45)$$

Let us assume that the culture does not produce product, or, if there is product formation, that it is directly linked with energy generation. If maintenance requirements can also be neglected, Eq. (13.43) can be simplified to:

$$\frac{ds}{dt} = D(s_i - s) - \frac{\mu x}{Y_{XS}} \quad (13.46)$$

At high cell density in the reactor, virtually all substrate entering the vessel is consumed immediately; therefore, $s \ll s_i$ and $\frac{ds}{dt} \approx 0$. Applying these relationships with $\mu \approx D$ to Eq. (13.46), we obtain:

$$x \approx Y_{XS} s_i \quad (13.47)$$

For product synthesis directly coupled with energy metabolism, Eq. (13.47) allows us to derive an approximate expression for product concentration in fed-batch reactors. Assuming the feed does not contain product:

$$p \approx Y_{PS} s_i \quad (13.48)$$

Even though cell concentration remains virtually unchanged with $\frac{dx}{dt} \approx 0$, because the liquid volume increases with time in fed-batch reactors, the total mass of cells also increases. Consider the rate of increase of total biomass in the reactor $\frac{dX}{dt}$, where X is equal to xV . Using the results of Eqs (13.34) and (13.47) with $\frac{dx}{dt} \approx 0$:

$$\frac{dX}{dt} = \frac{d(xV)}{dt} = x \frac{dV}{dt} + V \frac{dx}{dt} = Y_{XS} s_i F \quad (13.49)$$

Eq. (13.49) can now be integrated with initial condition $X = X_0$ at the start of liquid flow to give:

$$X = X_0 + (Y_{XS} s_i F) t_{fb} \quad (13.50)$$

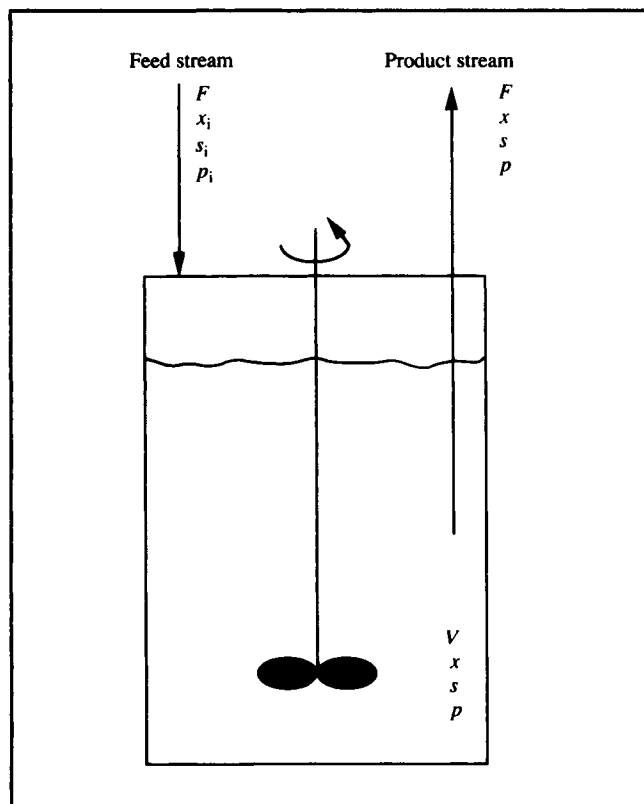
where t_{fb} is the fed-batch time after commencement of feeding. Eq. (13.50) indicates that, for Y_{XS} , s_i and F constant, total biomass in fed-batch fermenters increases as a linear function of time.

Under conditions of high biomass density and almost complete depletion of substrate, a *quasi-steady-state* condition prevails in fed-batch reactors where $\frac{dX}{dt} \approx 0$, $\frac{ds}{dt} \approx 0$ and $\frac{dp}{dt} \approx 0$. At quasi-steady state, Eqs (13.47) and (13.45) can be used to calculate biomass and substrate concentrations in reactors where cell death and maintenance requirements are negligible and product is either absent or directly coupled with energy metabolism. Eq. (13.48) allows calculation of product concentration for metabolites directly coupled with energy generation in the cell. At quasi-steady state, the specific growth rate μ and dilution rate F/V are approximately equal; therefore as V increases, the growth rate decreases. When fed-batch operation is used for production of biomass such as bakers' yeast, it is useful to be able to predict the total mass of cells in the reactor as a function of time. An expression for total biomass is given by Eq. (13.50). Note that under quasi-steady-state conditions, x , s and p are almost constant, but μ , V , D and X are changing. Further details of fed-batch operation are given by Pirt [37].

13.5.4 Continuous Operation of a Mixed Reactor

Bioreactors are operated continuously in a few bioprocess industries such as brewing, production of bakers' yeast and waste treatment; enzyme conversions can also be carried out using continuous systems. The flow sheet for a continuous

Figure 13.23 Flowsheet for a continuous stirred-tank fermenter.



reactor is shown in Figure 13.23. If the vessel is well mixed, the product stream has the same composition as the liquid in the reactor (see Section 6.4). Therefore, when continuous reactors are used with freely-suspended cells or enzymes, catalyst is continuously withdrawn from the vessel in the product stream. For enzymes this is a serious shortcoming as catalyst is not produced by the reaction; in cell culture, growth supplies additional cells to replace those removed. Continuous reactors are used with free enzymes only if the enzyme is inexpensive and can be added continuously to maintain the catalyst concentration. With more costly enzymes, continuous operation may be feasible if the enzyme is immobilised and retained inside the vessel. Well-mixed continuous reactors are often referred to using the abbreviation CSTR, meaning *continuous stirred-tank reactor*. The term CSTF is also sometimes used to denote *continuous stirred-tank fermenter*.

Different steady-state operating strategies are available for continuous fermenters. In a *chemostat* the liquid volume is kept constant by setting the inlet and outlet flow rates equal;

the dilution rate is therefore constant and steady state is achieved by concentrations in the chemostat adjusting themselves to the feed rate. In a *turbidostat*, the liquid volume is kept constant by setting the outlet flow rate equal to the inlet flow rate; however, the inlet flow rate is adjusted to keep the biomass concentration constant. Thus, in a turbidostat the dilution rate adjusts to its steady-state value corresponding to the set biomass concentration. Turbidostats require more complex monitoring and control systems than chemostats and are not used in large scale. Accordingly, we will concentrate here on chemostat operation.

Characteristic operating parameters for continuous reactors are the dilution rate D and the average *residence time* τ . These parameters are related as follows:

$$\tau = \frac{1}{D} = \frac{V}{F} \quad (13.51)$$

where D is defined by Eq. (13.39). In continuous reactor operation, the amount of material that can be processed over a given period of time is represented by the flow rate, F . Therefore, for a given throughput, the reactor size V and associated capital and operating costs are minimised when τ is made as small as possible. Continuous reactor theory allows us to determine relationships between τ (or D) and steady-state substrate, product and cell concentrations in the reactor. This theory is based on steady-state mass balances derived from Eq. (6.5).

13.5.4.1 Enzyme reaction

Let us apply Eq. (6.5) to the limiting substrate in a continuous enzyme reactor operated at steady state. The mass flow rate of substrate entering the reactor $\dot{M}_i$ is equal to Fs_i ; $\dot{M}_o$, the mass flow rate of substrate leaving, is Fs . As substrate is not generated by the reactor, $R_G = 0$. Rate of substrate consumption R_C is equal to the volumetric rate of reaction v multiplied by V ; v is given by Eq. (11.31). The left-hand side of Eq. (6.5) is zero because the system is at steady state. Therefore, the steady-state substrate mass-balance equation for continuous enzyme reaction is:

$$Fs_i - Fs - \frac{v_{\max}s}{K_m + s} V = 0 \quad (13.52)$$

where $v_{\max}$ is the maximum rate of reaction and K_m is the Michaelis constant. For reactions with free enzyme, we assume that enzyme lost in the product stream is replaced continuously

so that $v_{\max}$ remains constant and steady state is achieved. Dividing through by V and applying the definition of dilution rate from Eq. (13.39) gives:

$$D(s_i - s) = \frac{v_{\max}s}{K_m + s} \quad (13.53)$$

If $v_{\max}$, K_m and s_i are known, Eq. (13.53) can be used directly to calculate the dilution rate required to achieve a particular level of substrate conversion. Steady-state product concentration can then be evaluated from stoichiometry.

For reactions with immobilised enzymes, Eq. (13.53) must be modified to account for mass-transfer effects:

$$D(s_i - s) = \frac{\eta_T v_{\max}s}{K_m + s} \quad (13.54)$$

where η_T is the total effectiveness factor (see Section 12.5), s is the bulk substrate concentration, and $v_{\max}$ and K_m are intrinsic kinetic parameters. η_T can be calculated for constant s using the theory for heterogeneous reactions outlined in Chapter 12.

Example 13.4 Immobilised-enzyme reaction in a CSTR

Mushroom tyrosinase is immobilised in 2-mm spherical beads for conversion of tyrosine to DOPA in a continuous, well-mixed bubble column. The Michaelis constant for the immobilised enzyme is 2 gmol m^{-3} . A solution containing 15 gmol m^{-3} tyrosine is fed into the reactor; because of the high cost of the substrate, the desired conversion is 99%. The reactor is loaded with beads at a density of $0.25 \text{ m}^3 \text{ m}^{-3}$; all enzyme is retained within the reactor. The intrinsic $v_{\max}$ for the immobilised enzyme is $1.5 \times 10^{-2} \text{ gmol s}^{-1} \text{ m}^{-3}$ beads. The effective diffusivity of tyrosine in the beads is $7 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$; external mass-transfer effects are negligible. Immobilisation stabilises the enzyme so that deactivation is minimal over the operating period. Determine the reactor volume needed to treat 18 m^3 tyrosine solution per day.

Solution:

$K_m = 2 \text{ gmol m}^{-3}$; $v_{\max} = 1.5 \times 10^{-2} \text{ gmol s}^{-1} \text{ m}^{-3}$; $R = 10^{-3} \text{ m}$; $\mathcal{D}_{Ac} = 7 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$; $s_i = 15 \text{ gmol m}^{-3}$. Converting the feed flow rate to $\text{m}^3 \text{ s}^{-1}$:

$$F = 18 \text{ m}^3 \text{ d}^{-1} \cdot \left| \frac{1 \text{ d}}{24 \text{ h}} \right| \cdot \left| \frac{1 \text{ h}}{3600 \text{ s}} \right| = 2.08 \times 10^{-4} \text{ m}^3 \text{ s}^{-1}.$$

For 99% conversion, the outlet and therefore the internal substrate concentration $s = (0.01 s_i) = 0.15 \text{ gmol m}^{-3}$. As $s \ll K_m$, we can assume first-order kinetics (see Section 11.3.3) with $k_1 = v_{\max}/K_m = 7.5 \times 10^{-3} \text{ s}^{-1}$. The first-order Thiele modulus for spherical catalysts is calculated from Table 12.2:

$$\phi_1 = \frac{R}{3} \sqrt{\frac{k_1}{\mathcal{D}_{Ac}}} = \frac{10^{-3} \text{ m}}{3} \sqrt{\frac{7.5 \times 10^{-3} \text{ s}^{-1}}{7 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}}} = 1.09.$$

From Figure 12.8 or Table 12.3, $\eta_{i1} = 0.64$. As external mass-transfer resistance is negligible, $\eta_e = 1$ and, from Eq. (12.46), $\eta_T = 0.64$. Substituting values into Eq. (13.54) gives:

$$D(15 - 0.15) \text{ gmol m}^{-3} = \frac{0.64 (1.5 \times 10^{-2} \text{ gmol s}^{-1} \text{ m}^{-3}) (0.15 \text{ gmol m}^{-3})}{2 \text{ gmol m}^{-3} + 0.15 \text{ gmol m}^{-3}}$$

$$D = 4.51 \times 10^{-5} \text{ s}^{-1}.$$

From Eq. (13.51):

$$V = \frac{F}{D} = \frac{2.08 \times 10^{-4} \text{ m}^3 \text{ s}^{-1}}{4.51 \times 10^{-5} \text{ s}^{-1}} = 4.6 \text{ m}^3.$$

13.5.4.2 Cell culture

Let us consider the reactor of Figure 13.23 operated as a continuous fermenter and apply Eq. (6.5) for steady-state mass balances on biomass, substrate and product.

For biomass, $\hat{M}_i$ in Eq. (6.5) is the mass flow rate of cells entering the reactor; $\hat{M}_i = Fx_i$; $\hat{M}_o$ is the mass flow rate of cells leaving; $\hat{M}_o = Fx$. The other terms in Eq. (6.5) are the same as in Section 13.5.1.2; $R_G = \mu xV$ where μ is the specific growth rate and V the reactor liquid volume; $R_C = k_d xV$ where k_d is the specific death constant. At steady state, the left-hand side of Eq. (6.5) is zero. Therefore, the steady-state mass-balance equation for biomass is:

$$Fx_i - Fx + \mu xV - k_d xV = 0. \quad (13.55)$$

Usually, the feed stream in continuous culture is sterile so that $x_i = 0$. If in addition the rate of cell death is negligible compared with growth, $k_d \ll \mu$ and Eq. (13.55) becomes:

$$\mu xV = Fx. \quad (13.56)$$

Cancelling x from both sides, dividing by V , and applying the definition of dilution rate from Eq. (13.39) gives:

$$\mu = D. \quad (13.57)$$

As in the derivation of Eq. (13.45), applying Eq. (13.57) to the Monod expression of Eq. (11.60) gives an equation for the steady-state concentration of limiting substrate in the reactor:

$$s = \frac{DK_S}{\mu_{\max} - D}. \quad (13.58)$$

Let us now apply Eq. (6.5) at steady state to the limiting substrate. $\hat{M}_i = Fs_i$, $\hat{M}_o = Fs$, $R_G = 0$ and R_C is given by Eq. (13.21). Therefore:

$$Fs_i - Fs - \left(\frac{\mu}{Y_{XS}} + \frac{q_P}{Y_{PS}} + m_S \right) xV = 0 \quad (13.59)$$

where μ is the specific growth rate, Y_{XS} is the true biomass yield from substrate, q_P is the specific rate of product formation not directly linked with energy metabolism, Y_{PS} is the

true product yield from substrate and m_S is the maintenance coefficient. In Eq. (13.59) we can divide through by V , substitute the definition of dilution rate from Eq. (13.39) and replace μ with D according to Eq. (13.57). Rearrangement then gives the following expression for the steady-state cell concentration x :

$$x = \frac{D(s_i - s)}{\frac{D}{Y_{XS}} + \frac{q_P}{Y_{PS}} + m_S}. \quad (13.60)$$

Eq. (13.60) can be simplified if there is no product synthesis or if production is directly linked with energy metabolism:

$$x = \frac{D(s_i - s)}{\frac{D}{Y_{XS}} + m_S}. \quad (13.61)$$

If, in addition, maintenance effects can be ignored, Eq. (13.61) becomes:

$$x = (s_i - s) Y_{XS}. \quad (13.62)$$

Substituting for s from Eq. (13.58) we obtain an expression for the steady-state cell concentration in a CSTR in terms of D and kinetic and yield parameters only:

$$x = \left(s_i - \frac{DK_S}{\mu_{\max} - D} \right) Y_{XS}. \quad (13.63)$$

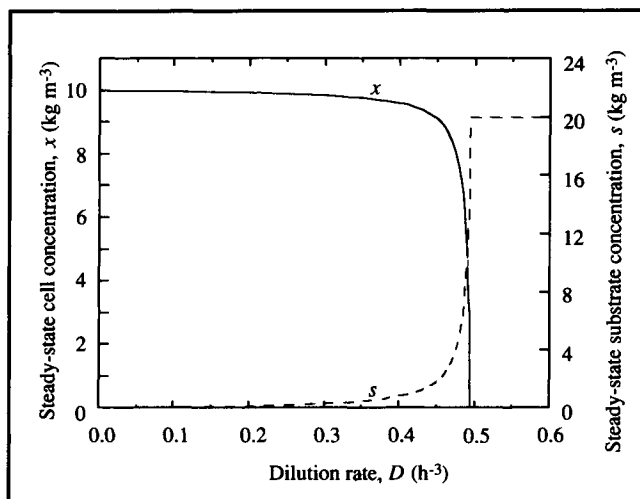
Eq. (13.63) is valid at steady state in the absence of maintenance requirements and when product synthesis is either absent or directly linked with energy metabolism.

We can also apply Eq. (6.5) for a steady-state mass balance on fermentation product. In this case, $\hat{M}_i = Fp_i$ and $\hat{M}_o = Fp$. R_G is given by Eq. (13.28); $R_C = 0$. Therefore, Eq. (6.5) becomes:

$$Fp_i - Fp + q_P xV = 0 \quad (13.64)$$

where q_P is the specific rate of formation for all classes of product. Dividing through by V , substituting the definition of

Figure 13.24 Steady-state cell and substrate concentrations as a function of dilution rate in a chemostat. Curves correspond to $\mu_{\max} = 0.5 \text{ h}^{-1}$, $K_S = 0.2 \text{ kg m}^{-3}$, $Y_{XS} = 0.5 \text{ kg kg}^{-1}$ and $s_i = 20 \text{ kg m}^{-3}$.



dilution rate from Eq. (13.39) and rearranging gives an expression for the steady-state product concentration as a function of biomass concentration x :

$$p = p_i + \frac{q_P x}{D} \quad (13.65)$$

x in Eq. (13.65) can be evaluated from Eq. (13.60), (13.61) or (13.62). The nature of q_P depends on the type of product formed (see Section 11.9).

Eq. (13.58) is an explicit expression for steady-state substrate concentration in a chemostat. The steady-state biomass concentration x can be evaluated from Eq. (13.60), (13.61) or (13.62); the choice of expression for biomass depends on the relative significance of maintenance metabolism and the type of product, if any, produced. If q_P is known, product concentration in the reactor can be evaluated from Eq. (13.65). In the simplest case when products are either absent or directly linked to energy generation and maintenance effects can be neglected, the chemostat is represented by Eqs (13.58) and (13.63). The form of these equations is shown in Figure 13.24. At low feed rates, i.e. $D \rightarrow 0$, nearly all the substrate is consumed at steady state so that, from Eq. (13.62), $x \approx s_i Y_{XS}$. As D increases, s increases slowly at first and then more rapidly as D approaches $\mu_{\max}$; correspondingly, x decreases so that $x \rightarrow 0$ as $D \rightarrow \mu_{\max}$. The condition at high dilution rate

whereby x reduces to zero is known as *washout*; washout of cells occurs when the rate of cell removal in the reactor outlet stream is greater than the rate of generation by growth. For systems with negligible maintenance requirements and either energy-associated or zero product formation, the critical dilution rate D_{crit} at which the steady-state biomass concentration just becomes zero can be estimated by substituting $x = 0$ into Eq. (13.63) and solving for D :

$$D_{\text{crit}} = \frac{\mu_{\max} s_i}{K_S + s_i} \quad (13.66)$$

For most cell cultures $K_S \ll s_i$; therefore $D_{\text{crit}} \approx \mu_{\max}$. To avoid washout of cells from the chemostat, the operating dilution rate must always be less than D_{crit} . Near washout the system is very sensitive to small changes in D which cause relatively large shifts in x and s .

The rate of biomass production in a CSTR is equal to the rate at which cells leave the reactor: Fx . The volumetric productivity is therefore equal to Fx divided by V :

$$Q_X = \frac{Fx}{V} = Dx \quad (13.67)$$

where Q_X is the volumetric rate of biomass production. Similarly the volumetric rate of product formation Q_P is:

$$Q_P = \frac{Fp}{V} = Dp \quad (13.68)$$

When maintenance requirements are negligible and product formation either absent or energy associated, we can substitute into Eq. (13.67) the expression for x from Eq. (13.63):

$$Q_X = D \left(s_i - \frac{K_S D}{\mu_{\max} - D} \right) Y_{XS} \quad (13.69)$$

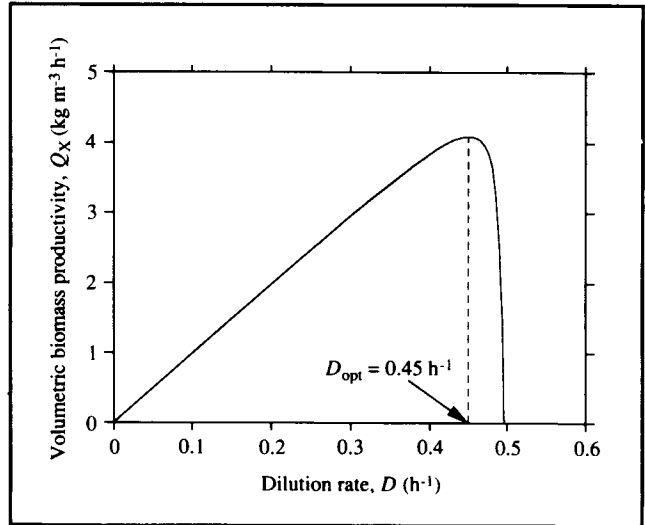
This relationship between Q_X and D is shown in Figure 13.25. Rate of biomass output reaches a sharp maximum at the optimum dilution rate for biomass productivity D_{opt} ; therefore, at D_{opt} the slope $dQ_X/dD = 0$. Differentiating Eq. (13.69) with respect to D and equating to zero provides an expression for D_{opt} :

$$D_{\text{opt}} = \mu_{\text{max}} \left(1 - \sqrt{\frac{K_S}{K_S + s_i}} \right). \quad (13.70)$$

Operation of a chemostat at D_{opt} gives the maximum rate of biomass production from the reactor. However, because D_{opt} is usually very close to D_{crit} , it may not be practical to operate at D_{opt} . Small variations of dilution rate in this region can cause large fluctuations in x and s and, unless the dilution rate is controlled very precisely, washout may occur.

Excellent agreement between chemostat theory and experimental results has been found for many culture systems. When deviations occur, they are due primarily to imperfect operation of the reactor. For example, if the vessel is not well mixed, some liquid will have higher residence time in the reactor than the rest and concentrations will not be uniform; under these conditions the equations derived in the section do not hold. Similarly, if cells adhere to glass or metal surfaces in the reactor and produce wall growth, biomass will be retained in the vessel and washout will not occur even at high dilution rates. Other deviations occur if inadequate time is allowed for the system to reach steady state.

Figure 13.25 Steady-state volumetric biomass productivity as a function of dilution rate in a chemostat. Curve corresponds to $\mu_{\text{max}} = 0.5 \text{ h}^{-1}$, $K_S = 0.2 \text{ kg m}^{-3}$, $Y_{\text{XS}} = 0.5 \text{ kg kg}^{-1}$ and $s_i = 20 \text{ kg m}^{-3}$.



Example 13.5 Steady-state concentrations in a chemostat

The *Zymomonas mobilis* cells of Example 13.3 are used for chemostat culture in a 60 m^3 fermenter. The feed contains 12 g l^{-1} glucose; K_S for the organism is 0.2 g l^{-1} .

- What flow rate is required for a steady-state substrate concentration of 1.5 g l^{-1} ?
- At the flow rate of (a), what is the cell density?
- At the flow rate of (a), what concentration of ethanol is produced?

Solution:

$Y_{\text{XS}} = 0.06 \text{ g g}^{-1}$; $Y_{\text{PX}} = 7.7 \text{ g g}^{-1}$; $\mu_{\text{max}} = 0.3 \text{ h}^{-1}$; $K_S = 0.2 \text{ g l}^{-1}$; $m_S = 2.2 \text{ g g}^{-1} \text{ h}^{-1}$; $s_i = 12 \text{ g l}^{-1}$; $V = 60 \text{ m}^3$. From Example 13.3, $q_p = 3.4 \text{ h}^{-1}$. From the general definition of yield in Section 11.6.1, $Y_{\text{PS}} = Y_{\text{PX}} Y_{\text{XS}} = 0.46 \text{ g g}^{-1}$.

- $s = 1.5 \text{ g l}^{-1}$. From Eq. (13.58):

$$D = \frac{\mu_{\text{max}} s}{K_S + s} = \frac{(0.3 \text{ h}^{-1})(1.5 \text{ g l}^{-1})}{0.2 \text{ g l}^{-1} + 1.5 \text{ g l}^{-1}} = 0.26 \text{ h}^{-1}.$$

From the definition of dilution rate Eq. (13.39):

$$F = DV = (0.26 \text{ h}^{-1})(60 \text{ m}^3) = 15.6 \text{ m}^3 \text{ h}^{-1}.$$

- When synthesis of product is coupled with energy metabolism as for ethanol, x is evaluated using Eq. (13.61). Therefore:

$$x = \frac{(0.26 \text{ h}^{-1})(12 - 1.5) \text{ g l}^{-1}}{\left(\frac{0.26 \text{ h}^{-1}}{0.06 \text{ g g}^{-1}} + 2.2 \text{ g g}^{-1} \text{ h}^{-1} \right)} = 0.42 \text{ g l}^{-1}.$$

(c) Assuming ethanol is not present in the feed, $p_i = 0$. Steady-state product concentration is given by Eq. (13.65):

$$p = \frac{(3.4 \text{ h}^{-1})(0.42 \text{ g l}^{-1})}{0.26 \text{ h}^{-1}} = 5.5 \text{ g l}^{-1}.$$

Example 13.6 Substrate conversion and biomass productivity in a chemostat

A 5 m³ fermenter is operated continuously with feed substrate concentration 20 kg m⁻³. The microorganism cultivated in the reactor has the following characteristics: $\mu_{\max} = 0.45 \text{ h}^{-1}$, $K_S = 0.8 \text{ kg m}^{-3}$, $Y_{XS} = 0.55 \text{ kg kg}^{-1}$.

- What feed flow rate is required to achieve 90% substrate conversion?
- How does the biomass productivity at 90% substrate conversion compare with the maximum possible?

Solution:

(a) For 90% substrate conversion, $s = (0.1 s_i) = 2 \text{ kg m}^{-3}$. From Eq. (13.58):

$$D = \frac{\mu_{\max} s}{K_S + s} = \frac{(0.45 \text{ h}^{-1})(2 \text{ kg m}^{-3})}{0.8 \text{ kg m}^{-3} + 2 \text{ kg m}^{-3}} = 0.32 \text{ h}^{-1}.$$

From Eq. (13.39):

$$F = DV = 0.32 \text{ h}^{-1} (5 \text{ m}^3) = 1.6 \text{ m}^3 \text{ h}^{-1}.$$

(b) Assuming maintenance requirements and product formation are negligible, from Eq. (13.69):

$$\begin{aligned} Q_X &= 0.32 \text{ h}^{-1} \left(20 \text{ kg m}^{-3} - \frac{(0.8 \text{ kg m}^{-3})(0.32 \text{ h}^{-1})}{(0.45 \text{ h}^{-1} - 0.32 \text{ h}^{-1})} \right) 0.55 \text{ kg kg}^{-1} \\ &= 3.17 \text{ kg m}^{-3} \text{ h}^{-1}. \end{aligned}$$

Maximum biomass productivity occurs at D_{opt} which can be evaluated using Eq. (13.70):

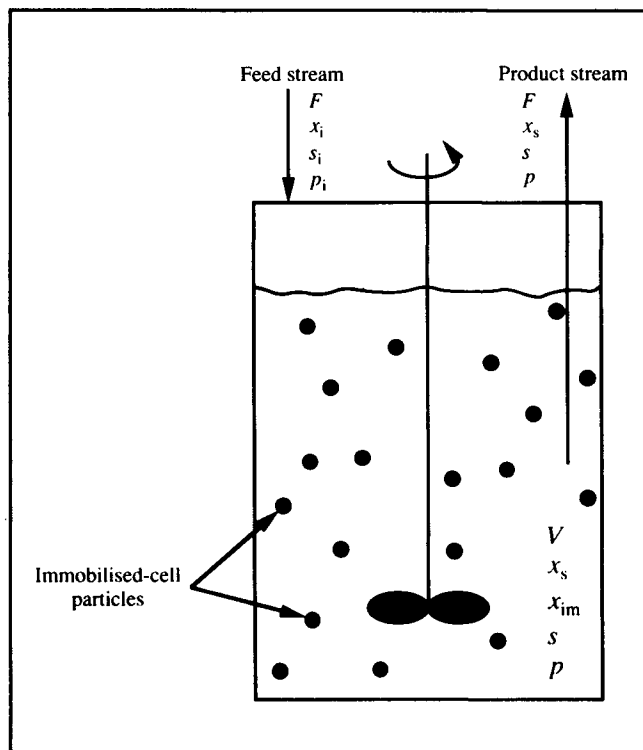
$$D_{\text{opt}} = 0.45 \text{ h}^{-1} \left(1 - \sqrt{\frac{0.8 \text{ kg m}^{-3}}{0.8 \text{ kg m}^{-3} + 20 \text{ kg m}^{-3}}} \right) = 0.36 \text{ h}^{-1}.$$

Maximum biomass productivity is determined from Eq. (13.69) with $D = D_{\text{opt}}$:

$$\begin{aligned} Q_{X,\max} &= 0.36 \text{ h}^{-1} \left(20 \text{ kg m}^{-3} - \frac{(0.8 \text{ kg m}^{-3})(0.36 \text{ h}^{-1})}{(0.45 \text{ h}^{-1} - 0.36 \text{ h}^{-1})} \right) 0.55 \text{ kg kg}^{-1} \\ &= 3.33 \text{ kg m}^{-3} \text{ h}^{-1}. \end{aligned}$$

Therefore, biomass productivity at 90% substrate conversion is $3.17/3.33 \times 100 = 95\%$ of the theoretical maximum.

Figure 13.26 Flowsheet for a continuous stirred-tank fermenter with immobilised cells.



13.5.5 Chemostat With Immobilised Cells

Consider the continuous stirred-tank immobilised-cell fermenter shown in Figure 13.26. Spherical particles containing cells are kept suspended and well mixed by the stirrer. The concentration of immobilised cells per unit volume of liquid in the reactor is x_{im} . Let us assume that x_{im} is constant; this is achieved if all particles are retained in the vessel and all cells produced by immobilised-cell growth are released into the medium. The concentration of suspended cells is x_s . We will assume that the intrinsic specific growth rates of suspended and immobilised cells are the same and equal to μ . Suspended cells are removed from the reactor in the product stream; immobilised cells are retained inside the vessel. For simplicity, let us assume that cell death and maintenance requirements are negligible, the reactor feed is sterile, and any product synthesis is directly coupled with energy metabolism.

The system shown in Figure 13.26 reaches steady state. Relationships between operating variables and concentrations inside the reactor can be determined using mass balances.

Let us consider a mass balance on suspended cells. At steady

state, the mass-balance equation is similar to Eq. (13.55) except that x_i is zero for sterile feed and cell death is assumed to be negligible. In addition, as suspended cells are produced by growth of both suspended- and immobilised-cell populations, the equation must contain two generation terms instead of one:

$$-Fx_s + \mu x_s V + \mu x_{im} V = 0. \quad (13.71)$$

If diffusional limitations affect the growth rate of the immobilised cells, μx_{im} must be replaced by $\eta_T \mu x_{im}$ where η_T is the total effectiveness factor defined in Section 12.5. Dividing through by V and applying the definition of dilution rate from Eq. (13.39) gives:

$$Dx_s = \mu(x_s + \eta_T x_{im}) \quad (13.72)$$

or

$$D = \mu \left(1 + \frac{\eta_T x_{im}}{x_s} \right). \quad (13.73)$$

For $x_{im} = 0$, Eq. (13.73) reduces to Eq. (13.57) for a chemostat containing suspended cells only: $\mu = D$.

The steady-state mass-balance equation for limiting substrate can be derived from Eq. (6.5) with $\hat{M}_i = Fs_i$, $\hat{M}_o = Fs$ and $R_G = 0$. Both cell populations consume substrate; in the absence of product- and maintenance-associated substrate requirements, rate of substrate consumption R_C can be related directly to the growth rates of immobilised and suspended cells using the biomass yield coefficient Y_{XS} . If we assume that the value of Y_{XS} is the same for all cells, by analogy with Eq. (13.59) the mass-balance equation for limiting substrate is:

$$Fs_i - Fs - \frac{\mu x_s}{Y_{XS}} V - \frac{\eta_T \mu x_{im}}{Y_{XS}} V = 0. \quad (13.74)$$

Dividing through by V , applying the definition of dilution rate from Eq. (13.39) and rearranging gives:

$$D(s_i - s) = \frac{\mu}{Y_{XS}} (x_s + \eta_T x_{im}). \quad (13.75)$$

By manipulating Eqs (13.73) and (13.75) and substituting the Monod expression for μ from Eq. (11.60), the following relationship between steady-state substrate concentration, dilution rate and immobilised-cell concentration is obtained:

$$\frac{\mu_{\max} s}{K_S + s} = \frac{D(s_i - s) Y_{XS}}{(s_i - s) Y_{XS} + \eta_T x_{im}} \quad (13.76)$$

The form of Eq. (13.76) is shown in Figure 13.27. For a chemostat with suspended cells only, i.e. $x_{im} = 0$, at steady state $D = \mu$ and the maximum operating dilution rate D_{crit} is limited by the maximum specific growth rate of the cells. From Eq. (13.73), for any $x_{im} > 0$, D at steady state in the immobilised-cell reactor is greater than μ . Accordingly, dilution rate is no longer limited by the maximum growth rate and, as shown in Figure 13.27, immobilised-cell chemostats can be operated at D considerably greater than D_{crit} without washout. At a given dilution rate, presence of immobilised cells also improves substrate conversion and reduces the amount of substrate lost in the product stream. However, reaction rates with immobilised cells can be reduced significantly by the effects of mass transfer in and around the particles. As illustrated in Figure 13.27, at the same concentration of immobilised cells, substrate conversion at $\eta_T = 1.0$ is greater than at lower values of η_T when mass-transfer limitations are significant.

13.5.6 Chemostat Cascade

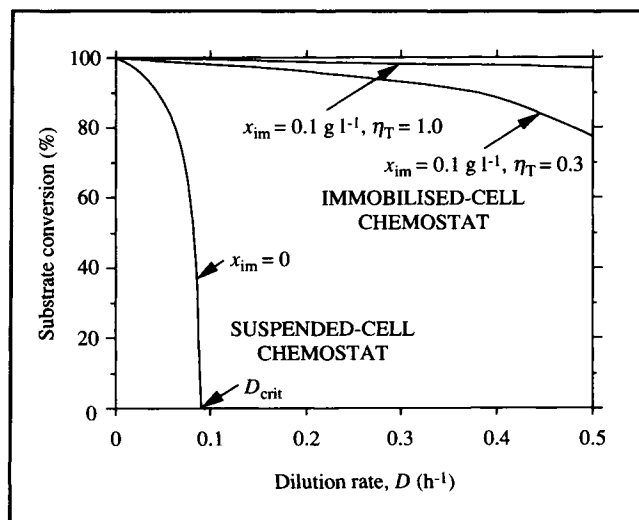
The joining together of two or more CSTRs in series produces a multi-stage process in which conditions such as pH, temperature and medium composition can be varied in each reactor. This is advantageous if reactor conditions required for growth are different from those for product synthesis, e.g. in production of recombinant proteins and many metabolites not directly linked with energy metabolism. One way of operating a two-stage chemostat cascade is shown in Figure 13.28; in this process the product stream from the first reactor feeds directly into the second. Substrate leaving the first reactor at concentration s_1 is converted in the second tank so that $s_2 < s_1$ and $p_2 > p_1$. In some applications, the second CSTR is supplemented with fresh medium containing nutrients, inducers or inhibitors for optimal product formation.

Design equations for cell and enzyme CSTR cascades can be derived as a simple extension of the theory developed in Section 13.5.4; the same mass-balance principles are applied

Figure 13.27 Steady-state substrate conversion as a function of dilution rate with and without immobilised cells.

Curves were calculated with the following parameter values:

$$\mu_{\max} = 0.1 \text{ h}^{-1}, K_S = 10^{-3} \text{ g l}^{-1}, Y_{XS} = 0.5 \text{ g g}^{-1}, s_i = 8 \times 10^{-3} \text{ g l}^{-1}.$$



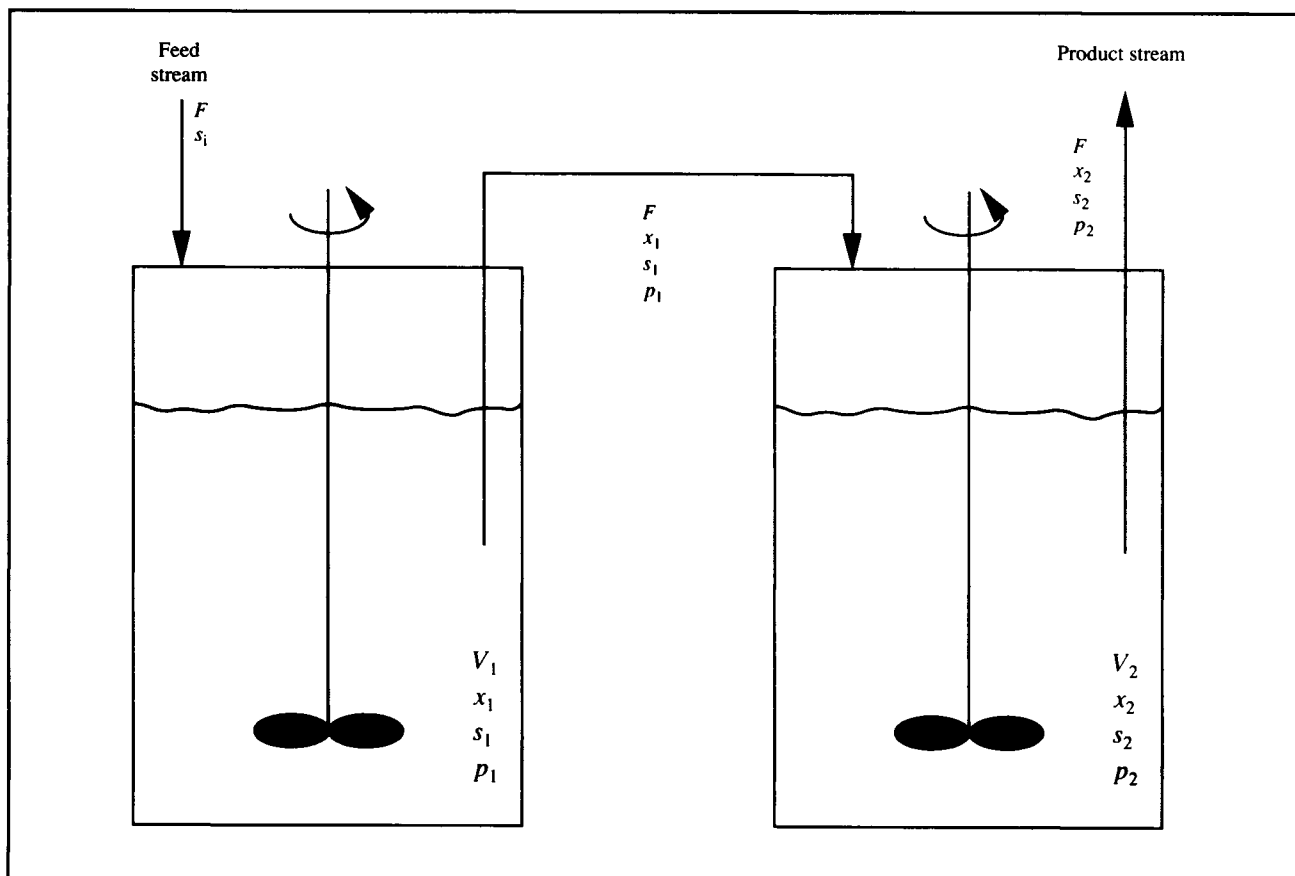
and steady state is assumed at each reactor stage. Details can be found in other references [5, 37, 38]. In fermenter cascades, cells entering the second and subsequent vessels may go through periods of unbalanced growth as they adapt to the new environmental conditions in each reactor. Therefore, use of simple unstructured metabolic models such as those outlined in this text does not always give accurate results. Nevertheless, it can be shown that the total reactor residence time required to achieve a given degree of substrate conversion is significantly smaller with two CSTRs in series than if only one CSTR were used. In other words, the total reactor volume required is reduced with two smaller tanks in series than with a single large tank. Usually however, only two to four reactors in series are justified as the benefits associated with adding successive stages diminish significantly [39].

13.5.7 Chemostat With Cell Recycle

Cell concentration in a single chemostat can be increased by recycling biomass from the product stream back to the reactor. With more catalyst present in the vessel, higher rates of substrate utilisation and product formation can be achieved. With cell recycle, the critical dilution rate for washout is also increased allowing greater operating flexibility.

There are several ways by which cells can be recycled in

Figure 13.28 Flowsheet for a cascade of two continuous stirred-tank fermenters.

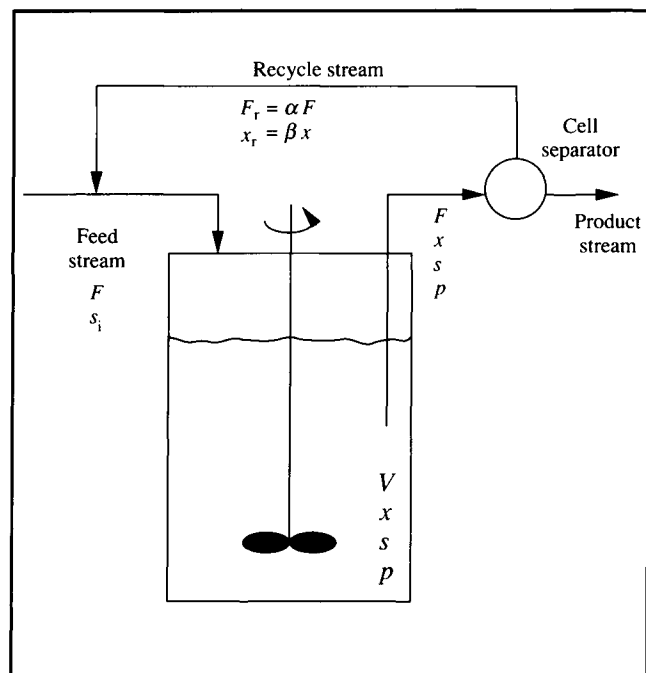


fermentation processes. *External biomass feedback* is illustrated in Figure 13.29; in this scheme, a cell separator such as a centrifuge or settling tank is used to concentrate biomass leaving the reactor. A portion of the concentrate is continuously recycled back to the CSTR with flow rate F_r and cell concentration x_r . Such systems can be operated under steady-state conditions and are used extensively in biological waste treatment. Another way of achieving biomass feedback is *perfusion culture* or *internal biomass feedback*. This operating scheme is often used for mammalian cell culture and is illustrated in Figure 13.30. Depletion of nutrients and accumulation of inhibitory products limit batch cell densities for many animal cell lines to about 10^6 cells ml^{-1} . Cell density and therefore the volumetric productivity of these cultures can be increased by retaining biomass in the reactor while fresh medium is continuously added and spent broth removed. As indicated in Figure 13.30, cells in a perfusion reactor are physically retained in the vessel by a mechanical device such as a filter.

Liquid throughput is thus achieved without continuous removal or dilution of the cells so that concentrations in excess of 10^7 cells ml^{-1} can be obtained. A common problem associated with perfusion systems is blocking or blinding of the filter.

With growing cells, if all the biomass is returned or retained in the reactor, the cell concentration will increase with time and steady state will not be achieved. Therefore, for steady-state operation, some proportion of the biomass must be removed from the system. Chemostat reactors with cell recycle can be analysed using the same mass-balance techniques applied in Section 13.5.4 [37, 38]. Typical results for biomass concentration and biomass productivity with and without cell recycle are shown in Figure 13.31. With cell feedback, because recycled cells are an additional source of biomass in the reactor, washout occurs at dilution rates greater than the maximum specific growth rate. If α is the recycle ratio:

Figure 13.29 Flowsheet for a continuous stirred-tank fermenter with external biomass feedback.



$$\alpha = \frac{F_r}{F}$$

(13.77)

and β is the biomass concentration factor:

$$\beta = \frac{x_r}{x}$$

(13.78)

the critical dilution rate with cell cycle is increased by a factor of $1/(1 + \alpha - \alpha\beta)$ relative to the simple chemostat. Figure 13.31 also shows that biomass productivity is greater in recycle systems by the same factor.

13.5.8 Continuous Operation of a Plug-Flow Reactor

Plug-flow operation is an alternative to mixed operation for continuous reactors. No mixing occurs in an ideal plug-flow reactor; liquid entering the reactor passes through as a discrete 'plug' and does not interact with neighbouring fluid elements. This is achieved at high flow rates which minimise backmixing and variations in liquid velocity. Plug flow is most readily achieved in column or tubular reactors such as that shown in Figure 13.32. Plug-flow reactors can be operated in upflow or

Figure 13.30 Flowsheet for a continuous perfusion reactor with internal biomass feedback.

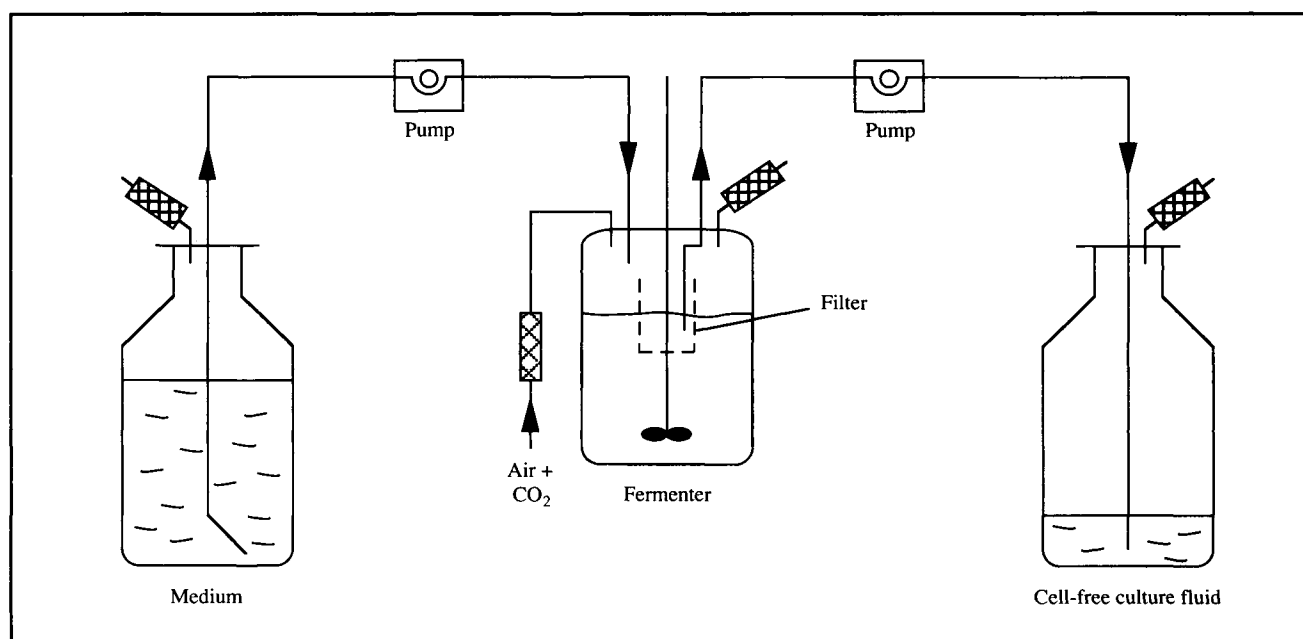
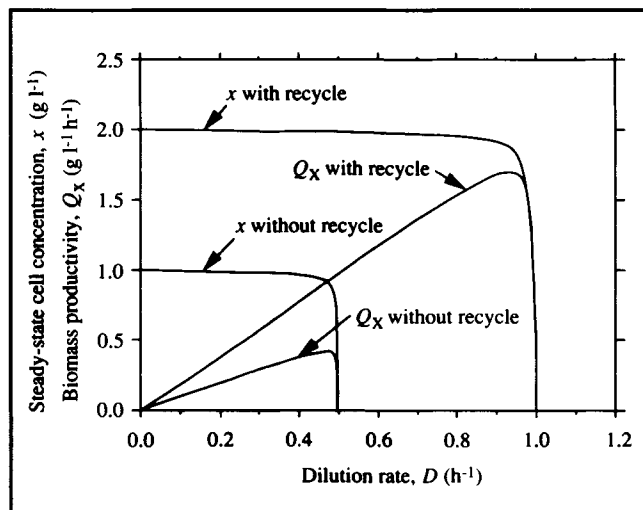


Figure 13.31 Steady-state biomass concentration x and volumetric biomass productivity Q_X for a chemostat with and without cell recycle. Curves were calculated with the following parameter values: $\mu_{\max} = 0.5 \text{ h}^{-1}$, $K_S = 0.01 \text{ g l}^{-1}$, $Y_{XS} = 0.5 \text{ g g}^{-1}$, $s_i = 2 \text{ g l}^{-1}$, $\alpha = 0.5$, $\beta = 2.0$.



downflow mode or, in some cases, horizontally. Plug-flow tubular reactors are known by the abbreviation PFTR.

Liquid in a PFTR flows at constant velocity; thus all parts of the liquid have identical residence time in the reactor. As reaction in the vessel proceeds, concentration gradients of substrate and product develop in the direction of flow. At the feed-end of the PFTR the substrate concentration will be high and the product concentration low because the reaction mixture has just entered the vessel; at the end of the tube the substrate concentration will be low and the product level high.

Consider operation of the PFTR shown in Figure 13.32. The volumetric liquid flow rate through the vessel is F ; the feed stream contains substrate at concentration s_i . At the reactor outlet, the substrate concentration is s_f . This exit concentration can be related to the inlet conditions and reactor residence time using mass-balance techniques. Let us first consider plug-flow operation for enzyme reaction.

13.5.8.1 Enzyme reaction

To develop equations for a plug-flow enzyme reactor, we must consider a small section of the reactor of length Δz as indicated in Figure 13.32. This section is located at distance z from the feed point. Let us perform a steady-state mass balance on substrate around the section using Eq. (6.5).

In Eq. (6.5), $\hat{M}_i$ is the mass flow rate of substrate entering the system; therefore $\hat{M}_i = F s|_z$ where F is the volumetric flow rate through the reactor and $s|_z$ is the substrate concentration at z . Similarly, $\hat{M}_o$, the mass flow rate of substrate leaving the section, is $F s|_{z+\Delta z}$. Substrate is not generated in the reaction; therefore $R_G = 0$. Rate of substrate consumption R_C is equal to the volumetric rate of reaction v multiplied by the volume of the section. v is given by Eq. (11.31); the section volume is equal to $A \Delta z$ where A is the cross-sectional area of the reactor. At steady state, the left-hand side of Eq. (6.5) is zero. Therefore, the mass-balance equation is:

$$F s|_z - F s|_{z+\Delta z} - \frac{v_{\max} s}{K_m + s} A \Delta z = 0 \quad (13.79)$$

where $v_{\max}$ is the maximum rate of enzyme reaction and K_m is the Michaelis constant. Dividing through by $A \Delta z$ and rearranging gives:

$$\frac{F(s|_{z+\Delta z} - s|_z)}{A \Delta z} = \frac{-v_{\max} s}{K_m + s} \quad (13.80)$$

The volumetric flow rate F divided by the reactor cross-sectional area A is equal to the superficial velocity through the column, u . Therefore:

$$\frac{u(s|_{z+\Delta z} - s|_z)}{\Delta z} = \frac{-v_{\max} s}{K_m + s} \quad (13.81)$$

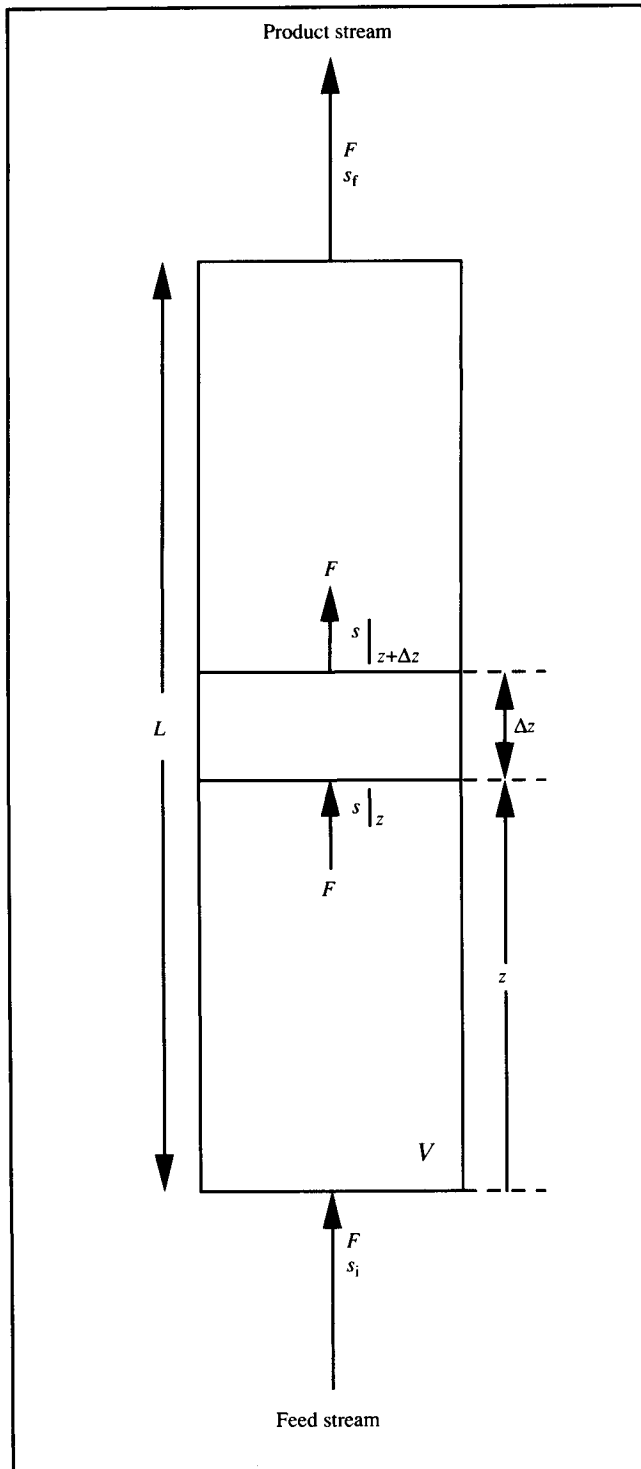
For F and A constant, u is also constant. Eq. (13.81) is valid for any section in the reactor of thickness Δz . For it to be valid at any point in the reactor, we must take the limit as $\Delta z \rightarrow 0$:

$$u \left(\lim_{\Delta z \rightarrow 0} \frac{s|_{z+\Delta z} - s|_z}{\Delta z} \right) = \frac{-v_{\max} s}{K_m + s} \quad (13.82)$$

and apply the definition of the differential from Eq. (D.13) in Appendix D:

$$u \frac{ds}{dz} = \frac{-v_{\max} s}{K_m + s} \quad (13.83)$$

Figure 13.32 Flowsheet for a continuous plug-flow tubular reactor.



Eq. (13.83) is a differential equation for the substrate concentration gradient through the length of a plug-flow reactor. Assuming u and the kinetic parameters are constant, Eq. (13.83) is ready for integration. Separating variables and integrating with boundary condition $s = s_i$ at $z = 0$ gives an expression for the reactor length L required to achieve an outlet concentration of s_f :

$$L = u \left[\frac{K_m}{v_{\max}} \ln \frac{s_i}{s_f} + \frac{s_i - s_f}{v_{\max}} \right]. \quad (13.84)$$

Residence time τ for continuous reactors is defined in Eq. (13.51). If we divide V and F in Eq. (13.51) by A , we can express the residence time for plug-flow reactors in terms of parameters L and u :

$$\tau = \frac{V}{F} = \frac{\left(\frac{V}{A} \right)}{\left(\frac{F}{A} \right)} = \frac{L}{u}. \quad (13.85)$$

Therefore, Eq. (13.84) can be written as:

$$\tau = \frac{K_m}{v_{\max}} \ln \frac{s_i}{s_f} + \frac{s_i - s_f}{v_{\max}}. \quad (13.86)$$

Eqs (13.84) and (13.86) allow us to calculate the reactor length and residence time required to achieve conversion of substrate from concentration s_i to s_f at flow rate u . The form of Eq. (13.86) is identical to that of Eq. (13.10) for batch reactors. As in batch reactors where substrate and product concentrations vary continuously during the reaction period, concentrations in plug-flow reactors change continuously as material moves from inlet to outlet. Thus, plug-flow operation can be seen as a way of simulating batch culture in a continuous flow system.

Plug-flow operation is generally impractical for enzyme conversions unless the enzyme is immobilised and retained inside the vessel. For immobilised-enzyme reactions affected by diffusion, Eq. (13.83) must be modified to account for mass-transfer effects:

$$u \frac{ds}{dz} = -\eta_T \frac{v_{\max} s}{K_m + s} \quad (13.87)$$

where η_T is the total effectiveness factor representing internal and external mass-transfer limitations (see Section 12.5), s is the bulk substrate concentration, and $v_{\max}$ and K_m are intrinsic kinetic parameters. Because η_T is a function of s , we cannot integrate Eq. (13.87) directly as s varies with z in plug-flow reactors.

Plug-flow operation with immobilised enzyme is most

likely to be approached in packed-bed reactors such as that shown in Figure 13.8. Packing in the column can cause substantial backmixing and axial dispersion of liquid, thus interfering with ideal plug flow. Nevertheless, application of the equations developed in this section can give satisfactory results for design of fixed-bed immobilised-enzyme reactors.

Example 13.7 Plug-flow reactor for immobilised enzymes

Immobilised lactase is used to hydrolyse lactose in dairy waste to glucose and galactose. Enzyme is immobilised in resin particles and packed into a 0.5 m^3 column. The total effectiveness factor for the system is close to unity; K_m for the immobilised enzyme is 1.32 kg m^{-3} ; $v_{\max}$ is $45 \text{ kg m}^{-3} \text{ h}^{-1}$. The lactose concentration in the feed stream is 9.5 kg m^{-3} ; a substrate conversion of 98% is required. The column is operated with plug flow for a total of 310 d per year.

- At what flow rate should the reactor be operated?
- How many tonnes of glucose are produced per year?

Solution:

- For 98% substrate conversion, $s_f = (0.02 s_i) = 0.19 \text{ kg m}^{-3}$. Substituting into Eq. (13.86) gives:

$$\tau = \frac{1.32 \text{ kg m}^{-3}}{45 \text{ kg m}^{-3} \text{ h}^{-1}} \ln \left(\frac{9.5 \text{ kg m}^{-3}}{0.19 \text{ kg m}^{-3}} \right) + \frac{(9.5 - 0.19) \text{ kg m}^{-3}}{45 \text{ kg m}^{-3} \text{ h}^{-1}} = 0.32 \text{ h}.$$

From Eq. (13.51):

$$F = \frac{V}{\tau} = \frac{0.5 \text{ m}^3}{0.32 \text{ h}} = 1.56 \text{ m}^3 \text{ h}^{-1}.$$

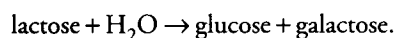
- The rate of lactose conversion is equal to the difference between inlet and outlet mass flow rates of lactose:

$$F(s_i - s_f) = 1.56 \text{ m}^3 \text{ h}^{-1} (9.5 - 0.19) \text{ kg m}^{-3} = 14.5 \text{ kg h}^{-1}.$$

Converting this to an annual rate based on 310 d per year and a molecular weight for lactose of 342:

$$\begin{aligned} \text{Lactose converted} &= 14.5 \text{ kg h}^{-1} \cdot \left| \frac{24 \text{ h}}{1 \text{ d}} \right| \cdot \left| \frac{310 \text{ d}}{1 \text{ yr}} \right| \cdot \left| \frac{1 \text{ kgmol}}{342 \text{ kg}} \right| \\ &= 315 \text{ kgmol yr}^{-1}. \end{aligned}$$

The enzyme reaction is:



Therefore, from reaction stoichiometry, 315 kgmol glucose are produced per year. The molecular weight of glucose is 180; therefore:

$$\begin{aligned}\text{Glucose produced} &= 315 \text{ kgmol yr}^{-1} \cdot \left| \frac{180 \text{ kg}}{1 \text{ kgmol}} \right| \cdot \left| \frac{1 \text{ tonne}}{1000 \text{ kg}} \right| \\ &= 56.7 \text{ tonne yr}^{-1}.\end{aligned}$$

13.5.8.2 Cell culture

Analysis of plug-flow reactors for cell culture follows the same procedure as for enzyme reaction. If the cell specific growth rate is constant and equal to $\mu_{\max}$ throughout the reactor and cell death can be neglected, the equations for reactor residence time are analogous to those derived in Section 13.5.1.2 for batch fermentation, e.g.:

$$\tau = \frac{1}{\mu_{\max}} \ln \frac{x_f}{x_i} \quad (13.88)$$

where τ is the reactor residence time defined in Eq. (13.51), x_i is the biomass concentration at the inlet and x_f is the biomass concentration at the outlet. The form of Eq. (13.88) is identical to that of Eq. (13.20) for batch reaction.

Plug-flow operation is not suitable for cultivation of suspended cells unless the biomass is recycled or there is continuous inoculation of the vessel. Plug-flow operation with cell recycle is used for large-scale wastewater treatment; however applications are limited. Plug-flow reactors are suitable for immobilised-cell reactions with catalyst packed into a fixed bed as shown in Figure 13.8. Even so, operating problems such as those mentioned in Section 13.2.5 mean that PFTRs are rarely employed for industrial fermentations.

13.5.9 Comparison Between Major Modes of Reactor Operation

The relative performance of batch, CSTR and PFTR reactors can be considered from a theoretical point of view in terms of the substrate conversion and product concentration obtained from vessels of the same size. Because the total reactor volume is not fully utilised at all times during fed-batch operation, it is difficult to include this mode of operation in a general comparison.

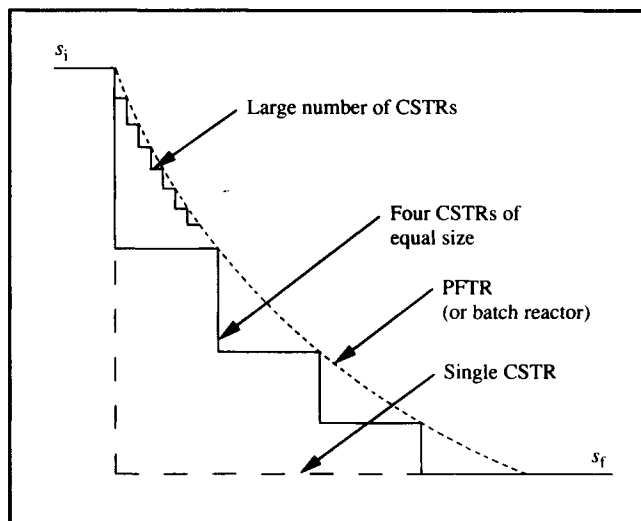
As indicated in Section 13.5.8, the kinetic characteristics of PFTRs are the same as batch reactors; the residence time required for conversion in a plug-flow reactor is therefore the same as in a mixed vessel operated in batch. It can also be shown theoretically that as the number of stages in a CSTR cascade increases, the conversion characteristics of the entire

system approach those of an ideal plug-flow or mixed batch reactor. This is shown diagrammatically in Figure 13.33. The smooth dashed curve represents the progressive decrease in substrate concentration with time spent in a PFTR or batch reactor; concentration is reduced from s_i at the inlet to s_f at the outlet. In a single well-mixed CSTR operated with the same inlet and outlet concentrations, because conditions in the vessel are uniform there is a step change in substrate concentration as soon as the feed enters the reactor. In a cascade of CSTRs, the concentration is uniform in each reactor but there is a step-wise drop in concentration between each stage. As illustrated in Figure 13.33, the larger the number of units in a CSTR cascade, the closer the concentration profile approaches plug-flow or batch behaviour.

The benefits associated with particular reactor designs or modes of operation depend on the kinetic characteristics of the reaction. For zero-order reactions there is no difference between single batch, CSTR and PFTR reactors in terms of overall conversion rate. However, for most reactions including first-order and Michaelis–Menten conversions, rate of reaction decreases as the concentration of substrate decreases. Reaction rate is therefore high at the start of batch culture or at the entrance to a plug-flow reactor because the substrate level is greatest. Subsequently, the reaction velocity falls gradually as substrate is consumed. In contrast, substrate entering a CSTR is immediately diluted to the final or outlet steady-state concentration so that the rate of reaction is comparatively low for the entire reactor. Accordingly, for first-order and Michaelis–Menten reactions, CSTRs achieve lower substrate conversions and lower product concentrations than batch reactors or PFTRs of the same volume. In practice, batch processing is much preferred to PFTR systems because of the operating problems mentioned in Section 13.2.5. However, as discussed in Section 13.5.2, the total time for batch operation depends on the duration of the downtime between batches as well as on the actual conversion time. Because the length of downtime varies considerably from system to system, we cannot account for it here in a general way. Downtime between batches should be minimised as much as possible to maintain high overall production rates.

The comparison between reactors yields a different result if the reaction is autocatalytic. Catalyst is produced by the reaction in fermentation processes; therefore, the volumetric rate of reac-

Figure 13.33 Concentration changes in PFTR, single CSTR and multiple CSTR vessels.



tion increases as the conversion proceeds because the amount of catalyst builds up. Volumetric reaction rate continues to increase until the substrate concentration becomes low, then it declines due to substrate depletion. At the beginning of batch culture, rate of substrate conversion is generally low because relatively few cells are present; it takes some time for cells to accumulate and the rate to pick up. However, in CSTR operation, substrate entering the vessel is immediately exposed to a relatively high biomass concentration so that the rate of conversion is also high. Rates of conversion in chemostats operated close to the optimum dilution rate for biomass productivity (see Section 13.5.4.2) are often 10–20 times greater than in PFTR or batch reactors. This rate advantage disappears if the steady-state substrate concentration in the CSTR is so small that, despite the higher biomass levels present, the conversion rate is lower than the average in a batch or PFTR device. For most fermentations however, CSTRs offer significant theoretical advantages over other modes of reactor operation.

Despite the productivity benefits associated with CSTRs, an overwhelming majority of commercial fermentations are conducted in batch. The reasons lie with the practical advantages associated with batch culture. Batch processes have a lower risk of contamination than continuous-flow reactors; equipment and control failures during long-term continuous operation are also potential problems. Continuous fermentation is feasible only when the cells are genetically stable; if developed strains revert to more rapidly-growing mutants the culture can become dominated over time by the revertant cells. In contrast, freshly-produced inocula are used in batch fermentations giving closer control over the genetic characteristics of the culture.

Continuous culture is not suitable for production of metabolites normally formed near stationary phase when the culture growth rate is low; as mentioned above, productivity in a batch reactor is likely to be greater than in a CSTR under these conditions. Continuous fermentations must be operated for lengthy periods to reap the full benefits of their high productivity. Production can be much more flexible with batch processing; for example, different products each with small market volumes can be made in different batches.

13.5.10 Evaluation of Kinetic and Yield Parameters in Chemostat Culture

In a steady-state chemostat with sterile feed and negligible cell death, the specific growth rate μ is equal to the dilution rate D . This relationship is useful for determining kinetic and yield parameters in cell culture. If growth can be modelling using Monod kinetics, for chemostat culture, Eq. (11.60) becomes:

$$D = \frac{\mu_{\max} s}{K_S + s} \quad (13.89)$$

where $\mu_{\max}$ is the maximum specific growth rate, K_S is the substrate constant and s is the steady-state substrate concentration in the reactor. Eq. (13.89) is analogous mathematically to the Michaelis–Menten expression for enzyme kinetics. If s is measured at various dilution rates, techniques described in Section 11.4 for determining $v_{\max}$ and K_m can be applied for evaluation of $\mu_{\max}$ and K_S . Rearrangement of Eq. (13.89) gives the following linearised equations which can be used for Lineweaver–Burk, Eadie–Hofstee and Langmuir plots, respectively:

$$\frac{1}{D} = \frac{K_S}{\mu_{\max} s} + \frac{1}{\mu_{\max}} \quad (13.90)$$

$$\frac{D}{s} = \frac{\mu_{\max}}{K_S} - \frac{D}{K_S} \quad (13.91)$$

and

$$\frac{s}{D} = \frac{K_S}{\mu_{\max}} + \frac{s}{\mu_{\max}} \quad (13.92)$$

For example, according to Eq. (13.90), $\mu_{\max}$ and K_S can be determined from the slope and intercept of a plot of $1/D$ versus $1/\mu$. The comments made in Sections 11.4.2–11.4.4 about distortion of experimental error apply also to Eqs (13.90)–(13.92).

Chemostat operation is also convenient for determining true yields and maintenance coefficients for cell cultures. An expression relating these parameters to the specific growth rate is given by Eq. (11.81). In chemostat culture with $\mu = D$, Eq. (11.81) becomes:

$$\frac{1}{Y'_{XS}} = \frac{1}{Y_{XS}} + \frac{m_S}{D} \quad (13.93)$$

where Y'_{XS} is the observed biomass yield from substrate, Y_{XS} is the true biomass yield from substrate and m_S is the maintenance coefficient. Therefore, as shown in Figure 13.34, a plot of $1/Y_{XS}$ versus $1/D$ gives a straight line with slope m_S and intercept $1/Y_{XS}$. In a chemostat with sterile feed, the observed biomass yield from substrate Y'_{XS} is obtained as follows:

$$Y'_{XS} = \frac{x}{s_i - s} \quad (13.94)$$

where x and s are steady-state cell and substrate concentrations, respectively, and s_i is the inlet substrate concentration.

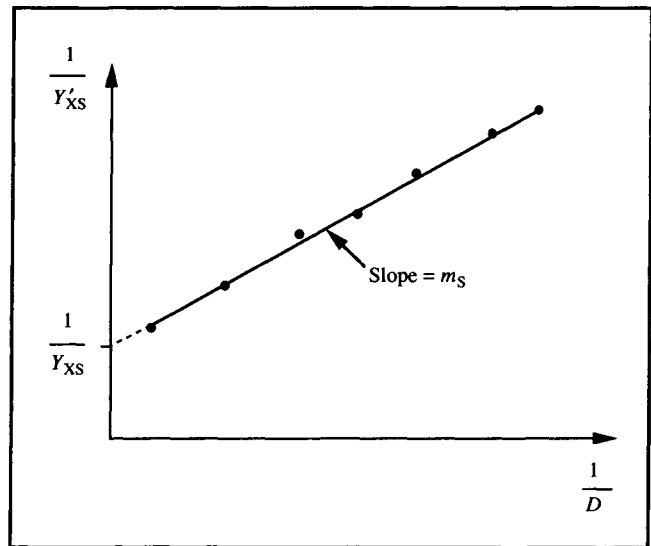
13.6 Sterilisation

Commercial fermentations typically require thousands of litres of liquid medium and millions of litres of air. For processes operated with axenic cultures, these raw materials must be provided free from contaminating organisms. Of all the methods available for sterilisation including chemical treatment, exposure to ultraviolet, gamma and X-ray radiation, sonication, filtration and heating, only the last two are used in large-scale operations. Aspects of fermenter design and construction for aseptic operation were considered in Sections 13.3.1 and 13.3.2. Here, we consider design of sterilisation systems for liquids and gases.

13.6.1 Batch Heat Sterilisation of Liquids

Liquid medium is most commonly sterilised in batch in the vessel where it will be used. The liquid is heated to sterilisation temperature by introducing steam into the coils or jacket of

Figure 13.34 Graphical determination of the maintenance coefficient m_S and true biomass yield Y_{XS} using data from chemostat culture.

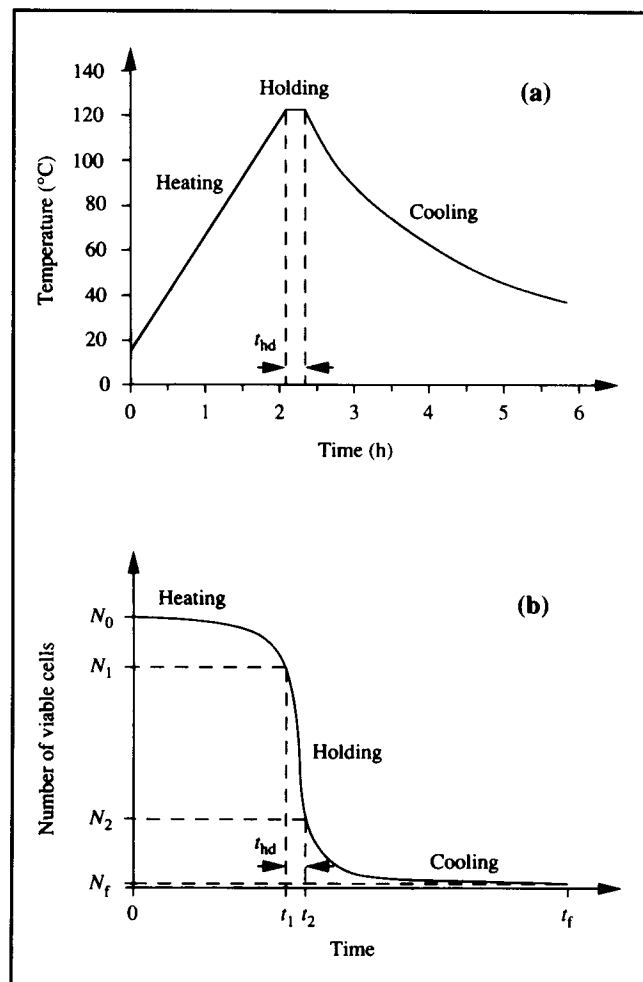


the vessel; alternatively, steam is bubbled directly into the medium, or the vessel is heated electrically. If direct steam injection is used, allowance must be made for dilution of the medium by condensate which typically adds 10–20% to the liquid volume; quality of the steam must also be sufficiently high to avoid contamination of the medium by metal ions or organics. A typical temperature–time profile for batch sterilisation is shown in Figure 13.35(a). Depending on the rate of heat transfer from the steam or electrical element, raising the temperature of the medium in large fermenters can take a significant period of time. Once the holding or sterilisation temperature is reached, the temperature is held constant for a period of time t_{hd} . Cooling water in the coils or jacket of the fermenter is then used to reduce the medium temperature to the required value.

For operation of batch sterilisation systems, we must be able to estimate the holding time required to achieve the desired level of cell destruction. As well as destroying contaminant organisms, heat sterilisation also destroys nutrients in the medium. To minimise this loss, holding times at the highest sterilisation temperature should be kept as short as possible. Cell death occurs at all times during batch sterilisation, including the heating-up and cooling-down periods. The holding time t_{hd} can be minimised by taking into account cell destruction during these periods.

Let us denote the number of contaminants present in the raw medium N_0 . As indicated in Figure 13.35(b), during the

Figure 13.35 (a) Variation of temperature with time for batch sterilisation of liquid medium. (b) Reduction in number of viable cells during batch sterilisation.



heating period this number is reduced to N_1 . At the end of the holding period, the cell number is N_2 ; the final number after cooling is N_f . Ideally, N_f is zero; at the end of the sterilisation cycle we want to have no contaminants present. However, because absolute sterility would require an infinitely-long sterilisation time, it is theoretically impossible to achieve. Normally, the target level of contamination is expressed as a fraction of a cell, which is related to the *probability* of contamination. For example, we could aim for an N_f value of 10^{-3} ; this means we accept the risk that one batch in 1000 will not be sterile at the end of the process. If N_0 and N_f are known, we can determine the holding time required to reduce the number of cells from N_1 to N_2 by considering the kinetics of cell death.

Rate of heat sterilisation is governed by the equations for thermal death outlined in Section 11.14. From Eq. (11.86) for first-order death kinetics, in a batch vessel where cell death is the only process affecting the number of viable cells:

$$\frac{dN}{dt} = -k_d N \quad (13.95)$$

where N is number of viable cells, t is time and k_d is the specific death constant. Eq. (13.95) applies to each stage of the batch sterilisation cycle: heating, holding and cooling. However, because k_d is a strong function of temperature, direct integration of Eq. (13.95) is valid only when the temperature is constant, i.e. during the holding period. The result is:

$$\ln \frac{N_1}{N_2} = k_d t_{hd} \quad (13.96)$$

or

$$t_{hd} = \frac{\ln \frac{N_1}{N_2}}{k_d} \quad (13.97)$$

where t_{hd} is the holding time, N_1 is the number of viable cells at the start of holding, and N_2 is the number of viable cells at the end of holding. k_d is evaluated as a function of temperature using the Arrhenius equation:

$$k_d = A e^{-E_d/RT} \quad (11.46)$$

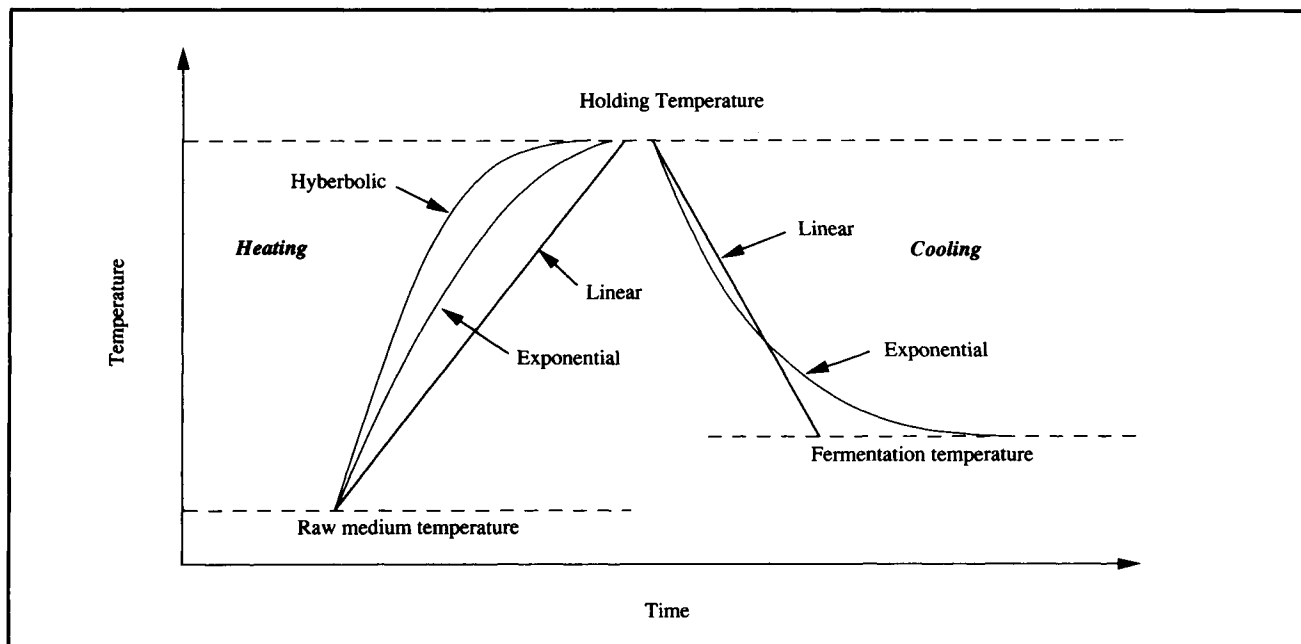
where A is the Arrhenius constant or frequency factor, E_d is the activation energy for the thermal cell death, R is the ideal gas constant and T is absolute temperature.

To use Eq. (13.97) we must know N_1 and N_2 . These numbers are determined by considering the extent of cell death during the heating and cooling periods when the temperature is not constant. Combining Eqs (13.95) and (11.46) gives:

$$\frac{dN}{dt} = -A e^{-E_d/RT} N \quad (13.98)$$

Integration of Eq. (13.98) gives for the heating period:

Figure 13.36 Generalised temperature–time profiles for the heating and cooling stages of a batch sterilisation cycle. (From F.H. Deindoerfer and A.E. Humphrey, 1959, Analytical method for calculating heat sterilization times. *Appl. Microbiol.* 7, 256–264.)



$$\ln \frac{N_0}{N_1} = \int_0^{t_1} A e^{-E_d/RT} dt \quad (13.99)$$

and, for the cooling period:

$$\ln \frac{N_2}{N_f} = \int_{t_2}^{t_f} A e^{-E_d/RT} dt \quad (13.100)$$

where t_1 is the time at the end of heating, t_2 is the time at the end of holding and t_f is the time at the end of cooling. We cannot complete integration of these equations until we know how the temperature varies with time during heating and cooling.

As outlined in Chapter 6, unsteady-state temperature profiles during heating and cooling can be determined from the heat transfer properties of the system. The general form of these equations is shown in Figure 13.36 and Table 13.3. Applying an appropriate expression for T in Eq. (13.99) from Table 13.3 allows us to evaluate the cell number N_1 at the start of the holding period. Similarly, substituting for T in Eq. (13.100) for cooling gives N_2 at the end of the holding period. Use of the resulting values for N_1 and N_2 in Eq. (13.97) completes the holding-time calculation.

Normally, cell death below about 100°C is minimal; however, when heating and cooling are relatively slow, temperatures remain close to the maximum for considerable periods of time and, as indicated in Figure 13.35(b), cell numbers can be reduced significantly outside of the holding period. Usually, holding periods are of the order of minutes whereas heating and cooling of large liquid volumes take hours. Sample design calculations for batch sterilisation are given by Aiba, Humphrey and Millis [40] and Richards [41].

The design procedures outlined in this section apply to batch sterilisation of medium when the temperature is uniform throughout the vessel. However, if the liquid contains contaminant particles in the form of flocs or pellets, temperature gradients may develop. Because heat transfer within solid particles is slower than in liquid, the temperature at the centre of the solid will be lower than that in the liquid for some proportion of the sterilising time. As a result, cell death inside the particles is not as effective as in the liquid. Longer holding times are required to treat solid-phase substrates and media containing particles.

When heat sterilisation is scaled up to larger volumes, longer treatment times are needed to achieve the same sterilisation results at the same holding temperature. For a given raw medium, the initial number of organisms N_0 is directly proportional to the liquid volume; therefore, to obtain the

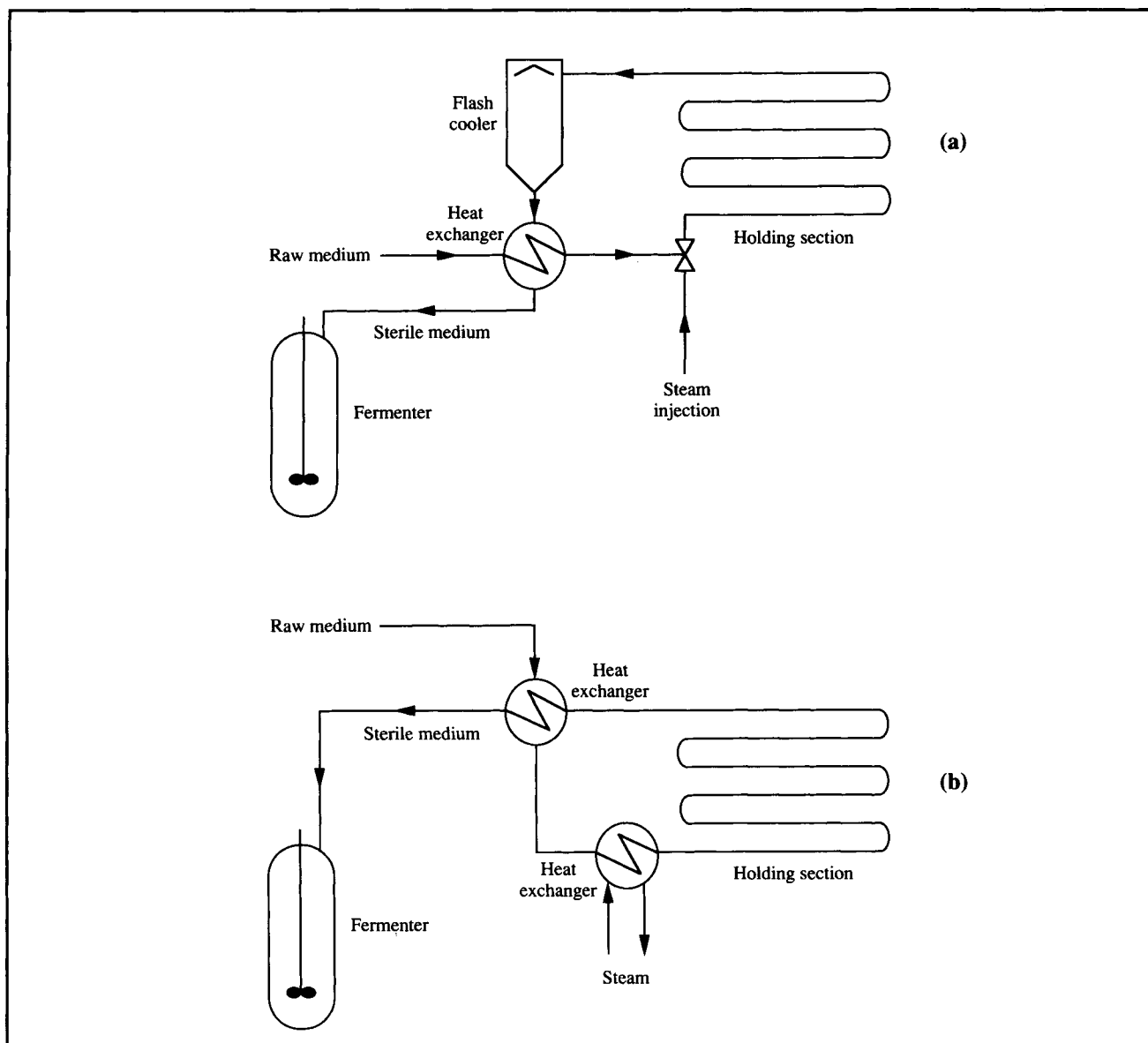
Table 13.3 General equations for temperature as a function of time during the heating and cooling periods of batch sterilisation

(From F.H. Deindoerfer and A.E. Humphrey, 1959, *Analytical method for calculating heat sterilization times*. Appl. Microbiol. 7, 256–264)

Heat transfer method	Temperature–time profile
<i>Heating</i>	
Direct sparging with steam	$T = T_0 \left(1 + \frac{\frac{h \hat{M}_s t}{M_m C_p T_0}}{1 + \frac{\hat{M}_s t}{M_m}} \right)$ (hyperbolic)
Electrical heating	$T = T_0 \left(1 + \frac{\hat{Q} t}{M_m C_p T_0} \right)$ (linear)
Heat transfer from isothermal steam	$T = T_S \left[1 + \frac{T_0 - T_S}{T_S} e^{\left(\frac{-UA}{M_m C_p} \right)} \right]$ (exponential)
<i>Cooling</i>	
Heat transfer to non-isothermal cooling water	$T = T_{ci} \left\{ 1 + \frac{T_0 - T_{ci}}{T_{ci}} e^{\left[\left(\frac{-\hat{M}_w C_{pw} t}{M_m C_p} \right) \left(1 - e^{\left[\frac{-UA}{\hat{M}_w C_{pw}} \right]} \right) \right]} \right\}$ (exponential)

A	=	surface area for heat transfer;
C_p	=	specific heat capacity of medium;
C_{pw}	=	specific heat capacity of cooling water;
h	=	specific enthalpy difference between steam and raw medium;
M_m	=	initial mass of medium;
$\hat{M}_s$	=	mass flow rate of steam;
$\hat{M}_w$	=	mass flow rate of cooling water;
$\hat{Q}$	=	rate of heat transfer;
T	=	temperature;
T_0	=	initial medium temperature;
T_{ci}	=	inlet temperature of cooling water;
T_S	=	steam temperature;
t	=	time; and
U	=	overall heat-transfer coefficient.

Figure 13.37 Continuous sterilising equipment: (a) continuous steam injection with flash cooling; (b) heat transfer using heat exchangers.



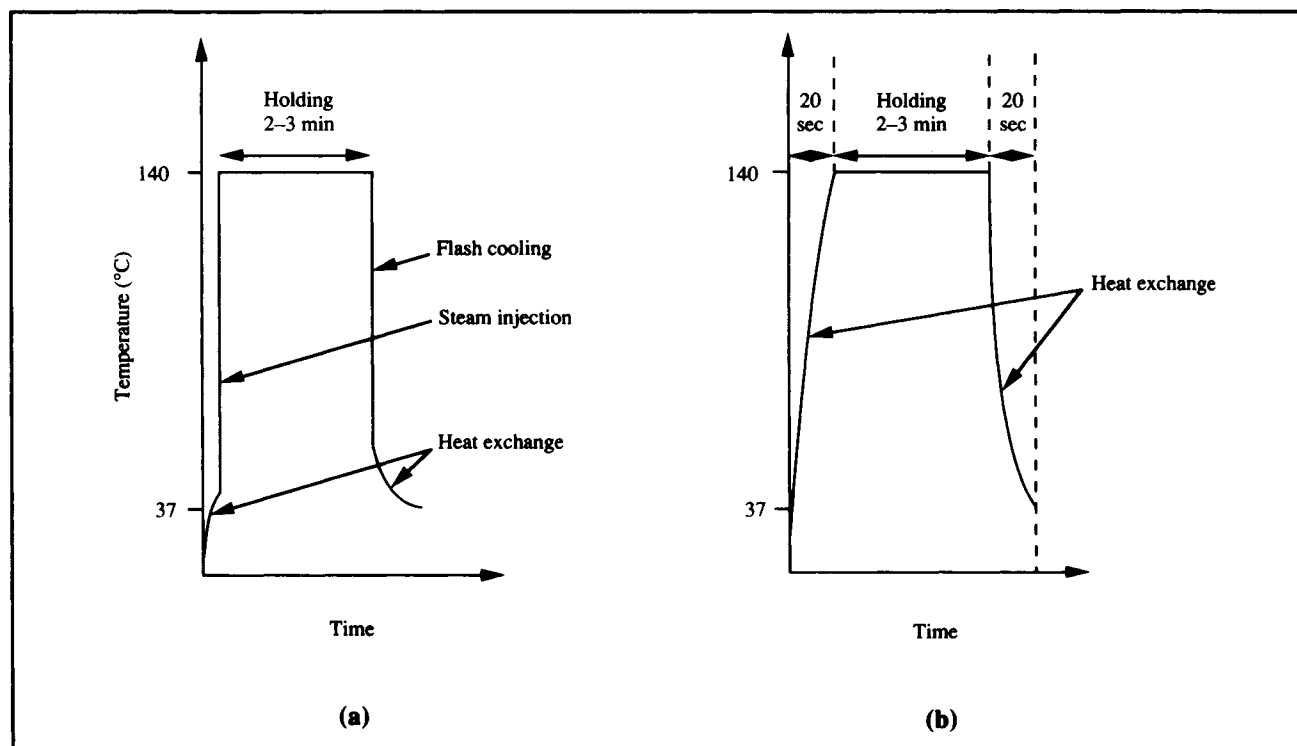
same final N_p a greater number of cells must be destroyed. Scale-up also affects the temperature–time profiles for heating and cooling. Heat-transfer characteristics depend on the equipment used; heating and cooling of larger volumes usually take more time. Sustained elevated temperatures during heating and cooling are damaging to vitamins, proteins and sugars in nutrient solutions and reduce the quality of the medium [42]. Because it is necessary to hold large volumes of medium

for longer periods of time, this problem is exacerbated with scale-up.

13.6.2 Continuous Heat Sterilisation of Liquids

Continuous sterilisation, particularly a high-temperature, short-exposure-time process, can significantly reduce damage

Figure 13.38 Variation of temperature with time in the continuous sterilisers of Figure 13.37.



to medium ingredients while achieving high levels of cell destruction. Other advantages include improved steam economy and more reliable scale-up. The amount of steam needed for continuous sterilisation is 20–25% that used in batch processes; the time required is also significantly reduced because heating and cooling are virtually instantaneous.

Typical equipment configurations for continuous sterilisation are shown in Figure 13.37. In Figure 13.37(a), raw medium entering the system is first pre-heated by hot, sterile medium in a heat exchanger; this economises on steam requirements for heating and cools the sterile medium. Steam is then injected directly into the medium as it flows through a pipe; the liquid temperature rises almost instantaneously to the desired sterilisation temperature. The time of exposure to this temperature depends on the length of pipe in the holding section of the steriliser. After sterilisation, the medium is cooled instantly by passing it through an expansion valve into a vacuum chamber; further cooling takes place in the heat exchanger where residual heat is used to pre-heat incoming medium. Figure 13.37(b) shows an alternative sterilisation scheme based on heat exchange between steam and medium. Raw medium is pre-heated with

hot, sterile medium in a heat exchanger then brought to the sterilisation temperature by further heat exchange with steam. The sterilisation temperature is maintained in the holding section before being reduced to the fermentation temperature by heat exchange with incoming medium. Heat-exchange systems are more expensive to construct than injection devices; fouling of the internal surfaces also reduces the efficiency of heat transfer between cleanings. On the other hand, a disadvantage associated with steam injection is dilution of the medium by condensate; foaming from direct steam injection can also cause problems with operation of the flash cooler. As indicated in Figure 13.38, rates of heating and cooling in continuous sterilisation are much more rapid than in batch; accordingly, in design of continuous sterilisers, contributions to cell death outside of the holding period are generally ignored.

An important variable affecting performance of continuous sterilisers is the nature of fluid flow in the system. Ideally, all fluid entering the equipment at a particular instant should spend the same time in the steriliser and exit the system at the same time; unless this occurs we cannot fully control the time spent in the steriliser by all fluid elements. No mixing should

Figure 13.39 Velocity distributions for flow in pipes. (a) In plug flow, fluid velocity is the same across the diameter of the pipe as indicated by the arrows of equal length. (b) In fully-developed turbulent flow, the velocity distribution approaches that of plug flow; however there is some reduction of flow speed at the walls. (c) In laminar flow, there is a continuous increase of velocity from the walls to the centre of the tube.

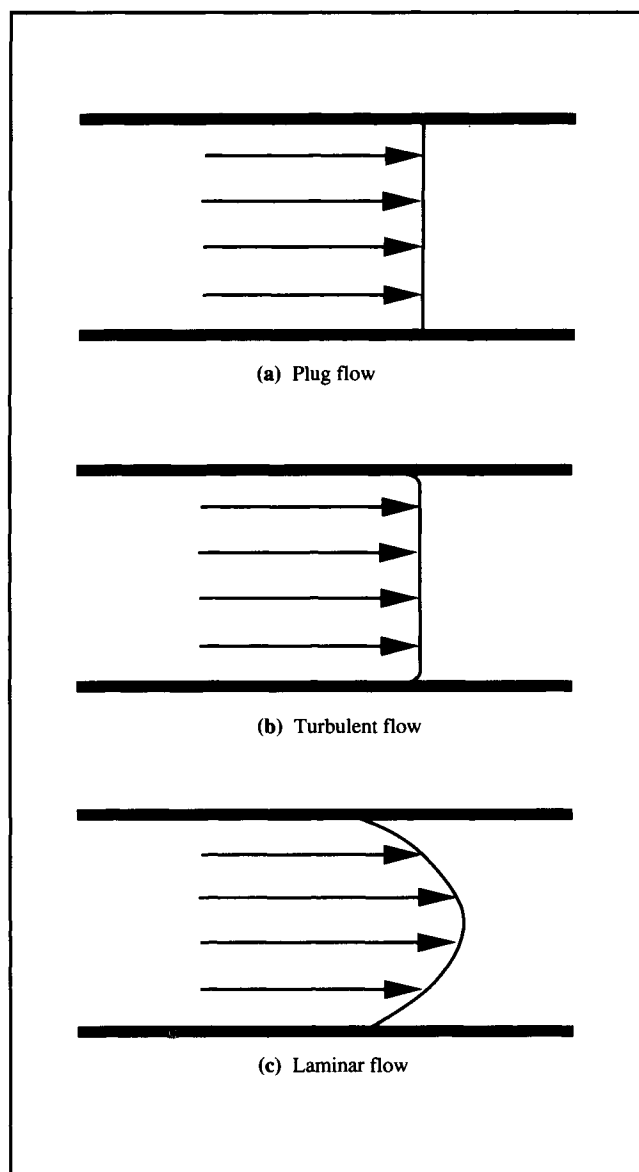
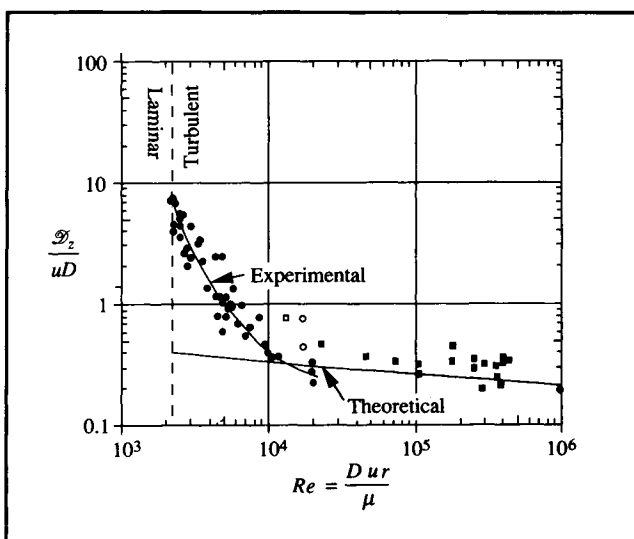


Figure 13.40 Correlation for determining the axial-dispersion coefficient in turbulent pipe flow. Re is Reynolds number, D is pipe diameter, u is average linear fluid velocity, ρ is fluid density, μ is fluid viscosity, and $\mathcal{D}_z$ is the axial-dispersion coefficient. Data were measured using single fluids in: (●) straight pipes; (■) pipes with bends; (□) artificially-roughened pipe; and (○) curved pipe. (From O. Levenspiel, 1958, Longitudinal mixing of fluids flowing in circular pipes. *Ind. Eng. Chem.* 50, 343–346.)



occur in the tubes; if fluid nearer the entrance of the pipe mixes with fluid ahead of it, there is a risk that contaminants will be transferred to the outlet of the steriliser. The type of flow in pipes where there is neither mixing nor variation in fluid velocity is called *plug flow* as already described in Section 13.5.8 for plug-flow reactors. Plug flow is an ideal flow pattern; in reality, fluid elements in pipes have a range of different velocities. As illustrated in Figure 13.39, flow tends to be faster through the centre of the tube than near the walls. However, plug flow is approached in pipes at turbulent Reynolds numbers (see Section 7.2.2) above about 2×10^4 ; operation at high Reynolds numbers minimises fluid mixing and velocity variation.

Deviation from plug-flow behaviour is characterised by the degree of *axial dispersion* in the system, i.e. the degree to which mixing occurs along the length or axis of the pipe. Axial dispersion is a critical factor affecting design of continuous sterilisers. The relative importance of axial dispersion and bulk flow in transfer of material through the pipe is represented by a dimensionless variable called the *Peclet number*:

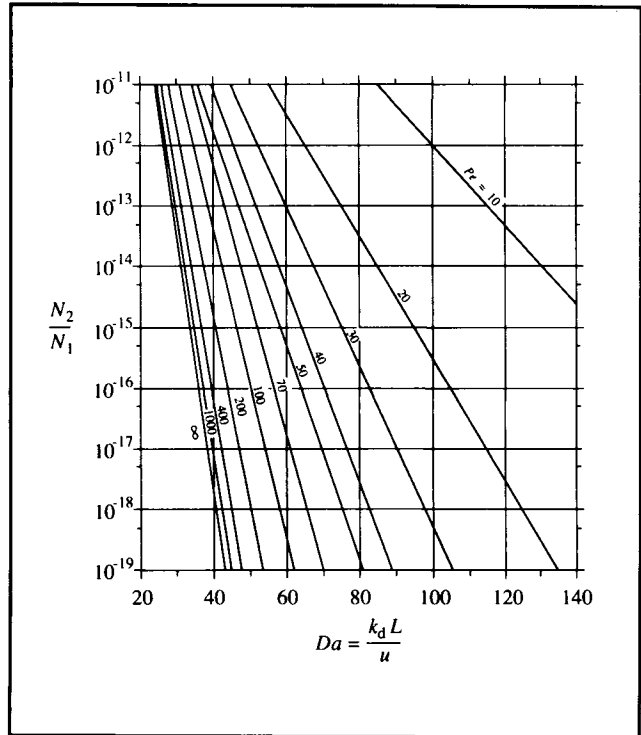
$$Pe = \frac{uL}{\mathcal{D}_z} \quad (13.101)$$

where Pe is the Peclet number, u is the average linear fluid velocity, L is the pipe length and $\mathcal{D}_z$ is the *axial-dispersion coefficient*. For perfect plug flow, $\mathcal{D}_z$ is zero and Pe is infinitely large; in practice, Peclet numbers between 3 and 600 are typical. The value of $\mathcal{D}_z$ for a particular system depends on the Reynolds number and pipe geometry; a correlation from the engineering literature for evaluating $\mathcal{D}_z$ is shown in Figure 13.40. Once the Peclet number has been calculated from Eq. (13.101), the extent of cell destruction in the steriliser can be related to the specific death constant k_d using Figure 13.41. In Figure 13.41, N_1 is the number of viable cells entering the system, N_2 is the number of cells leaving, Pe is the Peclet number as defined by Eq. (13.101), and Da is another dimensionless number called the *Damköhler number*:

$$Da = \frac{k_d L}{u} \quad (13.102)$$

where k_d is the specific death constant, L is the length of the holding pipe and u is the average linear liquid velocity. The lower the value of N_2/N_1 , the greater is the level of cell destruction. Figure 13.41 shows that, at any given sterilisation temperature defining the value of k_d and Da , performance of the steriliser declines significantly as the Peclet number decreases. Design calculations for a continuous steriliser are illustrated in Example 13.8.

Figure 13.41 Thermal destruction of contaminating organisms as a function of the Peclet number Pe and Damköhler number Da . N_1 is the number of viable cells entering the holding section of the steriliser; N_2 is the number of cells leaving. (From S. Aiba, A.E. Humphrey and N.F. Millis, 1965, *Biochemical Engineering*, Academic Press, New York.)



Example 13.8 Holding temperature in a continuous steriliser

Medium at a flow rate of $2 \text{ m}^3 \text{ h}^{-1}$ is to be sterilised by heat exchange with steam in a continuous steriliser. The liquid contains bacterial spores at a concentration of $5 \times 10^{12} \text{ m}^{-3}$; the activation energy and Arrhenius constant for thermal destruction of these contaminants are 283 kJ gmol^{-1} and $5.7 \times 10^{39} \text{ h}^{-1}$, respectively. A contamination risk of one organism surviving every 60 days' operation is considered acceptable. The steriliser pipe has an inner diameter of 0.1 m ; the length of the holding section is 24 m . The density of the medium is 1000 kg m^{-3} and the viscosity is $3.6 \text{ kg m}^{-1} \text{ h}^{-1}$. What sterilising temperature is required?

Solution:

The desired level of cell destruction is evaluated using a basis of 60 days. Ignoring any cell death in the heating and cooling sections, the number of cells entering the holding section over 60 d is:

$$N_1 = 2 \text{ m}^3 \text{ h}^{-1} (5 \times 10^{12} \text{ m}^{-3}) \cdot \left| \frac{24 \text{ h}}{1 \text{ d}} \right| \cdot (60 \text{ d}) = 1.44 \times 10^{16}.$$

N_2 , the acceptable number of cells leaving during this period is 1. Therefore:

$$\frac{N_2}{N_1} = \frac{1}{1.44 \times 10^{16}} = 6.9 \times 10^{-17}.$$

The linear velocity u in the steriliser is equal to the volumetric flow rate divided by the cross-sectional area of the pipe:

$$u = \frac{2 \text{ m}^3 \text{ h}^{-1}}{\pi \left(\frac{0.1 \text{ m}}{2} \right)^2} = 254.6 \text{ m h}^{-1}.$$

To calculate Pe we must first determine $\mathcal{D}_z$ using Figure 13.40:

$$Re = \frac{D u \rho}{\mu} = \frac{(0.1 \text{ m}) (254.6 \text{ m h}^{-1}) (1000 \text{ kg m}^{-3})}{3.6 \text{ kg m}^{-1} \text{ h}^{-1}} = 7.07 \times 10^3.$$

For $Re = 7.07 \times 10^3$ we can determine $\mathcal{D}_z$ from Figure 13.40 using either the experimental or theoretical curve. Let us choose the experimental curve as this gives a larger value of $\mathcal{D}_z$ and a smaller value of Pe ; the steriliser design will thus be more conservative. Therefore, $\mathcal{D}_z / uD = 0.65$:

$$\mathcal{D}_z = 0.65 (254.6 \text{ m h}^{-1}) (0.1 \text{ m}) = 16.6 \text{ m}^2 \text{ h}^{-1}.$$

From Eq. (13.101):

$$Pe = \frac{uL}{\mathcal{D}_z} = \frac{(254.6 \text{ m h}^{-1}) (24 \text{ m})}{16.6 \text{ m}^2 \text{ h}^{-1}} = 368.$$

Using Figure 13.41, we can determine the value of k_d for the desired level of cell destruction. Da corresponding to $N_2/N_1 = 6.9 \times 10^{-17}$ and $Pe = 368$ is about 42. Therefore:

$$k_d = \frac{u Da}{L} = \frac{(254.6 \text{ m h}^{-1}) (42)}{24 \text{ m}} = 445.6 \text{ h}^{-1}.$$

The sterilisation temperature can be evaluated after rearranging Eq. (11.46). Dividing both sides by A and taking natural logarithms gives:

$$\ln \frac{k_d}{A} = \frac{-E_d}{RT}.$$

Therefore:

$$T = \frac{\left(\frac{-E_d}{R} \right)}{\ln \left(\frac{k_d}{A} \right)}.$$

$E_d = 283 \text{ kJ gmol}^{-1} = 283 \times 10^3 \text{ J gmol}^{-1}$; $A = 5.7 \times 10^{39} \text{ h}^{-1}$; from Table 2.5, the ideal gas constant R is $8.3144 \text{ J K}^{-1} \text{ gmol}^{-1}$. Therefore:

$$T = \frac{\left(\frac{-283 \times 10^3 \text{ J gmol}^{-1}}{8.3144 \text{ J K}^{-1} \text{ gmol}^{-1}} \right)}{\ln \left(\frac{445.6 \text{ h}^{-1}}{5.7 \times 10^{39} \text{ h}^{-1}} \right)} = 398.4 \text{ K}.$$

Using the conversion between K and °C given in Eq. (2.24), $T = 125^\circ\text{C}$.

Heating and cooling in continuous sterilisers are so rapid that in design calculations they are considered instantaneous. While reducing nutrient deterioration, this feature of the process can cause problems if there are solids present in the medium. During heating, the temperature at the core of solid particles remains lower than in the medium. Because of the extremely short contact times in continuous sterilisers compared with batch systems, there is a greater risk that particles will not be properly sterilised. It is important therefore that raw medium be clarified as much as possible before it enters a continuous steriliser.

13.6.3 Filter Sterilisation of Liquids

Sometimes, fermentation media or selected ingredients are sterilised by filtration rather than heat. For example, media containing heat-labile components such as enzymes and serum are easily destroyed by heat and must be sterilised by other means. Typically, membranes used for filter sterilisation of liquids are made of cellulose esters or other polymers and have pores between 0.2 and 0.45 μm in diameter. The membranes themselves must be sterilised before use, usually by steam. As medium is passed through the filter, bacteria and other particles with dimensions greater than the pore size are screened out and collect on the surface of the membrane. The small pore sizes used in liquid filtration mean that the membranes are readily blocked unless the medium is pre-filtered to remove any large particles. To achieve high flow rates, large surface areas are required.

Liquid filtration is generally not as effective or reliable as heat sterilisation. Viruses and mycoplasma are able to pass through membrane filters; care must also be taken to prevent holes or tears in the membrane. Usually, filter-sterilised medium is incubated for a period of time before use to test its sterility.

13.6.4 Sterilisation of Air

The number of microbial cells in air is of the order 10^3 – 10^4 m^{-3} [40]. Filtration is the most common method for sterilising air in large-scale bioprocesses; heat sterilisation of gases is economically impractical. *Depth filters* consisting of compacted beds or pads of fibrous material such as glass wool have been used widely in the fermentation industry. Distances between the fibres in depth filters are typically 2–10 μm , about 10 times greater than the dimensions of the bacteria and spores to be removed. Air-borne particles penetrate the bed to various depths before their passage through the filter is arrested; the depth of the filter medium required to produce air of sufficient

quality depends on the operating flow rate and the incoming level of contamination. Cells are collected in depth filters by a combination of impaction, interception, electrostatic effects, and, for particles smaller than about 1.0 μm , diffusion to the fibres. Depth filters do not perform well if there are large fluctuations in flow rate or if the air is wet; liquid condensing in the filter increases the pressure drop, causes channelling of the gas flow, and provides a pathway for organisms to grow through the bed.

Increasingly, depth filters are being replaced for industrial applications by membrane cartridge filters. These filters use steam-sterilisable polymeric membranes which act as *surface filters* trapping contaminants as on a sieve. Membrane filter cartridges typically contain a pleated, hydrophobic filter with small and uniformly-sized pores 0.45 μm or less in diameter. The hydrophobic nature of the surface minimises problems with filter wetting while the pleated configuration allows a high filtration area to be packed into a small cartridge volume. Pre-filters built into the cartridge or up-stream reduce fouling of the membrane by removing large particles, oil, water droplets and foam from the incoming gas.

Filters are also used to sterilise effluent gases leaving fermenters. In this application, the objective is to prevent release into the atmosphere of any microorganisms entrained in aerosols in the headspace of the reactor. The concentration of cells in fermenter off-gas is several times greater than in air. Containment is particularly important when organisms used in fermentation are potentially harmful to plant personnel or the environment; companies operating fermentations with pathogenic or recombinant strains are required by regulatory authorities to prevent escape of the cells.

13.7 Summary of Chapter 13

Chapter 13 contains a variety of qualitative and quantitative information about design and operation of bioreactors. After studying this chapter, you should:

- (i) be able to assess in general terms the effect of reaction engineering on total production costs in bioprocessing;
- (ii) be familiar with a range of bioreactor configurations in addition to the standard *stirred tank*, including *bubble-column*, *airlift*, *packed-bed*, *fluidised-bed* and *trickle-bed* designs;
- (iii) understand the practical aspects of bioreactor construction, particularly those aimed at maintaining aseptic conditions;
- (iv) be familiar with measurements used in *fermentation monitoring* and the problems associated with lack of on-line methods for important fermentation parameters;

- (v) be familiar with established and modern approaches to *fermentation control*;
- (vi) be able to predict *batch reaction times* for enzyme and cells reactions;
- (vii) be able to predict the performance of *fed-batch reactors* operated under quasi-steady-state conditions;
- (viii) be able to predict and compare the performance of *continuous stirred-tank reactors* and *continuous plug-flow reactors*;
- (ix) know how to use steady-state chemostat data to determine kinetic and yield parameters for cell culture; and
- (x) know how batch and continuous systems are designed for heat sterilisation of liquid medium and methods for filter sterilisation of fermentation gases.

Problems

13.1 Economics of batch enzyme conversion

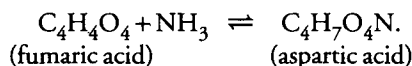
An enzyme is used to convert substrate to a commercial product in a 1600-litre batch reactor. $v_{\max}$ for the enzyme is $0.9 \text{ g l}^{-1} \text{ h}^{-1}$; K_m is 1.5 g l^{-1} . Substrate concentration at the start of the reaction is 3 g l^{-1} ; according to the stoichiometry of the reaction, conversion of 1 g substrate produces 1.2 g product. The cost of operating the reactor including labour, maintenance, energy and other utilities is estimated at \$4800 per day. The cost of recovering the product depends on the extent of substrate conversion and the resulting concentration of product in the final reaction mixture. For conversions between 70% and 100%, the cost of downstream processing can be approximated using the equation:

$$C = 155 - 0.33X$$

where C is cost in \$ per kg product treated and X is the percentage substrate conversion. The market price for the product is \$750 kg^{-1} . Currently, the enzyme reactor is operated with 75% substrate conversion; however it is proposed to increase this to 90%. Estimate the effect this will have on the economics of the process.

13.2 Batch production of aspartic acid using cell-bound enzyme

Aspartase enzyme is used industrially for manufacture of aspartic acid, a component of low-calorie sweetener. Fumaric acid and ammonia are converted to aspartic acid according to the equation:



Under investigation is a process using aspartase in intact *Bacillus cadaveris* cells. In the substrate range of interest, the conversion can be described using Michaelis–Menten kinetics with K_m 4.0 g l^{-1} . The substrate solution contains 15% (w/v) ammonium fumarate; enzyme is added in the form of lyophilised cells and the reaction stopped when 85% of the substrate is converted. At 32°C , $v_{\max}$ for the enzyme is $5.9 \text{ g l}^{-1} \text{ h}^{-1}$ and its half-life is 10.5 d. At 37°C , $v_{\max}$ increases to $8.5 \text{ g l}^{-1} \text{ h}^{-1}$ but the half-life is reduced to 2.3 d.

- (a) Which operating temperature would you recommend?
- (b) The average downtime between batch reactions is 28 h. At the temperature chosen in (a), calculate the reactor volume required to produce 5000 tonnes of aspartic acid per year.

13.3 Prediction of batch culture time

A strain of *Escherichia coli* has been genetically engineered to produce human protein. A batch culture is started by inoculating 12 g cells into a 100-litre bubble-column fermenter containing 10 g l^{-1} glucose. The maximum specific growth rate of the culture is 0.9 h^{-1} ; the biomass yield from glucose is 0.575 g g^{-1} .

- (a) Estimate the time required to reach stationary phase.
- (b) What will be the final cell density if the fermentation is stopped after only 70% of the substrate is consumed?

13.4 Fed-batch scheduling

Nicotiana tabacum cells are cultured to high density for production of polysaccharide gum. The reactor used is a stirred tank, containing initially 100 litres medium. The maximum specific growth rate of the culture is 0.18 d^{-1} and the yield of biomass from substrate is 0.5 g g^{-1} . The concentration of growth-limiting substrate in the medium is 3% (w/v). The reactor is inoculated with 1.5 g l^{-1} cells and operated in batch until the substrate is virtually exhausted; medium flow is then started at a rate of 4 l d^{-1} . Fed-batch operation occurs under quasi-steady-state conditions.

- (a) Estimate the batch culture time and final biomass concentration.
- (b) Fed-batch operation is carried out for 40 d. What is the final mass of cells in the reactor?
- (c) The fermenter is available 275 d per year with a downtime between runs of 24 h. How much plant cell biomass is produced annually?

13.5 Fed-batch production of cheese starter culture

Lactobacillus casei is propagated under essentially anaerobic conditions to provide a starter culture for manufacture of Swiss cheese. The culture produces lactic acid as a by-product of energy metabolism. The system has the following characteristics:

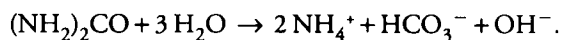
$$\begin{aligned} Y_{XS} &= 0.23 \text{ kg kg}^{-1} \\ K_S &= 0.15 \text{ kg m}^{-3} \\ \mu_{\max} &= 0.35 \text{ h}^{-1} \\ m_S &= 0.135 \text{ kg kg}^{-1} \text{ h}^{-1} \end{aligned}$$

A stirred fermenter is operated in fed-batch mode at quasi-steady state with a feed flow rate of $4 \text{ m}^3 \text{ h}^{-1}$ and feed substrate concentration of 80 kg m^{-3} . After 6 h, the liquid volume is 40 m^3 .

- What was the initial culture volume?
- What is the concentration of substrate at quasi-steady state?
- What is the concentration of cells at quasi-steady state?
- What mass of cells is produced after 6 h fed-batch operation?

13.6 Continuous enzyme conversion in a fixed-bed reactor

A system is being developed to remove urea from the blood of patients with renal failure. A prototype fixed-bed reactor is set up with urease immobilised in 2-mm gel beads; buffered urea solution is recycled rapidly through the bed so that the system is well mixed. The urease reaction is:



K_m for the immobilised urease is 0.54 g l^{-1} . The volume of beads in the reactor is 250 cm^3 , the total amount of urease is 10^{-4} g , and the turnover number is $11\,000 \text{ g NH}_4^+ (\text{g enzyme})^{-1} \text{ s}^{-1}$. The effective diffusivity of urease in the gel is $7 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$; external mass-transfer effects are negligible. The reactor is operated continuously with a liquid volume of 1 litre. The feed stream contains 0.42 g l^{-1} urea; the desired urea concentration is 0.02 g l^{-1} . Ignoring enzyme deactivation, what volume of urea solution can be treated in 30 min?

13.7 Batch and continuous biomass production

Pseudomonas methylotrophus is used to produce single-cell protein from methanol in a 1000 m^3 pressure-cycle airlift

fermenter. The biomass yield from substrate is 0.41 g g^{-1} , K_S is 0.7 mg l^{-1} , and the maximum specific growth rate is 0.44 h^{-1} . The medium contains 4% (w/v) methanol. A substrate conversion of 98% is desirable. The reactor may be operated either in batch or continuous mode. If operated in batch, an inoculum of 0.01% (w/v) is used and the downtime between batches is 20 h. If operated continuously, a downtime of 25 d is expected per year. Neglecting maintenance requirements, compare the annual biomass production from batch and continuous reactors.

13.8 Reactor design for immobilised enzymes

6-Aminopenicillanic acid used to produce semi-synthetic penicillins is prepared by enzymatic hydrolysis of fermentation-derived penicillin G. Penicillin acylase immobilised in alginate is being considered for the process; the immobilised-enzyme particles are small enough that mass-transfer effects can be ignored. The starting concentration of penicillin G is 10% (w/v); because of the high cost of the substrate, 99% conversion is required. Under these conditions, enzymatic conversion of penicillin G can be considered a first-order reaction. It has not been decided whether a batch, CSTR or plug-flow reactor would be most suitable. The downtime between batch reactions is expected to be 20 h. For the batch and CSTR reactors, the reaction rate constant is $0.8 \times 10^{-4} \text{ s}^{-1}$; in the PFTR, the packing density of enzyme beads can be up to four times greater than in the other reactors. Determine the smallest reactor required to treat 400 tonnes penicillin G per year.

13.9 Two-stage chemostat for secondary metabolite production

A two-stage chemostat system is used for production of secondary metabolite. The volume of each reactor is 0.5 m^3 ; the flow rate of feed is 50 l h^{-1} . Mycelial growth occurs in the first reactor; the second reactor is used for product synthesis. The concentration of substrate in the feed is 10 g l^{-1} . Kinetic and yield parameters for the organism are:

$$\begin{aligned} Y_{XS} &= 0.5 \text{ kg kg}^{-1} \\ K_S &= 1.0 \text{ kg m}^{-3} \\ \mu_{\max} &= 0.12 \text{ h}^{-1} \\ m_S &= 0.025 \text{ kg kg}^{-1} \text{ h}^{-1} \\ q_P &= 0.16 \text{ kg kg}^{-1} \text{ h}^{-1} \\ Y_{PS} &= 0.85 \text{ kg kg}^{-1} \end{aligned}$$

Assume that product synthesis is negligible in the first reactor and growth is negligible in the second reactor.

- Determine the cell and substrate concentrations entering the second reactor.

- (b) What is the overall substrate conversion?
 (c) What is the final concentration of product?

13.10 Kinetic analysis of bioremediating bacteria using a chemostat

A strain of *Ancylobacter* bacteria capable of growing on 1,2-dichloroethane is isolated from sediment in the river Rhine. The bacteria are to be used for on-site bioremediation of soil contaminated with chlorinated halogens. Kinetic parameters for the organism are determined using chemostat culture. A 1-litre fermenter is used with a 1,2-dichloroethane feed at a concentration of 100 μM . Steady-state substrate concentrations are measured as a function of flow rate.

Flow rate (ml h^{-1})	Substrate concentration (μM)
10	17.4
15	25.1
20	39.8
25	46.8
30	69.4
35	80.1
50	100

- (a) Determine μ_{max} and K_S for this organism.
 (b) Determine the maximum operating flow rate of the reactor.

13.11 Kinetic and yield parameters of an auxotrophic mutant

An *Enterobacter aerogenes* auxotroph capable of overproducing threonine has been isolated. The kinetic and yield parameters for this organism are investigated using a 2-litre chemostat with 10 g l^{-1} glucose in the feed. Steady-state cell and substrate concentrations are measured as a function of flow rate.

Flow rate (l h^{-1})	Cell concentration (g l^{-1})	Substrate concentration (g l^{-1})
1.0	3.15	0.010
1.4	3.22	0.038
1.6	3.27	0.071
1.7	3.26	0.066
1.8	3.21	0.095
1.9	3.10	0.477

Determine the maximum specific growth rate, the substrate constant K_S , the maintenance coefficient, and the true biomass yield for this culture.

13.12 Continuous sterilisation

A 15- m^3 chemostat is operated with dilution rate 0.1 h^{-1} . A continuous steriliser with steam injection and flash cooling delivers sterilised medium to the fermenter. Medium in the holding section of the steriliser is maintained at 130°C . The concentration of contaminants in the raw medium is 10^5 ml^{-1} ; an acceptable contamination risk is one organism every 3 months. The Arrhenius constant and activation energy for thermal death are estimated to be $7.5 \times 10^{39} \text{ h}^{-1}$ and $288.5 \text{ kJ gmol}^{-1}$, respectively. The steriliser pipe inner diameter is 12 cm. At 130°C the liquid density is 1000 kg m^{-3} and the viscosity is $4 \text{ kg m}^{-1} \text{ h}^{-1}$.

- (a) Assuming perfect plug flow, determine the length of the holding section.
 (b) What length is required if axial-dispersion effects are taken into account?
 (c) If the steriliser is constructed with length as determined in (a) and operated at 130°C as planned, what will be the rate of fermenter contamination?

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Suggestions for Further Reading

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Appendices

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A

Conversion Factors

Note on Tables:

Entries in the same row are equivalent. For example, in Table A.1, 1 m = 3.281 ft, 1 mile = 1.609×10^3 m, etc. Exact numerical values are printed in bold type; others are given to four significant figures.

Table A.1 Length (L)

metre (m)	inch (in.)	foot (ft)	mile	micrometre (μm) (formerly micron)	angstrom ($\text{\AA}$)
1	3.937×10^1	3.281	6.214×10^{-4}	10^6	10^{10}
2.54×10^{-2}	1	8.333×10^{-2}	1.578×10^{-5}	2.54×10^4	2.54×10^8
3.048×10^{-1}	1.2×10^1	1	1.894×10^{-4}	3.048×10^5	3.048×10^9
1.609×10^3	6.336×10^4	5.28×10^3	1	1.609×10^9	1.609×10^{13}
10^{-6}	3.937×10^{-5}	3.281×10^{-6}	6.214×10^{-10}	1	10^4
10^{-10}	3.937×10^{-9}	3.281×10^{-10}	6.214×10^{-14}	10^{-4}	1

Table A.2 Volume (L^3)

cubic metre (m^3)	litre (l or L) or cubic decimetre* (dm^3)	cubic foot (ft^3)	cubic inch (in.^3)	imperial gallon (UKgal)	US gallon (USgal)
1	10^3	3.531×10^1	6.102×10^4	2.200×10^2	2.642×10^2
10^{-3}	1	3.531×10^{-2}	6.102×10^1	2.200×10^{-1}	2.642×10^{-1}
2.832×10^{-2}	2.832×10^1	1	1.728×10^3	6.229	7.481
1.639×10^{-5}	1.639×10^{-2}	5.787×10^{-4}	1	3.605×10^{-3}	4.329×10^{-3}
4.546×10^{-3}	4.546	1.605×10^{-1}	2.774×10^2	1	1.201
3.785×10^{-3}	3.785	1.337×10^{-1}	2.31×10^2	8.327×10^{-1}	1

* The litre was defined in 1964 as 1 dm^3 exactly.

Table A.3 Mass (M)

kilogram (kg)	gram (g)	pound (lb)	ounce (oz)	tonne (t)	imperial ton (UK ton)	atomic mass unit* (u)
1	10^3	2.205	3.527×10^1	10^{-3}	9.842×10^{-4}	6.022×10^{26}
10^{-3}	1	2.205×10^{-3}	3.527×10^{-2}	10^{-6}	9.842×10^{-7}	6.022×10^{23}
4.536×10^{-1}	4.536×10^2	1	1.6×10^1	4.536×10^{-4}	4.464×10^{-4}	2.732×10^{26}
2.835×10^{-2}	2.835×10^1	6.25×10^{-2}	1	2.835×10^{-5}	2.790×10^{-5}	1.707×10^{25}
10^3	10^6	2.205×10^3	3.527×10^4	1	9.842×10^{-1}	6.022×10^{29}
1.016×10^3	1.016×10^6	2.240×10^3	3.584×10^4	1.016	1	6.119×10^{29}
1.661×10^{-27}	1.661×10^{-24}	3.661×10^{-27}	5.857×10^{-26}	1.661×10^{-30}	1.634×10^{-30}	1

* Atomic mass unit (unified); $1 \text{ u} = 1/12$ of the rest mass of a neutral atom of the nuclide ^{12}C in the ground state.

Table A.4 Force (LMT^{-2})

newton (N, kg m s^{-2})	kilogram-force (kg_f)	pound-force (lb_f)	dyne (dyn, g cm s^{-2})	poundal (pdl, lb ft s^{-2})
1	1.020×10^{-1}	2.248×10^{-1}	10^5	7.233
9.807	1	2.205	9.807×10^5	7.093×10^1
4.448	4.536×10^{-1}	1	4.448×10^5	3.217×10^1
10^{-5}	1.020×10^{-6}	2.248×10^{-6}	1	7.233×10^{-5}
1.383×10^{-1}	1.410×10^{-2}	3.108×10^{-2}	1.383×10^4	1

Table A.5 Pressure and stress ($\text{L}^{-1}\text{MT}^{-2}$)

pascal (Pa, N m^{-2} J m^{-3} $\text{kg m}^{-1} \text{s}^{-2}$)	pound force per inch ² (psi, $\text{lb}_f \text{in.}^{-2}$)	kilogram force per metre ² ($\text{kg}_f \text{m}^{-2}$)	standard atmosphere (atm)	dyne per cm^2 (dyn cm^{-2})	torr (Torr, mmHg)*	inches of water** (inH ₂ O)	bar
1	1.450×10^{-4}	1.020×10^{-1}	9.869×10^{-6}	10^1	7.501×10^{-3}	4.015×10^{-3}	10^{-5}
6.895×10^3	1	7.031×10^2	6.805×10^{-2}	6.895×10^4	5.171×10^1	2.768×10^1	6.895×10^{-2}
9.807	1.422×10^{-3}	1	9.678×10^{-5}	9.807×10^1	7.356×10^{-2}	3.937×10^{-2}	9.807×10^{-5}
1.013×10^5	1.470×10^1	1.033×10^4	1	1.013×10^6	7.6×10^2	4.068×10^2	1.013
10^{-1}	1.450×10^{-5}	1.020×10^{-2}	9.869×10^{-7}	1	7.501×10^{-4}	4.015×10^{-4}	10^{-6}
1.333×10^2	1.934×10^{-2}	1.360×10^1	1.316×10^{-3}	1.333×10^3	1	5.352×10^{-1}	1.333×10^{-3}
2.491×10^2	3.613×10^{-2}	2.540×10^1	2.458×10^{-3}	2.491×10^3	1.868	1	2.491×10^{-3}
10^5	1.450×10^1	1.020×10^4	9.869×10^{-1}	10^6	7.501×10^2	4.015×10^2	1

* mmHg refers to Hg at 0°C; 1 Torr = 1.00000 mmHg.

** in.H₂O refers to water at 4°C.**Table A.6 Surface tension (MT^{-2})**

newton per metre (N m^{-1} , kg s^{-2} , J m^{-2})	dyne per centimetre (dyn cm^{-1} , g s^{-2} , erg cm^{-2})	kilogram-force per metre ($\text{kg}_f \text{m}^{-1}$)
1	10^3	1.020×10^{-1}
10^{-3}	1	1.020×10^{-4}
9.807	9.807×10^3	1

Table A.7 Energy, work and heat (L^2MT^{-2})

joule (J, N m, Pa m^3 , W s, $\text{kg m}^2 \text{s}^{-2}$)	kilocalorie* (kcal)	British thermal unit (Btu)	foot pound- force (ft lb_f)	litre atmosphere (l atm)	kilowatt hour (kW h)	erg (dyn cm)
1	2.388×10^{-4}	9.478×10^{-4}	7.376×10^{-1}	9.869×10^{-3}	2.778×10^{-7}	10^7
4.187×10^3	1	3.968	3.088×10^3	4.132×10^1	1.163×10^{-3}	4.187×10^{10}
1.055×10^3	2.520×10^{-1}	1	7.782×10^2	1.041×10^1	2.931×10^{-4}	1.055×10^{10}
1.356	3.238×10^{-4}	1.285×10^{-3}	1	1.338×10^{-2}	3.766×10^{-7}	1.356×10^7
1.013×10^2	2.420×10^{-2}	9.604×10^{-2}	7.473×10^1	1	2.815×10^{-5}	1.013×10^9
3.6×10^6	8.598×10^2	3.412×10^3	2.655×10^6	3.553×10^4	1	3.6×10^{13}
10^{-7}	2.388×10^{-11}	9.478×10^{-11}	7.376×10^{-8}	9.869×10^{-10}	2.778×10^{-14}	1

* International Table kilocalorie (kcal_{IT})

Table A.8 Power (L^2MT^{-3})

watt (W, $J s^{-1}$, $kg m^2 s^{-3}$)	kilocalorie per min (kcal min^{-1})	foot pound- force per second (ft $lb_f s^{-1}$)	horsepower (British) (hp)	metric horsepower (-)	British thermal unit per minute (Bru min^{-1})	kilogram-force metre per second ($kg_f m s^{-1}$)
1	1.433×10^{-2}	7.376×10^{-1}	1.341×10^{-3}	1.360×10^{-3}	5.687×10^{-2}	1.020×10^{-1}
6.978×10^1	1	5.147×10^1	9.358×10^{-2}	9.487×10^{-2}	3.968	7.116
1.356	1.943×10^{-2}	1	1.818×10^{-3}	1.843×10^{-3}	7.710×10^{-2}	1.383×10^{-1}
7.457×10^2	1.069×10^1	5.5×10^2	1	1.014	4.241×10^1	7.604×10^1
7.355×10^2	1.054×10^1	5.425×10^2	9.863×10^{-1}	1	4.183×10^1	7.5×10^1
1.758×10^1	2.520×10^{-1}	1.297×10^1	2.358×10^{-2}	2.391×10^{-2}	1	1.793
9.807	1.405×10^{-1}	7.233	1.315×10^{-2}	1.333×10^{-2}	5.577×10^{-1}	1

Table A.9 Dynamic viscosity ($L^{-1}MT^{-1}$)

pascal second (Pa s, $N s m^{-2}$, $kg m^{-1} s^{-1}$)	poise ($g cm^{-1} s^{-1}$, $dyn s cm^{-2}$)	centipoise (cP)	$kg m^{-1} h^{-1}$	$lb ft^{-1} h^{-1}$
1	10^1	10^3	3.6×10^3	2.419×10^3
10^{-1}	1	10^2	3.6×10^2	2.419×10^2
10^{-3}	10^{-2}	1	3.6	2.419
2.778×10^{-4}	2.778×10^{-3}	2.778×10^{-1}	1	6.720×10^{-1}
4.134×10^{-4}	4.134×10^{-3}	4.134×10^{-1}	1.488	1

Table A.10 Plane angle (1)

radian (rad)	revolution (rev)	degree (°)	minute (')	second (")
1	1.592×10^{-1}	5.730×10^1	3.438×10^3	2.063×10^5
6.283	1	3.6×10^2	2.160×10^4	1.296×10^6
1.745×10^{-2}	2.778×10^{-3}	1	6×10^1	3.6×10^3
2.909×10^{-4}	4.630×10^{-5}	1.667×10^{-2}	1	6×10^1
4.848×10^{-6}	7.716×10^{-7}	2.778×10^{-4}	1.667×10^{-2}	1

Table A.11 Illuminance ($L^{-2}J$)

lux or lumen per metre ² (lx or $lm m^{-2}$)	foot-candle (fc, $lm ft^{-2}$)
1	9.290×10^{-2}
1.076×10^1	1

B

Physical and Chemical Property Data

Table B.1 Atomic weights and numbers

Based on the atomic mass of ^{12}C . Values for atomic weights apply to elements as they exist in nature.

Name	Symbol	Relative atomic mass	Atomic number
Actinium	Ac	—	89
Aluminium	Al	26.9815	13
Americium	Am	—	95
Antimony	Sb	121.75	51
Argon	Ar	39.948	18
Arsenic	As	74.9216	33
Astatine	At	—	85
Barium	Ba	137.34	56
Berkelium	Bk	—	97
Beryllium	Be	9.0122	4
Bismuth	Bi	208.98	83
Boron	B	10.811	5
Bromine	Br	79.904	35
Cadmium	Cd	112.40	48
Caesium	Cs	132.905	55
Calcium	Ca	40.08	20
Californium	Cf	—	98
Carbon	C	12.011	6
Cerium	Ce	140.12	58
Chlorine	Cl	35.453	17
Chromium	Cr	51.996	24
Cobalt	Co	58.9332	27
Copper	Cu	63.546	29
Curium	Cm	—	96
Dysprosium	Dy	162.50	66
Einsteinium	Es	—	99
Erbium	Er	167.26	68
Europium	Eu	151.96	63
Fermium	Fm	—	100
Fluorine	F	18.9984	9
Francium	Fr	—	87
Gadolinium	Gd	157.25	64
Gallium	Ga	69.72	31
Germanium	Ge	72.59	32
Gold	Au	196.967	79
Hafnium	Hf	178.49	72
Helium	He	4.0026	2
Holmium	Ho	164.930	67
Hydrogen	H	1.00797	1
Indium	In	114.82	49
Iodine	I	126.9044	53
Iridium	Ir	192.2	77
Iron	Fe	55.847	26
Krypton	Kr	83.80	36
Lanthanum	La	138.91	57

Name	Symbol	Relative atomic mass	Atomic number
Lawrencium	Lr	—	103
Lead	Pb	207.19	82
Lithium	Li	6.939	3
Lutetium	Lu	174.97	71
Magnesium	Mg	24.312	12
Manganese	Mn	54.938	25
Mendelevium	Md	—	101
Mercury	Hg	200.59	80
Molybdenum	Mo	95.94	42
Neodymium	Nd	144.24	60
Neon	Ne	20.183	10
Neptunium	Np	—	93
Nickel	Ni	58.71	28
Niobium	Nb	92.906	41
Nitrogen	N	14.0067	7
Nobelium	No	—	102
Osmium	Os	190.2	76
Oxygen	O	15.9994	8
Palladium	Pd	106.4	46
Phosphorus	P	30.9738	15
Platinum	Pt	195.09	78
Plutonium	Pu	—	94
Polonium	Po	—	84
Potassium	K	39.102	19
Praseodymium	Pr	140.907	59
Promethium	Pm	—	61
Protactinium	Pa	—	91
Radium	Ra	—	88
Radon	Rn	—	86
Rhenium	Re	186.2	75
Rhodium	Rh	102.905	45
Rubidium	Rb	85.47	37
Ruthenium	Ru	101.07	44
Samarium	Sm	150.35	62
Scandium	Sc	44.956	21
Selenium	Se	78.96	34
Silicon	Si	28.086	14
Silver	Ag	107.868	47
Sodium	Na	22.9898	11
Strontium	Sr	87.62	38
Sulphur	S	32.064	16
Tantalum	Ta	180.948	73
Technetium	Tc	—	43
Tellurium	Te	127.60	52
Terbium	Tb	158.924	65
Thallium	Tl	204.37	81
Thorium	Th	232.038	90
Thulium	Tm	168.934	69
Tin	Sn	118.69	50
Titanium	Ti	47.90	22
Tungsten	W	183.85	74
Uranium	U	238.03	92
Vanadium	V	50.942	23
Wolfram (Tungsten)	W	183.85	74
Xenon	Xe	131.30	54
Ytterbium	Yb	173.04	70
Yttrium	Y	88.905	39
Zinc	Zn	65.37	30
Zirconium	Zr	91.22	40

Table B.2 Degree of reduction of biological materials(Adapted from J.A. Roels, 1983, *Energetics and Kinetics in Biotechnology*, Elsevier Biomedical Press, Amsterdam)

Compound	Formula	Degree of reduction γ relative to NH_3	Degree of reduction γ relative to N_2
Acetaldehyde	$\text{C}_2\text{H}_4\text{O}$	5.00	5.00
Acetic acid	$\text{C}_2\text{H}_4\text{O}_2$	4.00	4.00
Acetone	$\text{C}_3\text{H}_6\text{O}$	5.33	5.33
Adenine	$\text{C}_5\text{H}_5\text{N}_5$	2.00	5.00
Alanine	$\text{C}_3\text{H}_7\text{O}_2\text{N}$	4.00	5.00
Ammonia	NH_3	0	3.00
Arginine	$\text{C}_6\text{H}_{14}\text{O}_2\text{N}_4$	3.67	5.67
Asparagine	$\text{C}_4\text{H}_8\text{O}_3\text{N}_2$	3.00	4.50
Aspartic acid	$\text{C}_4\text{H}_7\text{O}_4\text{N}$	3.00	3.75
<i>n</i> -Butanol	$\text{C}_4\text{H}_{10}\text{O}$	6.00	6.00
Butyraldehyde	$\text{C}_4\text{H}_8\text{O}$	5.50	5.50
Butyric acid	$\text{C}_4\text{H}_8\text{O}_2$	5.00	5.00
Carbon monoxide	CO	2.00	2.00
Citric acid	$\text{C}_6\text{H}_8\text{O}_7$	3.00	3.00
Cytosine	$\text{C}_4\text{H}_5\text{ON}_3$	2.50	4.75
Ethane	C_2H_6	7.00	7.00
Ethanol	$\text{C}_2\text{H}_6\text{O}$	6.00	6.00
Ethene	C_2H_4	6.00	6.00
Ethylene glycol	$\text{C}_2\text{H}_6\text{O}_2$	5.00	5.00
Ethyne	C_2H_2	5.00	5.00
Formaldehyde	CH_2O	4.00	4.00
Formic acid	CH_2O_2	2.00	2.00
Fumaric acid	$\text{C}_4\text{H}_4\text{O}_4$	3.00	3.00
Glucitol	$\text{C}_6\text{H}_{14}\text{O}_6$	4.33	4.33
Gluconic acid	$\text{C}_6\text{H}_{12}\text{O}_7$	3.67	3.67
Glucose	$\text{C}_6\text{H}_{12}\text{O}_6$	4.00	4.00
Glutamic acid	$\text{C}_5\text{H}_9\text{O}_4\text{N}$	3.60	4.20
Glutamine	$\text{C}_5\text{H}_{10}\text{O}_3\text{N}_2$	3.60	4.80
Glycerol	$\text{C}_3\text{H}_8\text{O}_3$	4.67	4.67
Glycine	$\text{C}_2\text{H}_5\text{O}_2\text{N}$	3.00	4.50
Graphite	C	4.00	4.00
Guanine	$\text{C}_5\text{H}_5\text{ON}_5$	1.60	4.60
Histidine	$\text{C}_6\text{H}_9\text{O}_2\text{N}_3$	3.33	4.83
Hydrogen	H_2	2.00	2.00
Isoleucine	$\text{C}_6\text{H}_{13}\text{O}_2\text{N}$	5.00	5.50
Lactic acid	$\text{C}_3\text{H}_6\text{O}_3$	4.00	4.00
Leucine	$\text{C}_6\text{H}_{13}\text{O}_2\text{N}$	5.00	5.50
Lysine	$\text{C}_6\text{H}_{14}\text{O}_2\text{N}_2$	4.67	5.67
Malic acid	$\text{C}_4\text{H}_6\text{O}_5$	3.00	3.00
Methane	CH_4	8.00	8.00
Methanol	CH_4O	6.00	6.00
Oxalic acid	$\text{C}_2\text{H}_2\text{O}_4$	1.00	1.00
Palmitic acid	$\text{C}_{16}\text{H}_{32}\text{O}_2$	5.75	5.75
Pentane	C_5H_{12}	6.40	6.40
Phenylalanine	$\text{C}_9\text{H}_{11}\text{O}_2\text{N}$	4.44	4.78
Proline	$\text{C}_5\text{H}_9\text{O}_2\text{N}$	4.40	5.00
Propane	C_3H_8	6.67	6.67
<i>iso</i> -Propanol	$\text{C}_3\text{H}_8\text{O}$	6.00	6.00
Propionic acid	$\text{C}_3\text{H}_6\text{O}_2$	4.67	4.67
Pyruvic acid	$\text{C}_3\text{H}_4\text{O}_3$	3.33	3.33
Serine	$\text{C}_3\text{H}_7\text{O}_3\text{N}$	3.33	4.33
Succinic acid	$\text{C}_4\text{H}_6\text{O}_4$	3.50	3.50

Compound	Formula	Degree of reduction γ relative to NH_3	Degree of reduction γ relative to N_2
Threonine	$\text{C}_4\text{H}_9\text{O}_3\text{N}$	4.00	4.75
Thymine	$\text{C}_5\text{H}_6\text{O}_2\text{N}_2$	3.20	4.40
Tryptophan	$\text{C}_{11}\text{H}_{12}\text{O}_2\text{N}_2$	4.18	4.73
Tyrosine	$\text{C}_9\text{H}_{11}\text{O}_3\text{N}$	4.22	4.56
Uracil	$\text{C}_4\text{H}_4\text{O}_2\text{N}_2$	2.50	4.00
Valeric acid	$\text{C}_5\text{H}_{10}\text{O}_2$	5.20	5.20
Valine	$\text{C}_5\text{H}_{11}\text{O}_2\text{N}$	4.80	5.40
Biomass	$\text{CH}_{1.8}\text{O}_{0.5}\text{N}_{0.2}$	4.20	4.80

Table B.3 Heat capacities

(Adapted from R.M. Felder and R.W. Rousseau, 1978, *Elementary Principles of Chemical Processes*, John Wiley and Sons, New York)

$$C_p (\text{J gmol}^{-1} \text{ } ^\circ\text{C}^{-1}) = a + bT + cT^2 + dT^3$$

Example. For acetone gas between 0°C and 1200°C :

$$C_p (\text{J gmol}^{-1} \text{ } ^\circ\text{C}^{-1}) = 71.96 + (20.10 \times 10^{-2}) T - (12.78 \times 10^{-5}) T^2 + (34.76 \times 10^{-9}) T^3, \text{ where } T \text{ is in } ^\circ\text{C}.$$

Note that some equations require T in K, as indicated.

State: g = gas; l = liquid; c = crystal.

Compound	State	Temperature unit	a	$b \cdot 10^2$	$c \cdot 10^5$	$d \cdot 10^9$	Temperature range (units of T)
Acetone	g	$^\circ\text{C}$	71.96	20.10	-12.78	34.76	0–1200
Air	g	$^\circ\text{C}$	28.94	0.4147	0.3191	-1.965	0–1500
	g	K	28.09	0.1965	0.4799	-1.965	273–1800
Ammonia	g	$^\circ\text{C}$	35.15	2.954	0.4421	-6.686	0–1200
Calcium hydroxide	c	K	89.5				276–373
Carbon dioxide	g	$^\circ\text{C}$	36.11	4.233	-2.887	7.464	0–1500
Ethanol	l	$^\circ\text{C}$	103.1				0
	l	$^\circ\text{C}$	158.8				100
	g	$^\circ\text{C}$	61.34	15.72	-8.749	19.83	0–1200
Formaldehyde	g	$^\circ\text{C}$	34.28	4.268	0.000	-8.694	0–1200
Hydrogen	g	$^\circ\text{C}$	28.84	0.00765	0.3288	-0.8698	0–1500
Hydrogen chloride	g	$^\circ\text{C}$	29.13	-0.1341	0.9715	-4.335	0–1200
Hydrogen sulphide	g	$^\circ\text{C}$	33.51	1.547	0.3012	-3.292	0–1500
Methane	g	$^\circ\text{C}$	34.31	5.469	0.3661	-11.00	0–1200
	g	K	19.87	5.021	1.268	-11.00	273–1500
Methanol	l	$^\circ\text{C}$	75.86				0
	l	$^\circ\text{C}$	82.59				40
	g	$^\circ\text{C}$	42.93	8.301	-1.87	-8.03	0–700
Nitric acid	l	$^\circ\text{C}$	110.0				25
Nitrogen	g	$^\circ\text{C}$	29.00	0.2199	0.5723	-2.871	0–1500
Oxygen	g	$^\circ\text{C}$	29.10	1.158	-0.6076	1.311	0–1500
Sulphur (rhombic)	c	K	15.2	2.68			273–368
(monoclinic)	c	K	18.3	1.84			368–392
Sulphuric acid	l	$^\circ\text{C}$	139.1	15.59			10–45
Sulphur dioxide	g	$^\circ\text{C}$	38.91	3.904	-3.104	8.606	0–1500
Water	l	$^\circ\text{C}$	75.4				0–100
	g	$^\circ\text{C}$	33.46	0.6880	0.7604	-3.593	0–1500

Table B.4 Mean heat capacities of gases(Adapted from D.M. Himmelblau, 1974, *Basic Principles and Calculations in Chemical Engineering*, 3rd edn, Prentice-Hall, New Jersey)Reference state: $T_{\text{ref}} = 0^\circ\text{C}$; $P_{\text{ref}} = 1 \text{ atm}$.

$T(^{\circ}\text{C})$	$C_{p,m} (\text{J g mol}^{-1} ^{\circ}\text{C}^{-1})$					
	Air	O ₂	N ₂	H ₂	CO ₂	H ₂ O
0	29.06	29.24	29.12	28.61	35.96	33.48
18	29.07	29.28	29.12	28.69	36.43	33.51
25	29.07	29.30	29.12	28.72	36.47	33.52
100	29.14	29.53	29.14	28.98	38.17	33.73
200	29.29	29.93	29.23	29.10	40.12	34.10
300	29.51	30.44	29.38	29.15	41.85	34.54
400	29.78	30.88	29.60	29.22	43.35	35.05
500	30.08	31.33	29.87	29.28	44.69	35.59

Table B.5 Specific heats of organic liquids(From R.H. Perry, D.W. Green and J.O. Maloney, Eds, 1984, *Chemical Engineers' Handbook*, 6th edn, McGraw-Hill, New York)

Compound	Formula	Temperature ($^{\circ}\text{C}$)	$C_p (\text{cal g}^{-1} ^{\circ}\text{C}^{-1})$
Acetic acid	C ₂ H ₄ O ₂	26 to 95	0.522
Acetone	C ₃ H ₆ O	3 to 22.6	0.514
		0	0.506
		24.2 to 49.4	0.538
		21 to 76	0.541
Acetonitrile	C ₂ H ₃ N	21 to 76	0.541
Benzaldehyde	C ₇ H ₆ O	22 to 172	0.428
Butyl alcohol (<i>n</i> -)	C ₄ H ₁₀ O	2.3	0.526
		19.2	0.563
		21 to 115	0.687
		30	0.582
		0	0.444
Butyric acid (<i>n</i> -)	C ₄ H ₈ O ₂	40	0.501
		20 to 100	0.515
		0	0.198
Carbon tetrachloride	CCl ₄	20	0.201
		30	0.200
		0	0.232
Chloroform	CHCl ₃	15	0.226
		30	0.234
		0	0.232
Cresol			
(<i>o</i> -)	C ₇ H ₈ O	0 to 20	0.497
(<i>m</i> -)		21 to 197	0.551
Dichloroacetic acid	C ₂ H ₂ Cl ₂ O ₂	0 to 20	0.477
		21 to 106	0.349
		21 to 196	0.348
Diethylamine	C ₄ H ₁₁ N	22.5	0.516
Diethyl malonate	C ₇ H ₁₂ O ₄	20	0.431
Diethyl oxalate	C ₆ H ₁₀ O ₄	20	0.431
Diethyl succinate	C ₈ H ₁₄ O ₄	20	0.450
Dipropyl malonate	C ₉ H ₁₆ O ₄	20	0.431
Dipropyl oxalate (<i>n</i> -)	C ₈ H ₁₄ O ₄	20	0.431
Dipropyl succinate	C ₁₀ H ₁₈ O ₄	20	0.450

Compound	Formula	Temperature (°C)	C_p (cal g ⁻¹ °C ⁻¹)
Ethanol	C ₂ H ₆ O	0 to 98	0.680
Ether	C ₄ H ₁₀ O	-5	0.525
		0	0.521
		30	0.545
		80	0.687
		120	0.800
		140	0.819
		180	1.037
Ethyl acetate	C ₄ H ₈ O ₂	20	0.457
		20	0.476
Ethylene glycol	C ₂ H ₆ O ₂	-11.1	0.535
		0	0.542
		2.5	0.550
		5.1	0.554
		14.9	0.569
		19.9	0.573
Formic acid	CH ₂ O ₂	0	0.436
		15.5	0.509
		20 to 100	0.524
Furfural	C ₅ H ₄ O ₂	0	0.367
		20 to 100	0.416
Glycerol	C ₃ H ₈ O ₃	15 to 50	0.576
Hexadecane (<i>n</i> -)	C ₁₆ H ₃₄	0 to 50	0.496
Isobutyl acetate	C ₆ H ₁₂ O ₂	20	0.459
Isobutyl alcohol	C ₄ H ₁₀ O	21 to 109	0.716
		30	0.603
Isobutyl succinate	C ₁₂ H ₂₂ O ₄	0	0.442
Isobutyric acid	C ₄ H ₈ O ₂	20	0.450
Lauric acid	C ₁₂ H ₂₄ O ₂	40 to 100	0.572
		57	0.515
Methanol	CH ₄ O	5 to 10	0.590
		15 to 20	0.601
Methyl butyl ketone	C ₆ H ₁₂ O	21 to 127	0.553
Methyl ethyl ketone	C ₄ H ₈ O	20 to 78	0.549
Methyl formate	C ₂ H ₄ O ₂	13 to 29	0.516
Methyl propionate	C ₄ H ₈ O ₂	20	0.459
Palmitic acid	C ₁₆ H ₃₂ O ₂	65 to 104	0.653
Propionic acid	C ₃ H ₆ O ₂	0	0.444
		20 to 137	0.560
Propyl acetate (<i>n</i> -)	C ₅ H ₁₀ O ₂	20	0.459
Propyl butyrate	C ₇ H ₁₄ O ₂	20	0.459
Propyl formate (<i>n</i> -)	C ₄ H ₈ O ₂	20	0.459
Pyridine	C ₅ H ₅ N	20	0.405
		21 to 108	0.431
		0 to 20	0.395
Quinoline	C ₉ H ₇ N	0 to 20	0.352
Salicylaldehyde	C ₇ H ₆ O ₂	18	0.382
Stearic acid	C ₁₈ H ₃₆ O ₂	75 to 137	0.550

Table B.6 Specific heats of organic solids(From R.H. Perry, D.W. Green and J.O. Maloney, Eds, 1984, *Chemical Engineers' Handbook*, 6th edn, McGraw-Hill, New York)

Compound	Formula	Temperature (°C) T	C_p (cal g ⁻¹ °C ⁻¹)
Acetic acid	C ₂ H ₄ O ₂	-200 to 25	0.330 + 0.00080 T
Acetone	C ₃ H ₆ O	-210 to -80	0.540 + 0.0156 T
Aniline	C ₆ H ₇ N		0.741
Capric acid	C ₁₀ H ₂₀ O ₂	8	0.695
Chloroacetic acid	C ₂ H ₃ ClO ₂	60	0.363
Crotonic acid	C ₄ H ₆ O ₂	38 to 70	0.520 + 0.00020 T
Dextrin	(C ₆ H ₁₀ O ₅) _x	0 to 90	0.291 + 0.00096 T
Diphenylamine	C ₁₂ H ₁₁ N	26	0.337
Erythritol	C ₄ H ₁₀ O ₄	60	0.351
Ethylene glycol	C ₂ H ₆ O ₂	-190 to -40	0.366 + 0.00110 T
Formic acid	CH ₂ O ₂	-22	0.387
		0	0.430
Glucose	C ₆ H ₁₂ O ₆	0	0.277
		20	0.300
Glutaric acid	C ₅ H ₈ O ₄	20	0.299
Glycerol	C ₃ H ₈ O ₃	0	0.330
Hexadecane	C ₁₆ H ₃₄		0.495
Lactose	C ₁₂ H ₂₂ O ₁₁	20	0.287
	C ₁₂ H ₂₂ O ₁₁ ·H ₂ O	20	0.299
Lauric acid	C ₁₂ H ₂₄ O ₂	-30 to 40	0.430 + 0.000027 T
Levoglucothane	C ₆ H ₁₀ O ₅	40	0.607
Levulose	C ₆ H ₁₂ O ₆	20	0.275
Malonic acid	C ₃ H ₄ O ₄	20	0.275
Maltose	C ₁₂ H ₂₂ O ₁₁	20	0.320
Mannitol	C ₆ H ₁₄ O ₆	0 to 100	0.313 + 0.00025 T
Oxalic acid	C ₂ H ₂ O ₄	-200 to 50	0.259 + 0.00076 T
	C ₂ H ₂ O ₄ ·2H ₂ O	0	0.338
		50	0.385
		100	0.416
Palmitic acid	C ₁₆ H ₃₂ O ₂	0	0.382
		20	0.430
Phenol	C ₆ H ₆ O	14 to 26	0.561
Succinic acid	C ₄ H ₆ O ₄	0 to 160	0.248 + 0.00153 T
Sucrose	C ₁₂ H ₂₂ O ₁₁	20	0.299
Sugar (cane)	C ₁₂ H ₂₂ O ₁₁	22 to 51	0.301
Tartaric acid	C ₄ H ₆ O ₆	36	0.287
	C ₄ H ₆ O ₆ ·H ₂ O	0	0.308
		50	0.366
Urea	CH ₄ N ₂ O	20	0.320

Table B.7 Normal melting points and boiling points, and standard heats of phase change(From R.M. Felder and R.W. Rousseau, 1978, *Elementary Principles of Chemical Processes*, John Wiley, New York).

All thermodynamic data are at 1 atm.

Compound	Molecular weight	Melting temperature (°C)	Δh_f at melting point (kJ gmol ⁻¹)	Normal boiling point (°C)	Δh_v at boiling point (kJ gmol ⁻¹)
Acetaldehyde	44.05	-123.7		20.2	25.1
Acetic acid	60.05	16.6	12.09	118.2	24.39
Acetone	58.08	-95.0	5.69	56.0	30.2
Ammonia	17.03	-77.8	5.653	-33.43	23.351
Benzaldehyde	106.12	-26.0		179.0	38.40
Carbon dioxide	44.01	-56.6	8.33	(sublimates at -78°C)	
Chloroform	119.39	-63.7		61.0	
Ethanol	46.07	-114.6	5.021	78.5	38.58
Formaldehyde	30.03	-92		-19.3	24.48
Formic acid	46.03	8.30	12.68	100.5	22.25
Glycerol	92.09	18.20	18.30	290.0	
Hydrogen	2.016	-259.19	0.12	-252.76	0.904
Hydrogen chloride	36.47	-114.2	1.99	-85.0	16.1
Hydrogen sulphide	34.08	-85.5	2.38	-60.3	18.67
Methane	16.04	-182.5	0.94	-161.5	8.179
Methanol	32.04	-97.9	3.167	64.7	35.27
Nitric acid	63.02	-41.6	10.47	86	30.30
Nitrogen	28.02	-210.0	0.720	-195.8	5.577
Oxalic acid	90.04			(decomposes at 186°C)	
Oxygen	32.00	-218.75	0.444	-182.97	6.82
Phenol	94.11	42.5	11.43	181.4	
Phosphoric acid	98.00	42.3	10.54		
Sodium chloride	58.45	808	28.5	1465	170.7
Sodium hydroxide	40.00	319	8.34	1390	
Sulphur					
(rhombic)	256.53	113	10.04	444.6	83.7
(monoclinic)	256.53	119	14.17	444.6	83.7
Sulphur dioxide	64.07	-75.48	7.402	-10.02	24.91
Sulphuric acid	98.08	10.35	9.87	(decomposes at 340°C)	
Water	18.016	0.00	6.0095	100.00	40.656

Table B.8 Heats of combustion(From *Handbook of Chemistry and Physics*, 1992, 73rd edn, CRC Press, Boca Raton; *Handbook of Chemistry and Physics*, 1976, 57th edn, CRC Press, Boca Raton; and R.M. Felder and R.W. Rousseau, 1978, *Elementary Principles of Chemical Processes*, John Wiley, New York)

Reference conditions: 1 atm and 25°C or 20°C; values marked with an asterisk refer to 20°C.

Products of combustion are taken to be CO₂ (gas), H₂O (liquid) and N₂ (gas); therefore, $\Delta h_c^\circ = 0$ for CO₂ (g), H₂O (l) and N₂ (g).

State: g = gas; l = liquid; c = crystal; s = solid.

Compound	Formula	Molecular weight	State	Heat of combustion Δh_c° (kJ gmol ⁻¹)
Acetaldehyde	C ₂ H ₄ O	44.053	l	-1166.9
			g	-1192.5
Acetic acid	C ₂ H ₄ O ₂	60.053	l	-874.2
			g	-925.9
Acetone	C ₃ H ₆ O	58.080	l	-1789.9
			g	-1820.7

Compound	Formula	Molecular weight	State	Heat of combustion Δh_c° (kJ gmol ⁻¹)
Acetylene	C ₂ H ₂	26.038	g	-1301.1
Adenine	C ₅ H ₅ N ₅	135.128	c	-2778.1
			g	-2886.9
Alanine (D-)	C ₃ H ₇ O ₂ N	89.094	c	-1619.7
Alanine (L-)	C ₃ H ₇ O ₂ N	89.094	c	-1576.9
			g	-1715.0
Ammonia	NH ₃	17.03	g	-382.6
Ammonium ion	NH ₄ ⁺		g	-383
Arginine (D-)	C ₆ H ₁₄ O ₂ N ₄	174.203	c	-3738.4
Asparagine (L-)	C ₄ H ₈ O ₃ N ₂	132.119	c	-1928.0
Aspartic acid (L-)	C ₄ H ₇ O ₄ N	133.104	c	-1601.1
Benzaldehyde	C ₇ H ₆ O	106.124	l	-3525.1
			g	-3575.4
Butanoic acid	C ₄ H ₈ O ₂	88.106	l	-2183.6
			g	-2241.6
1-Butanol	C ₄ H ₁₀ O	74.123	l	-2675.9
			g	-2728.2
2-Butanol	C ₄ H ₁₀ O	74.123	l	-2660.6
			g	-2710.3
Butyric acid	C ₄ H ₈ O ₂	88.106	l	-2183.6
			g	-2241.6
Caffeine	C ₈ H ₁₀ O ₂ N ₄		s	-4246.5*
Carbon	C	12.011	c	-393.5
Carbon monoxide	CO	28.010	g	-283.0
Citric acid	C ₆ H ₈ O ₇		s	-1962.0
Codeine	C ₁₈ H ₂₁ O ₃ N.H ₂ O		s	-9745.7*
Cytosine	C ₄ H ₅ ON ₃	111.103	c	-2067.3
Ethane	C ₂ H ₆	30.070	g	-1560.7
Ethanol	C ₂ H ₆ O	46.069	l	-1366.8
			g	-1409.4
Ethylene	C ₂ H ₄	28.054	g	-1411.2
Ethylene glycol	C ₂ H ₆ O ₂	62.068	l	-1189.2
			g	-1257.0
Formaldehyde	CH ₂ O	30.026	g	-570.7
Formic acid	CH ₂ O ₂	46.026	l	-254.6
			g	-300.7
Fructose (D-)	C ₆ H ₁₂ O ₆		s	-2813.7
Fumaric acid	C ₄ H ₄ O ₄	116.073	c	-1334.0
Galactose (D-)	C ₆ H ₁₂ O ₆		s	-2805.7
Glucose (D-)	C ₆ H ₁₂ O ₆		s	-2805.0
Glutamic acid (L-)	C ₅ H ₉ O ₄ N	147.131	c	-2244.1
Glutamine (L-)	C ₅ H ₁₀ O ₃ N ₂	146.146	c	-2570.3
Glutaric acid	C ₅ H ₈ O ₄	132.116	c	-2150.9
Glycerol	C ₃ H ₈ O ₃	92.095	l	-1655.4
			g	-1741.2
Glycine	C ₂ H ₅ O ₂ N	75.067	c	-973.1
Glycogen	(C ₆ H ₁₀ O ₅) _x per kg		s	-17530.1*
Guanine	C ₅ H ₅ ON ₅	151.128	c	-2498.2
Hexadecane	C ₁₆ H ₃₄	226.446	l	-10699.2
			g	-10780.5
Hexadecanoic acid	C ₁₆ H ₃₂ O ₂	256.429	c	-9977.9
			l	-10031.3
			g	-10132.3
Histidine (L-)	C ₆ H ₉ O ₂ N ₃	155.157	c	-3180.6
Hydrogen	H ₂	2.016	g	-285.8
Hydrogen sulphide	H ₂ S	34.08		-562.6
Inositol	C ₆ H ₁₂ O ₆		s	-2772.2*
Isoleucine (L-)	C ₆ H ₁₃ O ₂ N	131.175	c	-3581.1
Isoquinoline	C ₉ H ₇ N	129.161	l	-4686.5

Compound	Formula	Molecular weight	State	Heat of combustion Δh_c° (kJ g ⁻¹)
Lactic acid (D,L-)	C ₃ H ₆ O ₃		l	-1368.3
Lactose	C ₁₂ H ₂₂ O ₁₁		s	-5652.5
Leucine (D-)	C ₆ H ₁₃ O ₂ N	131.175	c	-3581.7
Leucine (L-)	C ₆ H ₁₃ O ₂ N	131.175	c	-3581.6
Lysine	C ₆ H ₁₄ O ₂ N ₂	146.189	c	-3683.2
Malic acid (L-)	C ₄ H ₆ O ₅		s	-1328.8
Malonic acid	C ₃ H ₄ O ₄		s	-861.8
Maltose	C ₁₂ H ₂₂ O ₁₁		s	-5649.5
Mannitol (D-)	C ₆ H ₁₄ O ₆		s	-3046.5*
Methane	CH ₄	16.043	g	-890.8
Methanol	CH ₄ O	32.042	l	-726.1
			g	-763.7
Morphine	C ₁₇ H ₁₉ O ₃ N.H ₂ O		s	-8986.6*
Nicotine	C ₁₀ H ₁₄ N ₂		l	-5977.8*
Oleic acid	C ₁₈ H ₃₄ O ₂		l	-11126.5
Oxalic acid	C ₂ H ₂ O ₄	90.036	c	-251.1
Papaverine	C ₂₀ H ₂₁ O ₄ N		s	-10375.8*
Pentane	C ₅ H ₁₂	72.150	l	-3509.0
			g	-3535.6
Phenylalanine (L-)	C ₉ H ₁₁ O ₂ N	165.192	c	-4646.8
Phthalic acid	C ₈ H ₆ O ₄	166.133	c	-3223.6
Proline (L-)	C ₅ H ₉ O ₂ N	115.132	c	-2741.6
Propane	C ₃ H ₈	44.097	g	-2219.2
1-Propanol	C ₃ H ₈ O	60.096	l	-2021.3
			g	-2068.8
2-Propanol	C ₃ H ₈ O	60.096	l	-2005.8
			g	-2051.1
Propionic acid	C ₃ H ₆ O ₂	74.079	l	-1527.3
			g	-1584.5
1,2-Propylene glycol	C ₃ H ₈ O ₂	76.095	l	-1838.2
			g	-1902.6
1,3-Propylene glycol	C ₃ H ₈ O ₂	76.095	l	-1859.0
			g	-1931.8
Pyridine	C ₅ H ₅ N	79.101	l	-2782.3
			g	-2822.5
Pyrimidine	C ₄ H ₄ N ₂	80.089	l	-2291.6
			g	-2341.6
Salicylic acid	C ₇ H ₆ O ₃	138.123	c	-3022.2
			g	-3117.3
Serine (L-)	C ₃ H ₇ O ₃ N	105.094	c	-1448.2
Starch	(C ₆ H ₁₀ O ₅) _x per kg		s	-17496.6*
Succinic acid	C ₄ H ₆ O ₄	118.089	c	-1491.0
Sucrose	C ₁₂ H ₂₂ O ₁₁		s	-5644.9
Thebaine	C ₁₉ H ₂₁ O ₃ N		s	-10221.7*
Threonine (L-)	C ₄ H ₉ O ₃ N	119.120	c	-2053.1
Thymine	C ₅ H ₆ O ₂ N ₂	126.115	c	-2362.2
Tryptophan (L-)	C ₁₁ H ₁₂ O ₂ N ₂	204.229	c	-5628.3
Tyrosine (L-)	C ₉ H ₁₁ O ₃ N	181.191	c	-4428.6
Uracil	C ₄ H ₄ O ₂ N ₂	112.088	c	-1716.3
			g	-1842.8
Urea	CH ₄ ON ₂	60.056	c	-631.6
			g	-719.4
Valine (L-)	C ₅ H ₁₁ O ₂ N	117.148	c	-2921.7
			g	-3084.5
Xanthine	C ₅ H ₄ O ₂ N ₄	152.113	c	-2159.6
Xylose	C ₅ H ₁₀ O ₅		s	-2340.5
Biomass	CH _{1.8} O _{0.5} N _{0.2}	25.9	s	-552

C

Steam Tables

(From R.W. Haywood, *Thermodynamic Tables in SI (Metric) Units*, 1972, 2nd edn, Cambridge University Press, Cambridge)

Table C.1 Enthalpy of saturated water and steam (Temperatures from 0.01°C to 100°C)

Reference state: Triple point of water: 0.01°C, 0.6112 kPa.

Temperature (°C)	Pressure (kPa)	Specific enthalpy (kJ kg ⁻¹)		
		Saturated liquid	Evaporation (Δh_v)	Saturated vapour
0.01 (Triple point)	0.611	+0.0	2501.6	2501.6
2	0.705	8.4	2496.8	2505.2
4	0.813	16.8	2492.1	2508.9
6	0.935	25.2	2487.4	2512.6
8	1.072	33.6	2482.6	2516.2
10	1.227	42.0	2477.9	2519.9
12	1.401	50.4	2473.2	2523.6
14	1.597	58.8	2468.5	2527.2
16	1.817	67.1	2463.8	2530.9
18	2.062	75.5	2459.0	2534.5
20	2.34	83.9	2454.3	2538.2
22	2.64	83.9	2454.3	2538.2
24	2.98	100.6	2444.9	2545.5
25	3.17	104.8	2442.5	2547.3
26	3.36	108.9	2440.2	2549.1
28	3.78	117.3	2435.4	2552.7
30	4.24	125.7	2430.7	2556.4
32	4.75	134.0	2425.9	2560.0
34	5.32	142.4	2421.2	2563.6
36	5.94	150.7	2416.4	2567.2
38	6.62	159.1	2411.7	2570.8
40	7.38	167.5	2406.9	2574.4
42	8.20	175.8	2402.1	2577.9
44	9.10	184.2	2397.3	2581.5
46	10.09	192.5	2392.5	2585.1
48	11.16	200.9	2387.7	2588.6
50	12.34	209.3	2382.9	2592.2
52	13.61	217.6	2378.1	2595.7
54	15.00	226.0	2373.2	2599.2
56	16.51	234.4	2368.4	2602.7
58	18.15	242.7	2363.5	2606.2
60	19.92	251.1	2358.6	2609.7
62	21.84	259.5	2353.7	2613.2
64	23.91	267.8	2348.8	2616.6
66	26.15	276.2	2343.9	2620.1
68	28.56	284.6	2338.9	2623.5
70	31.16	293.0	2334.0	2626.9

Temperature (°C)	Pressure (kPa)	Specific enthalpy (kJ kg ⁻¹)		
		Saturated liquid	Evaporation (Δh_v)	Saturated vapour
72	33.96	301.4	2329.0	2630.3
74	36.96	309.7	2324.0	2633.7
76	40.19	318.1	2318.9	2637.1
78	43.65	326.5	2313.9	2640.4
80	47.36	334.9	2308.8	2643.8
82	51.33	343.3	2303.8	2647.1
84	55.57	351.7	2298.6	2650.4
86	60.11	360.1	2293.5	2653.6
88	64.95	368.5	2288.4	2656.9
90	70.11	376.9	2283.2	2660.1
92	75.61	385.4	2278.0	2663.4
94	81.46	393.8	2272.8	2666.6
96	87.69	402.2	2267.5	2669.7
98	94.30	410.6	2262.2	2672.9
100 (Boiling point)	101.325	419.1	2256.9	2676.0

Table C.2 Enthalpy of saturated water and steam

(Pressures from 0.6112 kPa to 22 120 kPa)

Reference state: Triple point of water: 0.01°C, 0.6112 kPa.

Pressure (kPa)	Temperature (°C)	Specific enthalpy (kJ kg ⁻¹)		
		Saturated liquid	Evaporation (Δh_v)	Saturated vapour
0.6112 (Triple point)	0.01	+0.0	2501.6	2501.6
0.8	3.8	15.8	2492.6	2508.5
1.0	7.0	29.3	2485.0	2514.4
1.4	12.0	50.3	2473.2	2523.5
1.8	15.9	66.5	2464.1	2530.6
2.0	17.5	73.5	2460.2	2533.6
2.4	20.4	85.7	2453.3	2539.0
2.8	23.0	96.2	2447.3	2543.6
3.0	24.1	101.0	2444.6	2545.6
3.5	26.7	111.8	2438.5	2550.4
4.0	29.0	121.4	2433.1	2554.5
4.5	31.0	130.0	2428.2	2558.2
5.0	32.9	137.8	2423.8	2561.6
6	36.2	151.5	2416.0	2567.5
7	39.0	163.4	2409.2	2572.6
8	41.5	173.9	2403.2	2577.1
9	43.8	183.3	2397.9	2581.1
10	45.8	191.8	2392.9	2584.8
12	49.4	206.9	2384.3	2591.2
14	52.6	220.0	2376.7	2596.7
16	55.3	231.6	2370.0	2601.6
18	57.8	242.0	2363.9	2605.9
20	60.1	251.5	2358.4	2609.9

Pressure (kPa)	Temperature (°C)	Specific enthalpy (kJ kg ⁻¹)		
		Saturated liquid	Evaporation (Δh_v)	Saturated vapour
24	64.1	268.2	2348.6	2616.8
28	67.5	282.7	2340.0	2622.7
30	69.1	289.3	2336.1	2625.4
35	72.7	304.3	2327.2	2631.5
40	75.9	317.7	2319.2	2636.9
45	78.7	329.6	2312.0	2641.7
50	81.3	340.6	2305.4	2646.0
55	83.7	350.6	2299.3	2649.9
60	86.0	359.9	2293.6	2653.6
65	88.0	368.6	2288.3	2656.9
70	90.0	376.8	2283.3	2660.1
80	93.5	391.7	2274.1	2665.8
90	96.7	405.2	2265.6	2670.9
100	99.6	417.5	2257.9	2675.4
101.325 (Boiling point)	100.0	419.1	2256.9	2676.0
120	104.8	439.4	2244.1	2683.4
140	109.3	458.4	2231.9	2690.3
160	113.3	475.4	2220.9	2696.2
180	116.9	490.7	2210.8	2701.5
200	120.2	504.7	2201.6	2706.3
220	123.3	517.6	2193.0	2710.6
240	126.1	529.6	2184.9	2714.5
260	128.7	540.9	2177.3	2718.2
280	131.2	551.4	2170.1	2721.5
300	133.5	561.4	2163.2	2724.7
320	135.8	570.9	2156.7	2727.6
340	137.9	579.9	2150.4	2730.3
360	139.9	588.5	2144.4	2732.9
380	141.8	596.8	2138.6	2735.3
400	143.6	604.7	2133.0	2737.6
420	145.4	612.3	2127.5	2739.8
440	147.1	619.6	2122.3	2741.9
460	148.7	626.7	2117.2	2743.9
480	150.3	633.5	2112.2	2745.7
500	151.8	640.1	2107.4	2747.5
550	155.5	655.8	2095.9	2751.7
600	158.8	670.4	2085.0	2755.5
650	162.0	684.1	2074.7	2758.9
700	165.0	697.1	2064.9	2762.0
750	167.8	709.3	2055.5	2764.8
800	170.4	720.9	2046.5	2767.5
850	172.9	732.0	2037.9	2769.9
900	175.4	742.6	2029.5	2772.1
950	177.7	752.8	2021.4	2774.2
1000	179.9	762.6	2013.6	2776.2
1100	184.1	781.1	1998.5	2779.7
1200	188.0	798.4	1984.3	2782.7
1300	191.6	814.7	1970.7	2785.4
1400	195.0	830.1	1957.7	2787.8
1500	198.3	844.7	1945.2	2789.9
1600	201.4	858.6	1933.2	2791.7
1700	204.3	871.8	1921.5	2793.4
1800	207.1	884.6	1910.3	2794.8

Pressure (kPa)	Temperature (°C)	Specific enthalpy (kJ kg ⁻¹)		
		Saturated liquid	Evaporation (Δh_v)	Saturated vapour
1900	209.8	896.8	1899.3	2796.1
2000	212.4	908.6	1888.6	2797.2
2200	217.2	931.0	1868.1	2799.1
2400	221.8	951.9	1848.5	2800.4
2600	226.0	971.7	1829.6	2801.4
2800	230.0	990.5	1811.5	2802.0
3000	233.8	1008.4	1793.9	2802.3
3200	237.4	1025.4	1776.9	2802.3
3400	240.9	1041.8	1760.3	2802.1
3600	244.2	1057.6	1744.2	2801.7
3800	247.3	1072.7	1728.4	2801.1
4000	250.3	1087.4	1712.9	2800.3
4200	253.2	1101.6	1697.8	2799.4
4400	256.0	1115.4	1682.9	2798.3
4600	258.8	1128.8	1668.3	2797.1
4800	261.4	1141.8	1653.9	2795.7
5000	263.9	1154.5	1639.7	2794.2
5200	266.4	1166.8	1625.7	2792.6
5400	268.8	1178.9	1611.9	2790.8
5600	271.1	1190.8	1598.2	2789.0
5800	273.3	1202.3	1584.7	2787.0
6000	275.6	1213.7	1571.3	2785.0
6200	277.7	1224.8	1558.0	2782.9
6400	279.8	1235.7	1544.9	2780.6
6600	281.8	1246.5	1531.9	2778.3
6800	283.8	1257.0	1518.9	2775.9
7000	285.8	1267.4	1506.0	2773.5
7200	287.7	1277.6	1493.3	2770.9
7400	289.6	1287.7	1480.5	2768.3
7600	291.4	1297.6	1467.9	2765.5
7800	293.2	1307.4	1455.3	2762.8
8000	295.0	1317.1	1442.8	2759.9
8400	298.4	1336.1	1417.9	2754.0
8800	301.7	1354.6	1393.2	2747.8
9000	303.3	1363.7	1380.9	2744.6
10000	311.0	1408.0	1319.7	2727.7
11000	318.0	1450.6	1258.7	2709.3
12000	324.6	1491.8	1197.4	2689.2
13000	330.8	1532.0	1135.0	2667.0
14000	336.6	1571.6	1070.7	2642.4
15000	342.1	1611.0	1004.0	2615.0
16000	347.3	1650.5	934.3	2584.9
17000	352.3	1691.7	859.9	2551.6
18000	357.0	1734.8	779.1	2513.9
19000	361.4	1778.7	692.0	2470.6
20000	365.7	1826.5	591.9	2418.4
21000	369.8	1886.3	461.3	2347.6
22000	373.7	2011	185	2196
22120	374.15	2108	0	2108
(Critical point)				

Table C.3 Enthalpy of superheated steam

Reference state: Triple point of water: 0.01°C, 0.6112 kPa.

Pressure (kPa)	10	50	100	500	1000	2000	4000	6000	8000	10000	15000	20000	22120*	30000	50000
Saturation temperature (°C)	45.8	81.3	99.6	151.8	179.9	212.4	250.3	275.6	295.0	311.0	342.1	365.7	374.15	–	–
State	Specific enthalpy at saturation (kJ kg ⁻¹)														
Water	191.8	340.6	417.5	640.1	762.6	908.6	1087.4	1213.7	1317.1	1408.0	1611.0	1826.5	2108	–	–
Steam	2584.8	2646.0	2675.4	2747.5	2776.2	2797.2	2800.3	2785.0	2759.9	2727.7	2615.0	2418.4	2108	–	–
Temperature (°C)	Specific enthalpy (kJ kg ⁻¹)														
0	0.0	0.0	0.1	0.5	1.0	2.0	4.0	6.1	8.1	10.1	15.1	20.1	22.2	30.0	49.3
25	104.8	104.8	104.9	105.2	105.7	106.6	108.5	110.3	112.1	114.0	118.6	123.1	125.1	132.2	150.2
50	2593	209.3	209.3	209.7	210.1	211.0	212.7	214.4	216.1	217.8	222.1	226.4	228.2	235.0	251.9
75	2640	313.9	314.0	314.3	314.7	315.5	317.1	318.7	320.3	322.0	326.0	330.0	331.7	338.1	354.2
100	2688	2683	2676	419.4	419.7	420.5	422.0	423.5	425.0	426.5	430.3	434.0	435.7	441.6	456.8
125	2735	2731	2726	525.2	525.5	526.2	527.6	529.0	530.4	531.8	535.3	538.8	540.2	545.8	560.1
150	2783	2780	2776	632.2	632.5	633.1	634.3	635.6	636.8	638.1	641.3	644.5	645.8	650.9	664.1
175	2831	2829	2826	2800	741.1	741.7	742.7	743.8	744.9	746.0	748.7	751.5	752.7	757.2	769.1
200	2880	2878	2875	2855	2827	852.6	853.4	854.2	855.1	855.9	858.1	860.4	861.4	865.2	875.4
225	2928	2927	2925	2909	2886	2834	967.2	967.7	968.2	968.8	970.3	971.8	972.5	975.3	983.4
250	2977	2976	2975	2961	2943	2902	1085.8	1085.8	1085.8	1085.8	1086.2	1086.7	1087.0	1088.4	1093.6
275	3027	3026	3024	3013	2998	2965	2886	1210.8	1210.0	1209.2	1207.7	1206.6	1206.3	1205.6	1206.7
300	3077	3076	3074	3065	3052	3025	2962	2885	2787	1343.4	1338.3	1334.3	1332.8	1328.7	1323.7
325	3127	3126	3125	3116	3106	3083	3031	2970	2899	2811	1486.0	1475.5	1471.8	1461.1	1446.0
350	3177	3177	3176	3168	3159	3139	3095	3046	2990	2926	2695	1647.1	1636.5	1609.9	1576.3
375	3228	3228	3227	3220	3211	3194	3156	3115	3069	3019	2862	2604	2319	1791	1716
400	3280	3279	3278	3272	3264	3249	3216	3180	3142	3100	2979	2820	2733	2162	1878
425	3331	3331	3330	3325	3317	3303	3274	3243	3209	3174	3075	2957	2899	2619	2068
450	3384	3383	3382	3377	3371	3358	3331	3303	3274	3244	3160	3064	3020	2826	2293
475	3436	3436	3435	3430	3424	3412	3388	3363	3337	3310	3237	3157	3120	2969	2522
500	3489	3489	3488	3484	3478	3467	3445	3422	3399	3375	3311	3241	3210	3085	2723
600	3706	3705	3705	3702	3697	3689	3673	3656	3640	3623	3580	3536	3516	3443	3248
700	3929	3929	3928	3926	3923	3916	3904	3892	3879	3867	3835	3804	3790	3740	3610
800	4159	4159	4158	4156	4154	4149	4140	4131	4121	4112	4089	4065	4055	4018	3925

* Critical isobar.

D

Mathematical Rules

In this Appendix, some simple rules for logarithms, differentiation and integration are presented. Further details of mathematical functions can be found in handbooks, e.g. [1–4].

D.1 Logarithms

The *natural logarithm* ($\ln$ or $\log_e$) is the inverse of the exponential function. Therefore, if:

$$y = \ln x \quad (\text{D.1})$$

then

$$e^y = x \quad (\text{D.2})$$

where the number e is approximately 2.71828. It also follows that:

$$\ln(e^y) = y \quad (\text{D.3})$$

and

$$e^{\ln x} = x. \quad (\text{D.4})$$

Natural logarithms are related to *common logarithms*, or logarithms to the base 10 (written as $\lg$, $\log$ or $\log_{10}$), as follows:

$$\ln x = \ln 10 (\log_{10} x). \quad (\text{D.5})$$

Since $\ln 10$ is approximately 2.30259:

$$\ln x = 2.30259 \log_{10} x. \quad (\text{D.6})$$

Zero and negative numbers do not have logarithms.

Rules for taking logarithms of products and powers are illustrated below. The logarithm of the product of two numbers is equal to the sum of the logarithms:

$$\ln(ax) = \ln a + \ln x. \quad (\text{D.7})$$

When one term of the product involves an exponential function, application of Eqs (D.7) and (D.3) gives:

$$\ln(be^{ax}) = \ln b + ax. \quad (\text{D.8})$$

The logarithm of the quotient of two numbers is equal to the logarithm of the numerator minus the logarithm of the denominator:

$$\ln\left(\frac{a}{x}\right) = \ln a - \ln x. \quad (\text{D.9})$$

As an example of this rule, because $\ln 1 = 0$:

$$\ln\left(\frac{1}{x}\right) = -\ln x. \quad (\text{D.10})$$

The rule for taking the logarithm of a power function is as follows:

$$\ln(x^b) = b \ln x. \quad (\text{D.11})$$

D.2 Differentiation

The derivative of y with respect to x , dy/dx , is defined as the limit of $\Delta y/\Delta x$ as Δx approaches zero, provided this limit exists:

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}. \quad (\text{D.12})$$

That is:

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{y|_{x+\Delta x} - y|_x}{\Delta x} \quad (\text{D.13})$$

where $y|_x$ means the value of y evaluated at x , and $y|_{x+\Delta x}$ means the value of y evaluated at $x + \Delta x$. The operation of calculating the derivative is called differentiation.

There are simple rules for rapid evaluation of derivatives. Derivatives of various functions with respect to x are listed below; in all these equations A is a constant:

$$\frac{dA}{dx} = 0 \quad (\text{D.14})$$

$$\frac{dx}{dx} = 1 \quad (\text{D.15})$$

$$\frac{d}{dx} (e^x) = e^x \quad (\text{D.16})$$

$$\frac{d}{dx} (e^{Ax}) = Ae^{Ax} \quad (\text{D.17})$$

and

$$\frac{d}{dx} (\ln x) = \frac{1}{x}. \quad (\text{D.18})$$

When a function is multiplied by a constant, the constant can be taken out of the differential. For example:

$$\frac{d}{dx} (Ax) = A \frac{dx}{dx} = A \quad (\text{D.19})$$

and

$$\frac{d}{dx} (A \ln x) = A \frac{d}{dx} (\ln x) = \frac{A}{x}. \quad (\text{D.20})$$

When a function consists of a sum of terms, the derivative of the sum is equal to the sum of the derivatives. Therefore, if $f(x)$ and $g(x)$ are functions of x :

$$\frac{d}{dx} [f(x) + g(x)] = \frac{df}{dx} + \frac{dg}{dx}. \quad (\text{D.21})$$

To illustrate application of Eq. (D.21), for A and B constants:

$$\frac{d}{dx} [Ax + e^{Bx}] = \frac{d(Ax)}{dx} + \frac{d(e^{Bx})}{dx} = A + Be^{Bx}.$$

When a function consists of terms multiplied together, the *product rule* for derivatives is:

$$\frac{d}{dx} [f(x) \cdot g(x)] = f(x) \frac{dg}{dx} + g(x) \frac{df}{dx}. \quad (\text{D.22})$$

As an example of the product rule:

$$\begin{aligned} \frac{d}{dx} [(Ax) \cdot \ln x] &= Ax \cdot \frac{d(\ln x)}{dx} + \ln x \cdot \frac{d(Ax)}{dx} \\ &= Ax \cdot \frac{1}{x} + \ln x \cdot (A) \\ &= A(1 + \ln x). \end{aligned}$$

D.3 Integration

The integral of y with respect to x is indicated as $\int y \, dx$. The function to be integrated (y) is called the *integrand*; the symbol $\int$ is the *integral sign*. Integration is the opposite of differentiation; integration is the process of finding a function from its derivative.

From Eq. (D.14), if the derivative of a constant is zero, the integral of zero must be a constant:

$$\int 0 \, dx = K \quad (\text{D.23})$$

where K is a constant. K is called the *constant of integration*, and appears whenever a function is integrated. For example, the integral of constant A with respect to x is:

$$\int A \, dx = Ax + K \quad (\text{D.24})$$

We can check that Eq. (D.24) is correct by taking the derivative of the right-hand side and making sure it is equal to the integrand, A . Although the equation:

$$\int A \, dx = Ax$$

is also correct, addition of K in Eq. (D.24) makes solution of the integral complete. Addition of K accounts for the possibility that the answer we are looking for may have an added constant that disappears when the derivative is taken. Irrespective of the value of K , the derivative of the integral will always be the same because $dK/dx = 0$. Extra information is needed to evaluate the actual magnitude of K ; this point is considered further in Chapter 6 where integration is used to solve unsteady-state mass and energy problems.

The integral of dy/dx with respect to x is:

$$\int \frac{dy}{dx} dx = y + K. \quad (\text{D.25})$$

When a function is multiplied by a constant, the constant can be taken out of the integral. For example, for $f(x)$ a function of x , and A a constant:

$$\int A f(x) dx = A \int f(x) dx. \quad (\text{D.26})$$

Other rules of integration are:

$$\int \frac{dx}{x} = \int \frac{1}{x} dx = \ln x + K \quad (\text{D.27})$$

and, for A and B constants:

$$\int \left(\frac{dx}{A + Bx} \right) = \int \left(\frac{1}{A + Bx} \right) dx = \frac{1}{B} \ln(A + Bx) + K. \quad (\text{D.28})$$

The results of Eqs (D.27) and (D.28) can be confirmed by differentiating the right-hand sides of the equations with respect to x .

References

1. *CRC Standard Mathematical Tables*, CRC Press, Florida.
2. Cornish-Bowden, A. (1981) *Basic Mathematics for Biochemists*, Chapman and Hall, London.
3. Newby, J.C. (1980) *Mathematics for the Biological Sciences*, Oxford University Press, Oxford.
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E

List of Symbols

Symbol	Definition	Dimensions	SI units
Roman symbols			
a	Area	L^2	m^2
a	Area per unit volume	L^{-1}	m^{-1}
a	Mass of dry gel	M	kg
A	Amplitude	L	m
A	Arrhenius constant	T^{-1}	s^{-1}
A	Area	L^2	m^2
A_c	Cross-sectional area	L^2	m^2
A_i	Inner surface area	L^2	m^2
A_o	Outer surface area	L^2	m^2
b	Length of bowl	L	m
b	Thickness	L	m
B	Thickness	L	m
c	Mass of cake solids deposited per volume of filtrate	$L^{-3}M$	$kg\ m^{-3}$
C	Geometry parameter in Eq. (8.44)	1	—
C	Concentration	$L^{-3}M$	$kg\ m^{-3}$
C_A	Concentration of component A	$L^{-3}M$	$kg\ m^{-3}$
C_{Ab}	Concentration of component A in the bulk fluid	$L^{-3}M$	$kg\ m^{-3}$
C_{AG}	Concentration of component A in gas	$L^{-3}M$	$kg\ m^{-3}$
C_{Al}	Concentration of component A in the lower phase	$L^{-3}M$	$kg\ m^{-3}$
C_{AL}	Concentration of component A in liquid	$L^{-3}M$	$kg\ m^{-3}$
$\bar{C}_{AL}$	Steady-state concentration of component A in liquid	$L^{-3}M$	$kg\ m^{-3}$
C_{AL0}^*	Solubility of component A in liquid at zero solute concentration, Eq. (9.45)	$L^{-3}M$	$kg\ m^{-3}$
$C_{A,min}$	Minimum concentration of component A	$L^{-3}M$	$kg\ m^{-3}$
C_{Ao}	Concentration of component A in the bulk fluid	$L^{-3}M$	$kg\ m^{-3}$
C_{As}	Surface concentration of component A	$L^{-3}M$	$kg\ m^{-3}$
C_{AS}	Average concentration of component A in the solid phase	$L^{-3}M$	$kg\ m^{-3}$
C_{ASm}	Maximum concentration of component A in the solid phase	$L^{-3}M$	$kg\ m^{-3}$
C_{Au}	Concentration of component A in the upper phase	$L^{-3}M$	$kg\ m^{-3}$
C_{crit}	Critical concentration	$L^{-3}M$	$kg\ m^{-3}$
C_e	Concentration at reaction equilibrium	$L^{-3}M$	$kg\ m^{-3}$

C_{iL}	Concentration of ionic component i in liquid	$L^{-3}M$	$kg\ m^{-3}$
C_{jL}	Concentration of non-ionic component j in liquid	$L^{-3}M$	$kg\ m^{-3}$
C_p	Specific heat capacity	$L^2T^{-2}\Theta^{-1}$	$J\ kg^{-1}\ K^{-1}$
C_{pc}	Specific heat capacity of cold fluid	$L^2T^{-2}\Theta^{-1}$	$J\ kg^{-1}\ K^{-1}$
C_{ph}	Specific heat capacity of hot fluid	$L^2T^{-2}\Theta^{-1}$	$J\ kg^{-1}\ K^{-1}$
C_{pm}	Mean heat capacity	$L^2T^{-2}\Theta^{-1}$	$J\ kg^{-1}\ K^{-1}$
C_{pw}	Specific heat capacity of cooling water	$L^2T^{-2}\Theta^{-1}$	$J\ kg^{-1}\ K^{-1}$
D	Dilution rate	T^{-1}	s^{-1}
D	Diameter	L	m
D_b	Bubble diameter	L	m
D_{crit}	Critical dilution rate for washout	T^{-1}	s^{-1}
D_i	Impeller diameter	L	m
D_o	Orifice diameter	L	m
D_{opt}	Dilution rate for optimum biomass productivity	T^{-1}	s^{-1}
D_p	Particle diameter	L	m
Da	Damköhler number defined in Eq. (13.102)	1	—
e	Base of natural logarithms	1	—
e	Enzyme concentration	$L^{-3}M$	$kg\ m^{-3}$
e_a	Concentration of active enzyme	$L^{-3}M$	$kg\ m^{-3}$
e_k	Kinetic energy per unit mass	L^2T^{-2}	$J\ kg^{-1}$
e_p	Potential energy per unit mass	L^2T^{-2}	$J\ kg^{-1}$
E	Quantity of enzyme (Section 11.1.3)		
E	Molar activation energy	$L^2MT^{-2}N^{-1}$	$J\ mol^{-1}$
E	Energy	L^2MT^{-2}	J
E_d	Molar activation energy for deactivation reaction	$L^2MT^{-2}N^{-1}$	$J\ mol^{-1}$
E_k	Kinetic energy	L^2MT^{-2}	J
E_p	Potential energy	L^2MT^{-2}	J
f	Function		
F	Shear force	LMT^{-2}	N
F	Fraction of cells carrying plasmid	1	—
F	Volumetric flow rate	L^3T^{-1}	$m^3\ s^{-1}$
F	Volumetric flow rate of gas	L^3T^{-1}	$m^3\ s^{-1}$
F_L^g	Volumetric flow rate of liquid	L^3T^{-1}	$m^3\ s^{-1}$
F_r	Volumetric flow rate of the recycle stream	L^3T^{-1}	$m^3\ s^{-1}$
g	Gravitational acceleration	LT^{-2}	$m\ s^{-2}$
g_c	Force unity bracket	1	—
G	Specific gravity	1	—
G	Free energy of formation	L^2MT^{-2}	J
G°	Standard free energy of formation	L^2MT^{-2}	J
ΔG_{rxn}°	Change in molar free energy for reaction under standard conditions	$L^2MT^{-2}N^{-1}$	$J\ mol^{-1}$
Gr	Grashof number defined by Eq. (8.41) or (12.51)	1	—
h	Height	L	m
h	Specific enthalpy	L^2T^{-2}	$J\ kg^{-1}$
h	Heat-transfer coefficient	$MT^{-3}\Theta^{-1}$	$W\ m^{-2}\ K^{-1}$
h_c	Heat-transfer coefficient for cold fluid	$MT^{-3}\Theta^{-1}$	$W\ m^{-2}\ K^{-1}$
Δh_c	Molar heat of combustion	$L^2MT^{-2}N^{-1}$	$J\ mol^{-1}$
Δh_c°	Molar heat of combustion under standard conditions	$L^2MT^{-2}N^{-1}$	$J\ mol^{-1}$

Δh_f	Specific latent heat of fusion	L^2T^{-2}	$J\ kg^{-1}$
h_{fc}	Fouling factor for cold fluid	$MT^{-3}\Theta^{-1}$	$W\ m^{-2}\ K^{-1}$
h_{fh}	Fouling factor for hot fluid	$MT^{-3}\Theta^{-1}$	$W\ m^{-2}\ K^{-1}$
h_h	Heat-transfer coefficient for hot fluid	$MT^{-3}\Theta^{-1}$	$W\ m^{-2}\ K^{-1}$
Δh_m	Molar integral heat of mixing	$L^2MT^{-2}N^{-1}$	$J\ mol^{-1}$
Δh_{rxn}	Specific heat of reaction	L^2T^{-2}	$J\ kg^{-1}$
Δh_s	Specific latent heat of sublimation	L^2T^{-2}	$J\ kg^{-1}$
Δh_v	Specific latent heat of vaporisation	L^2T^{-2}	$J\ kg^{-1}$
H	Henry's constant	L^2T^{-2}	$m^2\ s^{-2}$
H	Height	L	m
H	Enthalpy	L^2MT^{-2}	J
H_A	Enthalpy of component A	L^2MT^{-2}	J
H_B	Enthalpy of component B	L^2MT^{-2}	J
H_i	Parameter in Eq. (9.45)	L^3N^{-1}	$m^3\ mol^{-1}$
ΔH_m	Heat of mixing	L^2MT^{-2}	J
H_{ref}	Enthalpy of the reference state	L^2MT^{-2}	J
ΔH_{rxn}	Heat of reaction	L^2MT^{-2}	J
ΔH_{rxn}^o	Heat of reaction under standard conditions	L^2MT^{-2}	J
$\Delta \hat{H}_{rxn}$	Rate of heat absorption or liberation by reaction	L^2MT^{-3}	W
J_A	Mass flux of component A	$L^{-2}MT^{-1}$	$kg\ m^{-2}\ s^{-1}$
k	Thermal conductivity	$LMT^{-3}\Theta^{-1}$	$W\ m^{-1}\ K^{-1}$
k	Geometry parameter in Eq. (7.11)	1	—
k	Mass-transfer coefficient	LT^{-1}	$m\ s^{-1}$
k	Capacity factor defined in Eq. (10.37)	1	—
k	Rate constant		
k_0	Zero-order rate constant	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
k'_0	Specific zero-order rate constant for enzyme reaction	T^{-1}	s^{-1}
k''_0	Specific zero-order rate constant for cell reaction	T^{-1}	s^{-1}
k_1	First-order rate constant	T^{-1}	s^{-1}
k_1	Proportionality constant in Eq. (7.19)	1	—
k_{-1}	Reverse-reaction rate constant		
k_2	Turnover number	T^{-1}	s^{-1}
k_d	First-order deactivation rate constant	T^{-1}	s^{-1}
k_{fb}	Thermal conductivity of bulk fluid	$LMT^{-3}\Theta^{-1}$	$W\ m^{-1}\ K^{-1}$
k_G	Gas-phase mass-transfer coefficient	LT^{-1}	$m\ s^{-1}$
k_L	Liquid-phase mass-transfer coefficient	LT^{-1}	$m\ s^{-1}$
k_S	Liquid-phase mass-transfer coefficient for transfer to or from a solid	LT^{-1}	$m\ s^{-1}$
K	Constant of integration		
K	Consistency index for power-law fluids	$L^{-1}MT^{n-2}$	$Pa\ s^n$
K	Reaction equilibrium constant		
K	Partition coefficient	1	—
K_1	Parameter defined in Eq. (10.13)	$L^{-6}T$	$m^{-6}\ s$
K_2	Parameter defined in Eq. (10.14)	$L^{-3}T$	$m^{-3}\ s$
K_A	Constant in Eq. (10.30)	L^3M^{-1}	$m^3\ kg^{-1}$
K_F	Parameter in Eq. (10.31)		
K_G	Overall gas-phase mass-transfer coefficient	LT^{-1}	$m\ s^{-1}$
K_j	Parameter in Eq. (9.45)	L^3N^{-1}	$m^3\ mol^{-1}$
K_L	Overall liquid-phase mass-transfer coefficient	LT^{-1}	$m\ s^{-1}$
K_m	Michaelis constant	$L^{-3}M$	$kg\ m^{-3}$
K_p	Constant in Eqs (7.9) and (7.10)		
K_p	Partition coefficient	1	—
K_S	Substrate constant	$L^{-3}M$	$kg\ m^{-3}$

K_v	Shape factor	1	—
L	Length	L	m
m	Distribution or partition coefficient	1	—
m_p	Specific rate of product formation due to maintenance activity	T^{-1}	s^{-1}
m_s	Maintenance coefficient	T^{-1}	s^{-1}
M	Torque	L^2MT^{-2}	N m
$\hat{M}$	Mass	M	kg
$\hat{M}$	Mass flow rate	MT^{-1}	$kg\ s^{-1}$
M_A	Mass of component A	M	kg
$\hat{M}_c$	Mass of cake solids	M	kg
$\hat{M}_c$	Mass flow rate of cold fluid	MT^{-1}	$kg\ s^{-1}$
$\hat{M}_h$	Mass flow rate of hot fluid	MT^{-1}	$kg\ s^{-1}$
$\hat{M}_m$	Initial mass of medium	M	kg
$\hat{M}_s$	Mass flow rate of steam	MT^{-1}	$kg\ s^{-1}$
$\hat{M}_v$	Mass of liquid evaporated	M	kg
$\hat{M}_v$	Mass flow rate of evaporated liquid	MT^{-1}	$kg\ s^{-1}$
$\hat{M}_w$	Mass flow rate of cooling water	MT^{-1}	$kg\ s^{-1}$
n	Mole	N	mol
n	Flow behaviour index for power-law fluids	1	—
n	Number of impellers	1	—
n	Number of generations	1	—
n	Adsorption parameter in Eq. (10.31)	1	—
N	Number of viable cells	1	—
N	Number of discs	1	—
N	Number of passes	1	—
N	Number of theoretical plates	1	—
N_1	Number of viable cells at the beginning of the holding period	1	—
N_2	Number of viable cells at the end of the holding period	1	—
N_A	Rate of mass transfer of component A	MT^{-1}	$kg\ s^{-1}$
N_A	Volumetric rate of mass transfer of component A	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
N_i	Rotational speed	T^{-1}	s^{-1}
N_i^*	Minimum stirrer speed for suspension of solids	T^{-1}	s^{-1}
N_{iB}	Minimum stirrer speed for complete dispersion of gas	T^{-1}	s^{-1}
N_{iR}	Minimum stirrer speed for gross recirculation of gas	T^{-1}	s^{-1}
N_p	Power number defined by Eq. (7.17)	1	—
N'_p	Constant value of the power number in the turbulent regime	1	—
Nu	Nusselt number defined in Eq. (8.37)	1	—
p	Probability of plasmid loss	1	—
p	Pressure	$L^{-1}MT^{-2}$	Pa
p	Product concentration	$L^{-3}M$	$kg\ m^{-3}$
p_1	Product concentration in the first reactor	$L^{-3}M$	$kg\ m^{-3}$
p_2	Product concentration in the second reactor	$L^{-3}M$	$kg\ m^{-3}$
p_{AG}	Partial pressure of component A in gas	$L^{-1}MT^{-2}$	Pa
p_T	Total pressure	$L^{-1}MT^{-2}$	Pa
P	Power	L^2MT^{-3}	W
P	Mass of product	M	kg
P_0	Power consumption without sparging	L^2MT^{-3}	W
P_g	Power consumption with sparging	L^2MT^{-3}	W
Pe	Peclet number defined in Eq. (13.101)	1	—
Pr	Prandtl number defined in Eq. (8.40)	1	—

q	Heat evolved per mole of available electrons transferred to oxygen	$L^2MT^{-2}N^{-1}$	$J\ mol^{-1}$
$\hat{q}$	Heat flux	MT^{-3}	$W\ m^{-2}$
q_O	Specific oxygen-uptake rate	T^{-1}	s^{-1}
q_P	Specific rate of product formation	T^{-1}	s^{-1}
q_S	Specific rate of substrate consumption	T^{-1}	s^{-1}
Q	Volumetric flow rate	L^3T^{-1}	$m^3\ s^{-1}$
Q	Heat	L^2MT^{-2}	J
$\hat{Q}$	Rate of heat flow	L^2MT^{-3}	W
$\hat{Q}_c$	Rate of heat transfer to cold fluid	L^2MT^{-3}	W
$\hat{Q}_h$	Rate of heat transfer from hot fluid	L^2MT^{-3}	W
Q_O	Volumetric rate of oxygen uptake	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
Q_P	Volumetric rate of product formation	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
Q_X	Volumetric rate of biomass production	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
$Q_{X,max}$	Maximum volumetric rate of biomass production	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
r	Radius	L	m
r	Volumetric rate of reaction	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
r_1	Inner radius of centrifuge disc	L	m
r_1	Radius of the liquid surface in a tubular centrifuge	L	m
r_2	Outer radius of centrifuge disc	L	m
r_2	Inner-wall radius of a tubular centrifuge	L	m
r_A	Volumetric rate of reaction with respect to component A	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
r_{Ab}^*	Volumetric rate of reaction with respect to component A at the bulk concentration	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
$r_{A,obs}$	Observed volumetric rate of reaction with respect to component A	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
r_{As}^*	Volumetric rate of reaction with respect to component A at the surface concentration	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
r_C	Volumetric rate of consumption by reaction	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
r_d	Volumetric rate of deactivation	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
r_m	Filter medium resistance	L^{-1}	m^{-1}
r_P	Volumetric rate of product formation	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
r_S	Volumetric rate of substrate consumption	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
r_X	Volumetric rate of cell growth	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
r_{X^+}	Volumetric rate of growth of plasmid-carrying cells	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
r_{X^-}	Volumetric rate of growth of plasmid-free cells	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
r_Z	Volumetric rate of reaction with respect to component Z	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
R	Ideal gas constant	$L^2MT^{-2}\Theta^{-1}N^{-1}$	$J\ mol^{-1}\ K^{-1}$
R	Radius	L	m
R	Thermal resistance	$L^{-2}M^{-1}T^3\Theta$	$K\ W^{-1}$
R	Amount of protein released in a homogeniser	M	kg
R	Total rate of reaction	MT^{-1}	$kg\ s^{-1}$
R_0	Radius at which substrate is depleted	L	m
R_A	Total rate of reaction with respect to component A	MT^{-1}	$kg\ s^{-1}$
R_c	Thermal resistance in cold fluid	$L^{-2}M^{-1}T^3\Theta$	$K\ W^{-1}$
R_C	Total rate of consumption by reaction	MT^{-1}	$kg\ s^{-1}$
R_G	Total rate of generation by reaction	MT^{-1}	$kg\ s^{-1}$
R_h	Thermal resistance in hot fluid	$L^{-2}M^{-1}T^3\Theta$	$K\ W^{-1}$
R_i	Inner radius	L	m

R_m	Resistance to mass transfer	T	s
R_m	Maximum amount of protein available for release in a homogeniser	M	kg
$R_{\max}$	Maximum particle radius	L	m
R_N	Resolution in chromatography	1	—
R_o	Outer radius	L	m
R_w	Thermal resistance of a wall	$L^{-2}M^{-1}T^3\Theta$	KW^{-1}
Re	Reynolds number defined in Eq. (7.1)	1	—
Re_i	Impeller Reynolds number defined in Eq. (7.2)	1	—
$Re_{\max}$	Maximum Reynolds number	1	—
Re_p	Particle Reynolds number defined in Eq. (12.48)	1	—
RQ	Respiratory quotient	1	—
s	Cake compressibility	1	—
s	Substrate concentration	$L^{-3}M$	$kg\ m^{-3}$
s_1	Substrate concentration in the first reactor	$L^{-3}M$	$kg\ m^{-3}$
s_2	Substrate concentration in the second reactor	$L^{-3}M$	$kg\ m^{-3}$
s_b	Bulk substrate concentration	$L^{-3}M$	$kg\ m^{-3}$
S	Mass of substrate	M	kg
S	Molar entropy	$L^2MT^{-2}\Theta^{-1}N^{-1}$	$J\ K^{-1}\ mol^{-1}$
S_G	Mass of substrate consumed for growth	M	kg
ΔS_{rxn}°	Molar entropy change during reaction under standard conditions	$L^2MT^{-2}\Theta^{-1}N^{-1}$	$J\ K^{-1}\ mol^{-1}$
S_R	Mass of substrate consumed other than for growth	M	kg
S_T	Total mass of substrate consumed	M	kg
S_x	External surface area	L^2	m^2
Sc	Schmidt number defined by Eq. (12.49)	1	—
Sh	Sherwood number defined by Eq. (12.50)	1	—
t	Time	T	s
t_1	Time at the end of the heating period	T	s
t_2	Time at the end of the holding period	T	s
t_b	Batch reaction time	T	s
t_c	Circulation time	T	s
t_d	Doubling time	T	s
t_{dn}	Total downtime	T	s
t_{fb}	Fed-batch time	T	s
t_h	Half-life	T	s
t_{hd}	Holding time	T	s
t_{hv}	Time taken to harvest culture	T	s
t_l	Lag time	T	s
t_m	Mixing time	T	s
t_p	Reactor-preparation time	T	s
t_T	Total batch reaction time	T	s
T	Temperature	Θ	K
ΔT_A	Arithmetic-mean temperature difference	Θ	K
T_c	Cold-fluid temperature	Θ	K
T_{cw}	Cold-fluid temperature at the wall	Θ	K
T_F	Fermenter temperature	Θ	K
T_h	Hot-fluid temperature	Θ	K
T_{hw}	Hot fluid temperature at the wall	Θ	K
ΔT_L	Logarithmic-mean temperature difference	Θ	K
T_{ref}	Reference temperature	Θ	K
T_S	Steam temperature	Θ	K
u	Linear velocity	LT^{-1}	$m\ s^{-1}$
u	Interstitial velocity	LT^{-1}	$m\ s^{-1}$
u	Specific internal energy	L^2T^{-2}	$J\ kg^{-1}$
u_c	Sedimentation velocity in a centrifuge	LT^{-1}	$m\ s^{-1}$
u_g	Sedimentation velocity under gravity	LT^{-1}	$m\ s^{-1}$

u_G	Gas superficial velocity	LT^{-1}	$m\ s^{-1}$
u_L	Liquid linear velocity	LT^{-1}	$m\ s^{-1}$
u_L	Liquid superficial velocity	LT^{-1}	$m\ s^{-1}$
u_L^*	Liquid velocity at which reaction rate becomes independent of liquid velocity	LT^{-1}	$m\ s^{-1}$
u_{pL}	Velocity of a particle relative to liquid	LT^{-1}	$m\ s^{-1}$
U	Internal energy	L^2MT^{-2}	J
U	Overall heat-transfer coefficient	$MT^{-3}\Theta^{-1}$	$W\ m^{-2}\ K^{-1}$
v	Specific volume	L^3M^{-1}	$m^3\ kg^{-1}$
v	Velocity	MT^{-1}	$m\ s^{-1}$
v	Volumetric rate of reaction	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
v_{max}	Maximum volumetric rate of reaction	$L^{-3}MT^{-1}$	$kg\ m^{-3}\ s^{-1}$
V	Volume	L^3	m^3
V_1	Volume of the first reactor	L^3	m^3
V_2	Volume of the second reactor	L^3	m^3
V_e	Volume of eluant	L^3	m^3
V_f	Volume of filtrate	L^3	m^3
V_G	Volume of gas	L^3	m^3
V_i	Internal volume	L^3	m^3
V_l	Volume of lower phase	L^3	m^3
V_L	Volume of liquid	L^3	m^3
V_o	Void volume	L^3	m^3
V_p	Particle volume	L^3	m^3
V_s	Volume of solid	L^3	m^3
V_T	Total volume	L^3	m^3
V_u	Volume of upper phase	L^3	m^3
w	Baseline width	L	m
W_i	Impeller blade width	L	m
W_r	Water regain value	L^3M^{-1}	$m^3\ kg^{-1}$
W_s	Shaft work	L^2MT^{-2}	J
$\dot{W}_s$	Rate of shaft work	L^2MT^{-3}	W
W_f	Flow work	L^2MT^{-2}	J
x	Distance	L	m
x	Cell concentration	$L^{-3}M$	$kg\ m^{-3}$
x^+	Concentration of plasmid-carrying cells	$L^{-3}M$	$kg\ m^{-3}$
x^-	Concentration of plasmid-free cells	$L^{-3}M$	$kg\ m^{-3}$
$\bar{x}$	Mean value of x		
x_1	Cell concentration in the first reactor	$L^{-3}M$	$kg\ m^{-3}$
x_2	Cell concentration in the second reactor	$L^{-3}M$	$kg\ m^{-3}$
x_C	Number of carbon atoms in the molecular formula	1	—
x_{im}	Concentration of immobilised cells	$L^{-3}M$	$kg\ m^{-3}$
x_{max}	Maximum cell concentration	$L^{-3}M$	$kg\ m^{-3}$
x_r	Concentration of cells in the recycle stream	$L^{-3}M$	$kg\ m^{-3}$
x_s	Concentration of suspended cells	$L^{-3}M$	$kg\ m^{-3}$
X	Mass of cells	M	kg
y	Distance	L	m
y	Wave displacement	L	m
y_{AG}	Mole fraction of component A in gas	1	—
y_C	Weight fraction of cells	1	—
Y_l	Yield in the lower phase	1	—
Y_{PS}	True yield of product from substrate	1	—
Y'_{PS}	Observed yield of product from substrate	1	—
Y_{PX}	True yield of product from biomass	1	—
Y'_{PX}	Observed yield of product from biomass	1	—
Y_u	Yield in the upper phase	1	—
Y_{XO}	Yield of biomass from oxygen	1	—
Y_{XS}	True yield of biomass from substrate	1	—

Y'_{XS}	Observed yield of biomass from substrate	1	–
$Y_{XS,max}$	Maximum yield of biomass from substrate	1	–
z	Distance	L	m
z	Concentration of cellular constituent Z	$L^{-3}M$	$kg\ m^{-3}$
z_i	Valency of ionic component i	1	–
Z	g-number in centrifugation	1	–

Script symbols

$\mathcal{D}$	Diffusion coefficient	L^2T^{-1}	$m^2\ s^{-1}$
$\mathcal{D}_{AB}$	Binary diffusion coefficient of component A in component B	L^2T^{-1}	$m^2\ s^{-1}$
$\mathcal{D}_{Ac}$	Effective diffusivity of component A	L^2T^{-1}	$m^2\ s^{-1}$
$\mathcal{D}_{AL}$	Binary diffusion coefficient of component A in liquid	L^2T^{-1}	$m^2\ s^{-1}$
$\mathcal{D}_{Aw}$	Binary diffusion coefficient of component A in water	L^2T^{-1}	$m^2\ s^{-1}$
$\mathcal{D}_{Sc}$	Effective diffusivity of substrate	L^2T^{-1}	$m^2\ s^{-1}$
$\mathcal{D}_z$	Axial-dispersion coefficient	L^2T^{-1}	$m^2\ s^{-1}$

Greek symbols

α	Specific cake resistance	LM^{-1}	$m\ kg^{-1}$
α	Exponent in Eq. (10.23)	1	–
α	Ratio defined in Eq. (11.65)	1	–
α	Recycle ratio defined in Eq. (13.77)	1	–
α'	Parameter defined in Eq. (10.2)	1	–
β	Thermal coefficient of linear expansion	Θ^{-1}	K^{-1}
β	Dimensionless parameter equal to K_m/C_{As}	1	–
β	Biomass concentration factor defined in Eq. (13.78)	1	–
γ	Degree of reduction	1	–
$\dot{\gamma}$	Shear rate	T^{-1}	s^{-1}
$\dot{\gamma}_{av}$	Average shear rate	T^{-1}	s^{-1}
γ_B	Degree of reduction of biomass	1	–
γ_P	Degree of reduction of product	1	–
γ_S	Degree of reduction of substrate	1	–
δ	Selectivity in chromatography	1	–
δ_c	Concentration factor	1	–
Δ	Difference	1	–
ε	Fractional gas hold-up	1	–
ε	Porosity	1	–
ε	Void fraction	1	–
ε	Rate of turbulent energy dissipation per mass of fluid	L^2T^{-3}	$W\ kg^{-1}$
ε	Time interval	T	s
ζ_B	Fraction of available electrons transferred to biomass	1	–
η_e	External effectiveness factor	1	–
η_{e0}	External effectiveness factor for zero-order reaction	1	–
η_{e1}	External effectiveness factor for first-order reaction	1	–
η_{em}	External effectiveness factor for Michaelis–Menten reaction	1	–
η_i	Internal effectiveness factor	1	–
η_{i0}	Internal effectiveness factor for zero-order reaction	1	–
η_{i1}	Internal effectiveness factor for first-order reaction	1	–
η_{im}	Internal effectiveness factor for Michaelis–Menten reaction	1	–
η_T	Total effectiveness factor	1	–

η_{T1}	Total effectiveness factor for first-order reaction	1	—
θ	Half-cone angle	1	rad
λ	Kolmogorov scale	L	m
μ	Specific growth rate	T^{-1}	s^{-1}
μ^+	Specific growth rate of plasmid-carrying cells	T^{-1}	s^{-1}
μ^-	Specific growth rate of plasmid-free cells	T^{-1}	s^{-1}
μ	Viscosity (dynamic)	$L^{-1}MT^{-1}$	Pa s
μ_a	Apparent viscosity	$L^{-1}MT^{-1}$	Pa s
μ_b	Bulk-fluid viscosity	$L^{-1}MT^{-1}$	Pa s
μ_f	Filtrate viscosity	$L^{-1}MT^{-1}$	Pa s
μ_L	Liquid viscosity	$L^{-1}MT^{-1}$	Pa s
μ_{max}	Maximum specific growth rate	T^{-1}	s^{-1}
μ_w	Fluid viscosity at the wall	$L^{-1}MT^{-1}$	Pa s
ν	Kinematic viscosity	L^2T^{-1}	m^2s^{-1}
ν_L	Liquid kinematic viscosity	L^2T^{-1}	m^2s^{-1}
π	3.14159	1	—
ρ	Density	$L^{-3}M$	$kg\,m^{-3}$
ρ_f	Fluid density	$L^{-3}M$	$kg\,m^{-3}$
ρ_g	Density of wet gel	$L^{-3}M$	$kg\,m^{-3}$
ρ_G	Gas density	$L^{-3}M$	$kg\,m^{-3}$
ρ_L	Liquid density	$L^{-3}M$	$kg\,m^{-3}$
ρ_p	Particle density	$L^{-3}M$	$kg\,m^{-3}$
ρ_w	Density of water	$L^{-3}M$	$kg\,m^{-3}$
σ	Surface tension	MT^{-2}	$N\,m^{-1}$
σ	Standard deviation		
Σ	Summation		
Σ	Centrifuge sigma factor defined in Eq. (10.18)	L^2	m^2
τ	Average residence time	T	s
τ	Shear stress	$L^{-1}MT^{-2}$	Pa
τ_0	Yield stress	$L^{-1}MT^{-2}$	Pa
ϕ	Angle	1	rad
ϕ	Thiele modulus	1	—
ϕ_0	Thiele modulus for zero-order reaction	1	—
ϕ_1	Thiele modulus for first-order reaction	1	—
ϕ_m	Thiele modulus for Michaelis–Menten reaction	1	—
Φ	Observable Thiele modulus	1	—
ψ	Volume fraction of solids	1	—
Ψ	Parameter defined in Table 12.3	1	—
ω	Angular velocity	T^{-1}	$rad\,s^{-1}$
Ω	Angular velocity	T^{-1}	$rad\,s^{-1}$
Ω	Observable modulus for external mass transfer	1	—

Subscripts

0	Initial
f	Final
i	Inlet
i	Interface
L	Logarithmic mean
o	Outlet

Superscripts

*	Equilibrium with prevailing value in the other phase
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